NETWORK FOR ELECTRONIC TRANSFERS (S) PTE LTD

NETS ECR II
Terminal Interface

Technical Specification Document
V1.27

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Change History

Version No.	Revision No.	Description	Approval Date
1.01	1	Initial Draft	
1.02	1	• Added function codes: ○ Reversal (14) ○ Get Last Approved Transaction (56) ○ Set Parameter (82) ○ Get Parameter (83) ○ TMS (84) ○ Cancel (XX) • Added Annex D for terminal parameter list • Added Card Entry Mode on: ○ NETS Purchase ○ NETS Void ○ Loyalty Enquiry ○ UPI Purchase ○ UPI Void ○ UPI Refund ○ NCC Topup ○ NFP Topup ○ Credit Card Sale ○ Credit Card Refund ○ Credit Card Void Sale • Modified Section 4.2 Function Codes • Editorial changes on selection callback • Editorial changes on protocol illustration • Added Use Case section • ECR Reference Number is added to the response: ○ NETS Purchase ○ NETS Void ○ Loyalty Enquiry ○ UPI Purchase ○ UPI Void ○ UPI Refund ○ NCC Purchase ○ NCC Topup ○ NFP Purchase ○ NFP Topup ○ BCA Purchase ○ BCA Void ○ Credit Card Sale ○ Credit Card Refund ○ Credit Card Void Sale	15-Apr-16
1.02	2	• Editorial change on the EFTPOS Use Case (3.4.1) • Editorial change on the selection callback (4.10.3) • Added “PE” communication type on the parameter setting. (Annex D)	19 April 2016
1.02	3	• Added message description for ETH connection (2.1.4.1)	3 May 2016
1.02	4	• Added NFP and NCC count and amount for Settlement. (Function Code 81)	3 June 2016

Version No.	Revision No.	Description	Approval Date
1.03	1	• Editorial changes for Credit Card Settlement • Added new field codes for Credit Sale o D7 – Processing Gateway o D8 – Card Loyalty Name	17 August 2016
1.03	2	• D8 – Card Loyalty Name change to Card Description	19 August 2016
1.04	1	• Added acquirer name on I0 • I5 credit card settlement o IPP sale count change from G0 to N0 o IPP sale grand total change from G1 to G0 • Changed Field code HC as card balance to HE	9 September 2016
1.05	1	• Added Pre-Settlement Command for NETS • Added Pre-Settlement Command for Credit Card • Added Loyalty Data to Credit Card Sale (dynamic) 4.9.1 o Field code 90 Loyalty Award o Field code 91 Loyalty Redeem o Field code 92 Loyalty Balance • Added Acquirer Name for credit card settlement 4.9.4	20 September 2016
1.06	1	• Modify 9G from 10 to 20 characters for I0 • Added topup for NCC and NFP count and totals for settlement command 81 • Added cashback count and total to settlement command 81 • Editorial change for pre-settlement (86) • Added topup for NCC and NFP count and totals for pre-settlement command 86 • Modify Settlement for Credit Card: double field code 9K. 9P is added for IPP host count. • Editorial changes on field code list Annex B 9K is Host ID • Added 9P, NB, GB, NC, GC, ND, GD, NE, GE, NF, GF on the field code list Annex B. • Credit Card Settlement Merchant ID field changed from field code 33 to D1 for consistency	28 September 2016
1.06	2	• Added STAN (65) for UPI Purchase transaction	21 November, 2016
1.07	1	• Increase version number for Credit Card API to include Category code.	03 March 2017
1.07	2	• Added Annex E : List of Acquirer Name • Added Annex F : List of Scheme Category	09 March 2017
1.08	1	• Added 4.5.4 Cash Deposit	21 March 2017
1.08	2	• Add “Get Card Info (Function Code S1)” to support POS to get the card information • Removed Field Code 90, 91 and 92. Converge the loyalty information into 2 fields to support multiple loyalty provider. Field 93 to indicate loyalty type, Field 94 to indicate loyalty information. • Added a new flag under NETS Purchase to indicate QR Code payment mode. This flag is added in order to minimize the impact to existing EFTPOS integrated merchant	31 March 2017

Version No.	Revision No.	Description	Approval Date
1.08	3	• Added field “Card Type” under “Credit Card Sale” response. This is to allow the terminal to return card type to POS whether Visa/MasterCard/Amex/JCB • Added field “Card Type” under “Get Card Info” request. This is to allow the POS to specifically decide which type of card the terminal should get the info from.	04 April 2017
1.08	4	• Add a section, Annex G to describe how an external device (POS) may pass through ECR command for non-NETS services	06 April 2017
1.08	5	• Added response structure for 3rd party command	10 April 2017
1.08	6	• Editorials changes • Added Acquirer Logon (Function Code A1) • Added Acquirer Settlement (Function Code A2) • Added CEPAS Debit (Function Code 26) • Added CEPAS Top Up (Function Code 47) • Added Prepaid Balance Inquiry (Function Code P0) • Added Prepaid Purchase (Function Code P1) • Added FEVO Load/Reload (Function Code F0) • Added FEVO Cash Out with Purchase (Function Code F1) • Added additional fields to support new added functions. Refer to Annex B	27 April 2017
1.08	7	• Added UOB loyalty structure in the Credit Sale response	13th July 2017
1.08	8	• Added Diners loyalty structure in Credit Sale response	17th Aug 2017
1.09	1	• Added RSVP functions section 4.11 o RSVP Get Group RID o RSVP Parameter Download o RSVP Issuance o RSVP Redemption o RSVP Balance Enquiry o RSVP Balance Enquiry with NFP card status	13th April 2018
1.10	1	• Added Connectivity test section (8A) 4.3.13 • Added Retrieve last signature image (IA) section 4.10.6 • Added Single Entry Mode	8th May 2018
1.10	2	• Editorial change on the retrieve last signature	14th May 2018
1.11	1	• Added IPP Commands: o IPP Sale (I7) o IPP Void (I8)	7th June, 2018
1.11	2	• Changed I0 version code from 03 to 01	27th June 2018
1.11	3	• Removed 3.1.4 retry transaction using ECR reference number.	3rd July 2018
1.11	4	• Added cash deposit count and total for settlement	14th Aug 2018

Version No.	Revision No.	Description	Approval Date
1.11	5	• Added Annex H. RSVP Response Code	2nd Oct 2018
1.12	1	• Added Cashback amount for NETS Purchase (Function Code 30) response	2nd Oct 2018
1.13	1	• Added supported functions on the Get Last Approved Transactions (Function Code 56) • Added Receipt Text Format as an optional field for the Acquirer Settlement (Function Code A2) • Added Acquirer Pre-Settlement (Function Code A3) • Added value for CEPAS Version 3.0 for CEPAS functions (CEPAS Debit, Topup and Void) • Added Stan as an optional field for the CEPAS Debit for use on CEPAS Void (Function Code 48) and Get Debit Transaction Status (Function Code 49) • Added Get Debit Transaction Status (Function Code 49) • Added Field Code “CP” to be used for Transaction Status	
1.14	1	• Updated NETS Purchase response • Added Section 4.17 for NPC o Added Balance Inquiry for NPC o Added Topup for NPC • No before balance for purchase and topup NPC.	03 Mar 2021
1.14	2	• Updated Credit Sale (I0) o Added Offline Txn Type (O1) Field Code for offline transaction types. • Added Field Code Card Entry Mode 2 (CQ)	01 April 2021
1.14	3	• Updated UOB Loyalty data	05 April 2021
1.14	4	• Updated 4.3.4 Settlement return message for NPC topup via NETS and Cash count and totals. • Updated 4.3.8 Pre-Settlement to include NPC topup via NETS and Cash count and totals.	25 May 2021

Version No.	Revision No.	Description	Approval Date
1.15	1	• Removed FEVO section • Removed RSVP section • Removed Merchant Prepaid section • Removed CEPAS section • Removed NETS Loyalty Inquiry (70) • Added sample message for Request and Response message structure • Updated NETS Function code 30 ○ Removed loyalty redemption field ○ Removed QR payment option in transaction type indicator ○ Removal of response Field code L1, L2, L3, L4, L8, L9 ○ Response Field 02 will contain CAN, before balance and after balance ○ Updated other fields to reflect NFP and Cashcard info • Removed Cash card purchase (51) • Removed NFP Purchase (24) • Updated Connectivity Test to exclude RSVP and Merchant Prepaid host.	12 Aug 2021
1.15	2	• Removal of NPC Balance and placed it in 02 instead. • Revert back Loyalty fields L1, L2, L3, L4, L8, L9 • Update section 4.2	23 Aug 2021
1.15	3	• Added Section 4.14 Rupay Transactions • Added Section 4.15 MyDebit Transactions • Editorial change for Function 00 • Deprecated 4.10.1 and 4.10.3	16 Sep 2021
1.15	4	• Added note for 4.10.3 regarding ACK and timeout. • Added Paynow functions Section 4.16 • Added additional field code supported by Paynow	7 Oct 2021
1.16	2	• Added new message in display info callback.	14 Dec 2021
1.16	3	• Removed Extra C2 on NPC Topup command	29 Dec 2021
1.17	1	• Added Pre-auth, Pre-auth void and Pre-auth Completion • Added NPC Activation	25 Jan 2022
1.17	2	• Added Paynow Pre-Summary Totals, Summary Totals and Last Summary Totals Command	9 Feb 2022
1.17	3	Editorial • Removed extra Field 31	3 Mar 2022
1.18	1	Added Sections: 1. 4.8.2 NFP Card Enquiry 2. 4.8.3 NFP Transaction Log 3. 4.8.4 Debit Statement Fee 4. 4.8.5 NFP ATU Activation	22 Mar 2022

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1.18	2	Added Qwikcilver commands in 3rd Party passthrough	23 Mar 2022
1.19	1	• Added CEPAS Section for EzLink ○ Acquirer Logon ○ Acquirer Settlement ○ Acquirer Pre-Settlement ○ CEPAS Purchase ○ CEPAS Top Up ○ CEPAS Void ○ CEPAS Balance Inquiry • Added NPC Activation count on NETS Settlement	12 May 2022
1.19	2	• Updated Annex G to include OCBC, EZLink and Diners version code.	8 June 2022
1.20	1	• Editorial Changes in section 4.3.2 to add paynow and single entry mode in get last approved. • Added response data for Function Code 00 in section 4.12.1 • Removed I0 • Editorial changes for topup option	25 Aug 2022
1.21	1	• Removed Paynow commands	7 Sep 2022
1.22	1	• Return I0 command	21 Oct 2022
1.22	2	• Added field code ED for NPC topup for previous balance that may contain negative sign. • Added field code EF – negative indicator for previous balance for NPC	28 NOV 2022
1.22	3	• Added field code EF for NPC Balance Inquiry.	5 JAN 2023
1.22	4	• Added Field Code EG for Current Balance Negative Indicator	28 FEB 2023
1.22	5	• Added back Transaction Type Indicator ‘04’ for NETS QR Purchase	27 JUN 2023
1.23	1	• Removed unsupported legacy functions • Updated CEPAS void to C1 and CEPAS transaction status to C2. • Added 2 NFP Refund (FC: 4C) and NFP Card Sale Activation (FC: 4D) • Added ECR on POS, Request QR payload (FC: Q1) and QR payment (FC: Q2)	3 OCT 2023
1.23	2	• Added ECR Reference Number as mandatory • Change NETS Purchase Field code L7 length to 20. • Change Card Balance Field code to HE from HC for NFP card enquiry.	19 APR 2024
1.23	3	• Added RP on Credit Settlement (I5) and Pre-Settlement (I6)	12 JUN 2024

Version No.	Revision No.	Description	Approval Date
1.24	1	• Consumer Choice Routing (**CCR**) function codes • Function Code P0 – CCR Purchase green button • Function Code P1 – CCR Purchase with selection • Function Code P2 – CCR Proceed as NETS • Function Code P3 – CCR Proceed as ICS (Credit)	22 Sep 2024
1.24	2	• Updated field codes for CCR Functions • Updated response for Function Codes P2 and P3	09 Oct 2024
1.24	3	• Updated CCR Implementation Diagram Flow	10 Oct 2024
1.24	4	• Added field code 9R to differentiate card type returned by host • Added 9R in response of Function Codes I0, P0, P1, P3	10 Dec 2024
1.25	1	• Added NETS Refund • Added Acquirer Name in ECR request of CCR Functions	21 Feb 2025
1.25	2	• Updated NETS Refund request and response	05 Mar 2025
1.26	1	• Added NETS Offline via UPOS NPX	26 Mar 2025
1.26	2	• Added DFI Read TFM LID	04 Apr 2025
1.27	1	• Added New Loyalty Type for DBS Instant Rewards • Moved Loyalty Information in the Annex section	19 Aug 2025
1.27	2	• Added new ECR function for Merchant QR or Tap • Remove BCA, MyDebit and Rupay functions from the specification as no longer supported • Adding note for response approval validation to use response code + approval code (Annex C) • Updated list of mandatory functions for merchant POS implementation (Annex I)	8 Dec 2025
1.27	3	• Added Acquirer TMS Update	15 Jan 2026
1.27	4	• Added Pre-authorization, Pre-authorization Completion, Pre-authorization Void	16 Feb 2026
1.27	5	• Editorial and terminology update • Removal of P0 command • Updated CCR Implementation Diagram Flow	12 May 2026

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Terminology and Convention

Term	Description
ASC	ASCII character. Represent 8 bits.
BCD	Binary Coded Decimal format. Represent 4 bits
HEX	Hexadecimal format. Represent 4 bits.
NFP	NETS FlashPay
NCC	NETS Cash Card
UPI	Union Pay International
BCA	Bank Central Asia
IPP	Installment Payment Plan

1. Introduction

This document provides information on how an external device, such as a cash register, can handle the requirements of Network for Electronic Transfer Singapore (NETS) in terms of a POS interface and communicate effectively with secure terminal module for the purpose of financial debit transactions involving a NETS EFTPOS DEBIT CARD (ATM Card) and Contactless Card. The interface specifications will include communications protocol, as well as payment procedures using different methods. Initialization and settlement procedures of the secure terminal module will be performed independently from the external device.

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2. Determine System Design Strategy

2.1 Message Exchange Sequence

The external module initiates the message exchange sequence. The device requests the terminal to perform a particular task. This is acknowledged at the link level by the terminal, which then performs the task. On completion of the task, the terminal responds to the external module with data pertinent to the task performed and the external module returns an acknowledgement at the link level.

2.1.1 Communication Protocol

External module can interface to the terminal via:

1. RS232
2. Ethernet

2.1.2 RS232 Communication Parameter

The terminal model interfaces through an asynchronous serial communications port with the following port parameters:

Baud Rate	:	9600
Parity	:	None
Data	:	8 Bits, 1 Stop Bit
Retries	:	3

Please note that the terminal ECR connection shall only be connected to standard RS-232 serial communication port.

2.1.3 Ethernet Communication Parameter

External module must ensure that firewall rules are allowed for the external module to communicate to the terminal.

The terminal IP address is pre-defined and the external module must know what the terminal IP is. In this case, static IP is suggested to ensure that IP address is fixed for the terminal.

Terminal should be able to set the port number to listen for the commands.

2.1.3.1 Ethernet Communication Message Protocol

Note: Ethernet mode will not include ACK and NACK mechanism.

Message Length	Message Body	LRC

Message Data Structure

The message structure is consisting of:

Name	Attribute	Len (Bytes)	Remark
Message Length	BCD	2	Total length of the message body and CRC
Message Body	VAR	VAR	Message body is consisting of message header and message data see below for further details.
LRC	BIN	1	XOR of every byte of message body. 00193030303030303032353233353530313 01C2B

2.1.4 Control Character

The following four control characters are used to control the communication protocol. All the proposed values are in hexadecimal format.

Start of Text	STX	0x02
End of Text	ETX	0x03
Acknowledgement	ACK	0x06
Negative Acknowledgement	NACK	0x15

2.1.5 Message Checksum

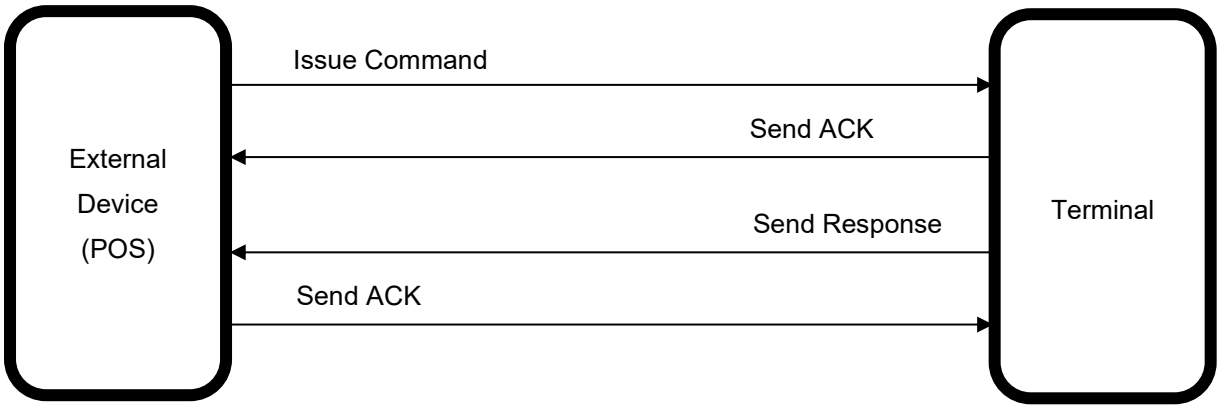
All messages have a Long Range Checksum (LRC) appended at the end of the message, following the ETX control character. The LRC is an exclusive OR of all data bytes contained in between STX and ETX. The LRC operation includes the ETX control character, but not the STX. The Start value for computing the LRC is hexadecimal 0x00.

STX	Length of Message	Message Data bytes ...	ETX	LRC

Following the message checksum rule, The Long Range Checksum of the above sample message is computed by performing an exclusive OR of the grey-shaded data. The Length of Message field is a 2-byte BCD format value, starting with the most significant byte. For example, a message of 128 bytes would have this content: **01 28** in the Length of Message field.

2.2 Message Request and Response Protocol

2.2.1 Communication Protocol between external device (POS) and terminal



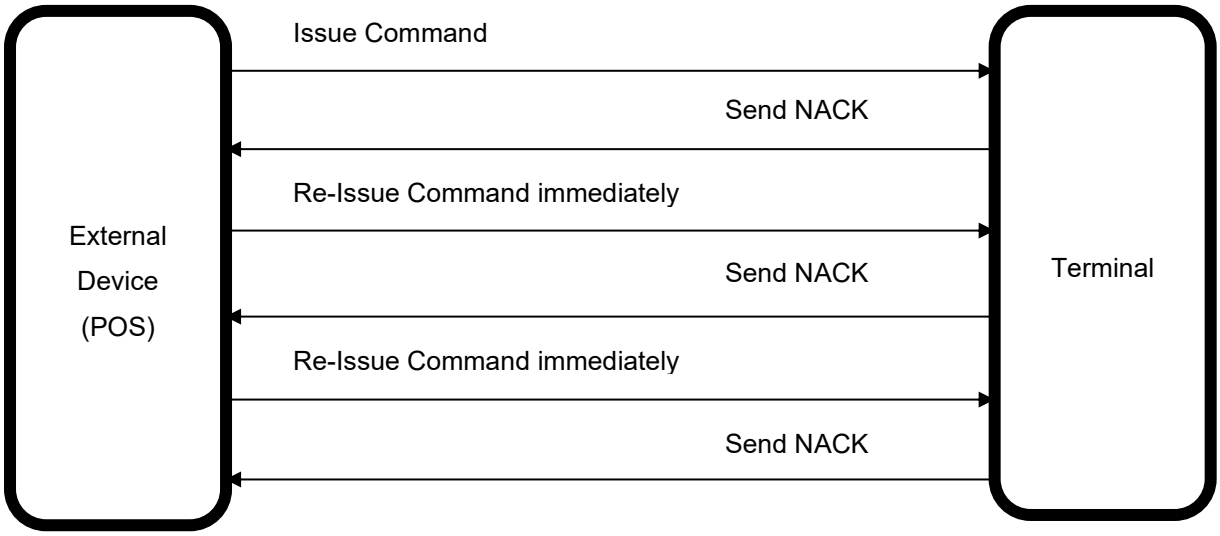
Under normal scenario, when the external device issues a command to terminal; it waits for the acknowledgement from the terminal. Upon received the command, the terminal should validates the validity of the message by checking its' command data length and LRC. If the data length and the message LRC are correct, the terminal responses to the external device with an acknowledgement (ACK).

With the external device received acknowledgement from the terminal, it waits for the message response from the terminal. At the terminal side, it processes the request message and then sends the response message to the external device. Depending on the nature of the request message, the terminal may needs to send online message to NETS host for approval which takes longer time for processing, compare to CASH CARD debit transaction which is offline transaction that take lesser time.

When the external device had received the message response from the terminal, it validates the validity of the response message by checking the message LRC. If the LRC is correct, it acknowledges to terminal to confirm that the response message had been received successfully.

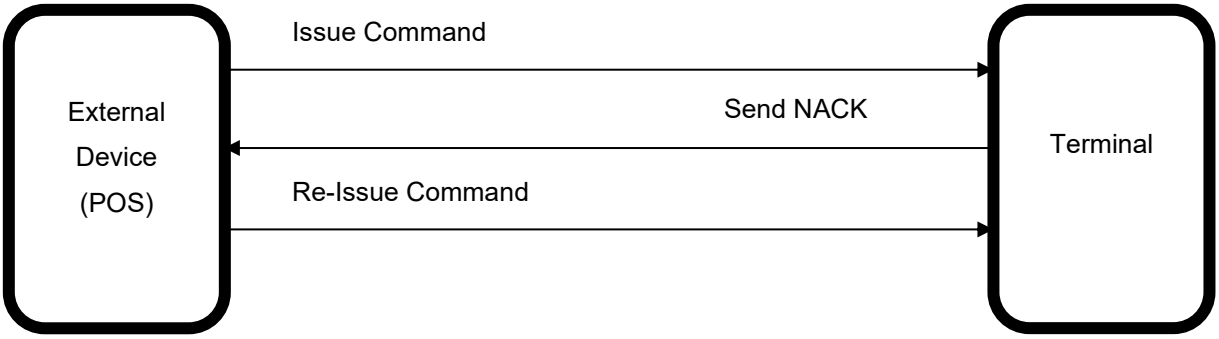
2.2.2 Communication Protocol between external device (POS) and terminal for exception handling

2.2.2.1 NACK is received by the external device



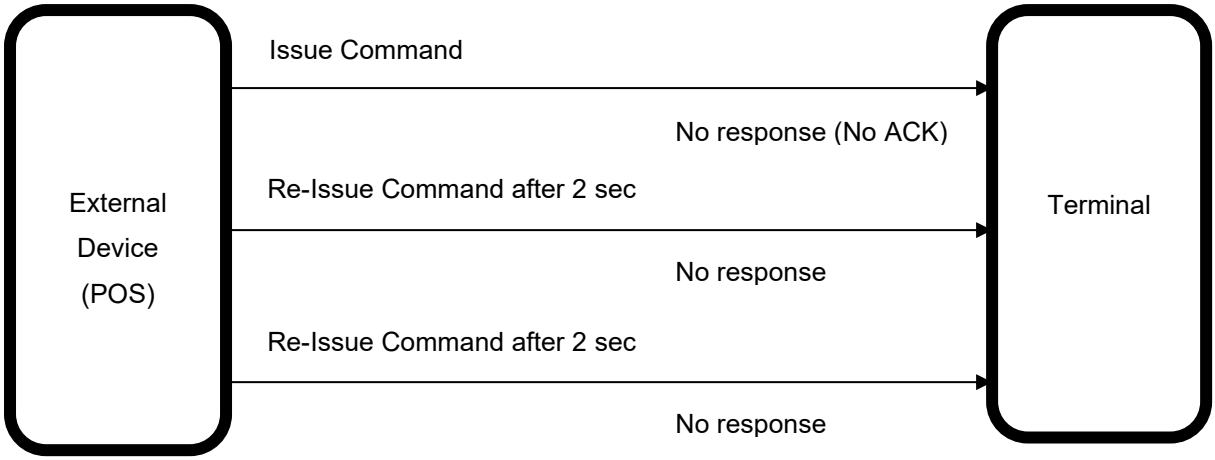
When the terminal validates that the request message (command) received is corrupted such as it fails the LRC checking, the terminal will send a negative acknowledgement (NACK) to the external device to request for message re-send. The external device, upon received the negative acknowledgement should re-send the request command again to terminal immediately. The external device only serves the NACK for two times, by re-send the command twice. If a NACK is received after the External device has made the final re-send, the transaction is consider unsuccessful and the external device should check for any error in the communication setup, fix the error and then re-issue another request message again.

2.2.2.2 NACK on Ethernet Mode



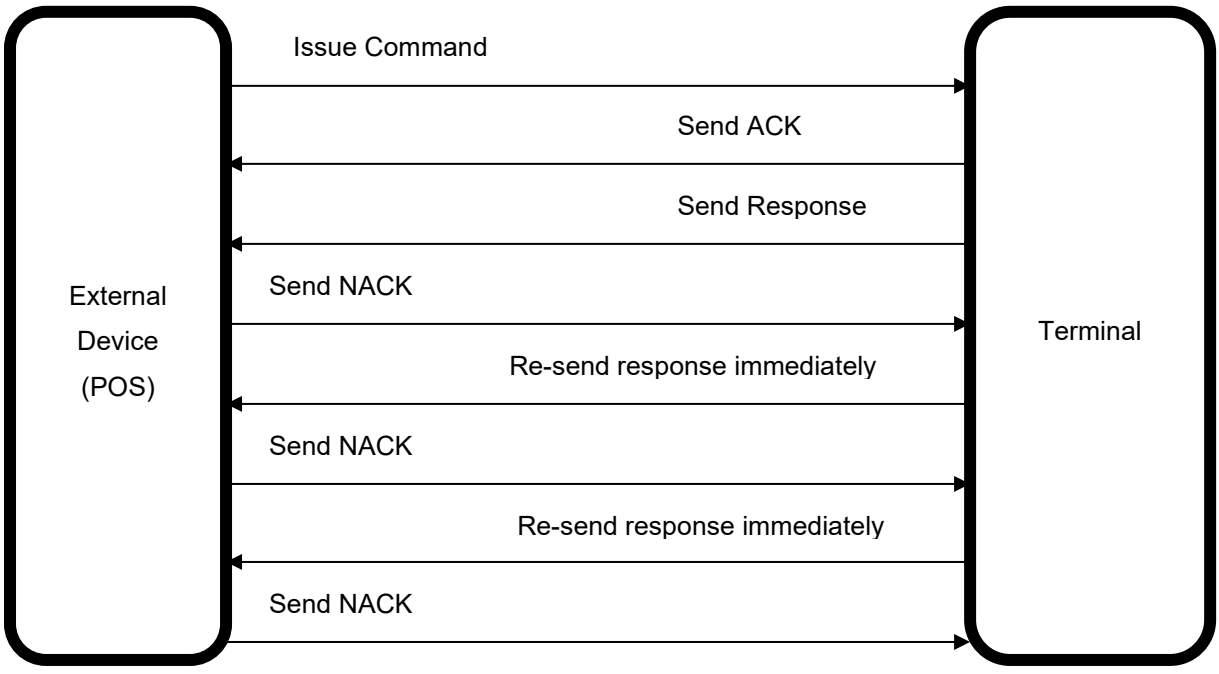
For Ethernet Mode (ECR via LAN), there is a parameter to enable the terminal to send a negative acknowledgement when terminal is busy and is unable to accept a new request for the time being. By default, this parameter is disabled. Upon receiving the negative acknowledgement, external device will need to re-issue a new command to retry the transaction.

2.2.2.3 No response from terminal after External Device has issued command



When the request message (command) is sent to terminal, but no acknowledgement is received by the external device from the terminal within a wait period, the external device should re-send the request command again to terminal. With each re-sends, the external device waits for acknowledgement for 2s. The external device only re-sends the command twice. After it re-send the command message for second time and waits for 2s, if no acknowledgement is received from terminal, the transaction is considered unsuccessful and the external device should check for any error in the communication setup, fix the error and then re-issue another request message again.

2.2.2.4 NACK is received by the terminal



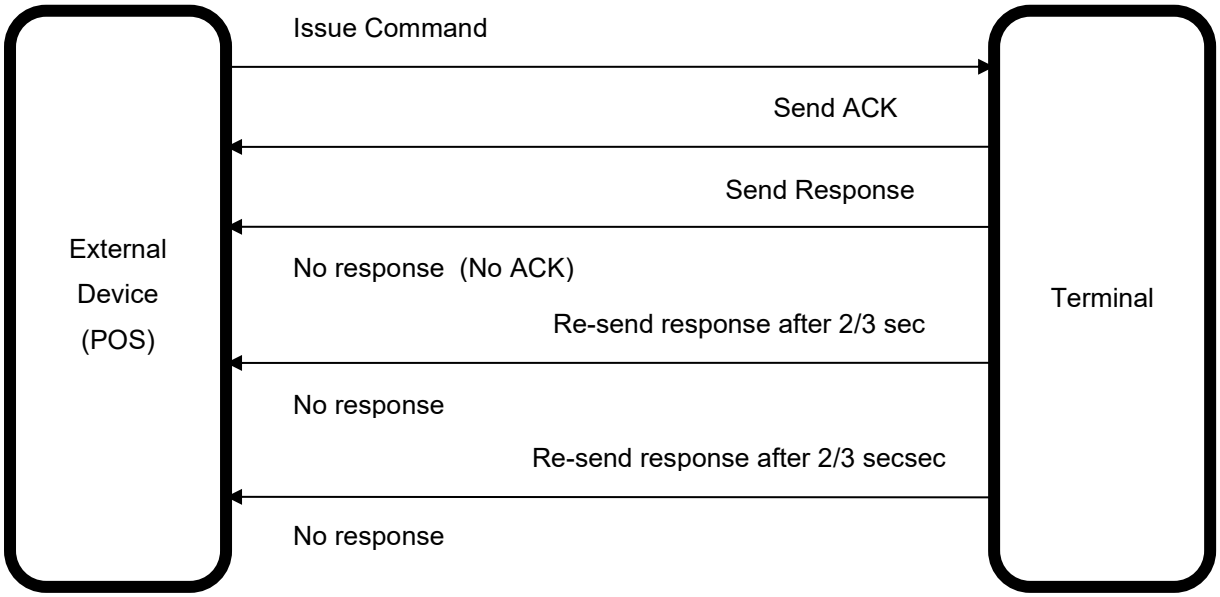
When the external device validates that the response message form the terminal is corrupted such that it fails the LRC check, the external device should send a negative acknowledgement (NACK) to the terminal to request the terminal to re-send the response message. The terminal, upon received the negative acknowledgement should re-send the response message to external device immediately. The

terminal will only re-sends the response message twice in response to the NACK received. If the 2nd resend message received by the external device still is corrupted,

For such situations it is recommended to perform the following:

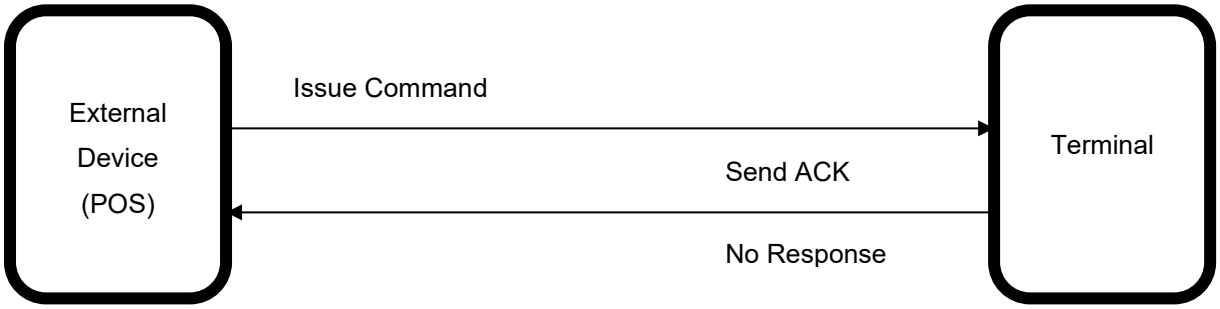
1) Check on terminal receipt for the payment status and POS close and process accordingly.
2) Repeat same payment with the same ECN
3) **Check on last approved transaction**
4) **Check on the response code** and if it's approved, POS proceed to close the payment.

2.2.2.5 No response from External Device after terminal has sent the response message



When the terminal sends response message to the external device and is expecting acknowledgement from the external device, but no acknowledgement is received within a wait period; the terminal should re-send the response message again to the external device, and waits for the acknowledgement for 2s. The terminal only re-sends the response message twice. After the terminal makes the final re-sends and has waited for 2s, if no acknowledgement is received from external device, the transaction is considered unsuccessful and the external device should check for any error in the communication setup.

2.2.2.6 No response from terminal after Acknowledgement is received by the External Device



After the External Device has received Acknowledgement from the terminal, it should set a timeout period while waiting for the response message from the terminal. The recommended time-out duration for the external device is:

a) 2 Minutes for Online Transactions.
b) 1 Minute for Offline Transactions.

When the External Device is unable to receive the response message from the terminal and time-out, user is advised to check on the NETS transaction receipt to see if the transaction is approved or declined.

If the transaction is approved, the application should take necessary step to close the transaction on the External Device's application side. If the transaction is declined, the user can then retry the transaction again.

In case of terminals with no receipt printing or receipt printing is turned off, the External Device can retry sending the same command with the same **ECN** (describe below). If the previous transaction has been processed, terminal will not process the transaction again, instead terminal will just return the response immediately. If the previous transaction was not processed at all, terminal will proceed to process it as a new transaction.

If external device supports the **ECR Reference Number** for every financial transaction, the reference number can also be used with the same function as the ECN. When the last ECR Reference number (ECN) is approved but external device did not receive the response, sending the same ECR Reference number (ECN) will allow terminal to resend the response without performing a new transaction. However, if the previous transaction was not performed, terminal will perform the transaction as a new request.

Get last approved transaction is also available if external device does not wish to perform a transaction retry and just wanted to query the last approved transaction.

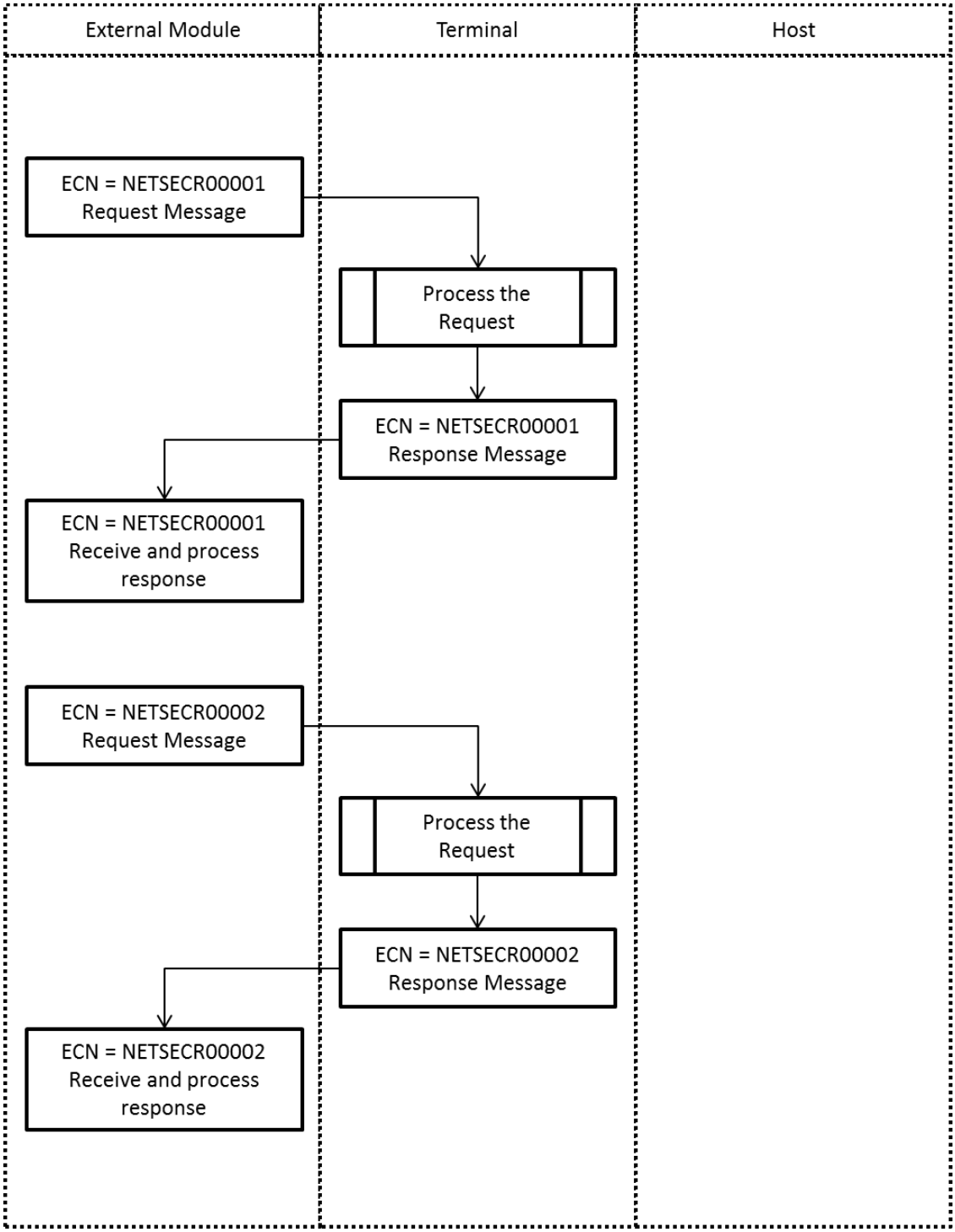
3. ECN Use Case Scenario

This section describes the use cases for various transaction and maintenance functions.

3.1 General Use Cases

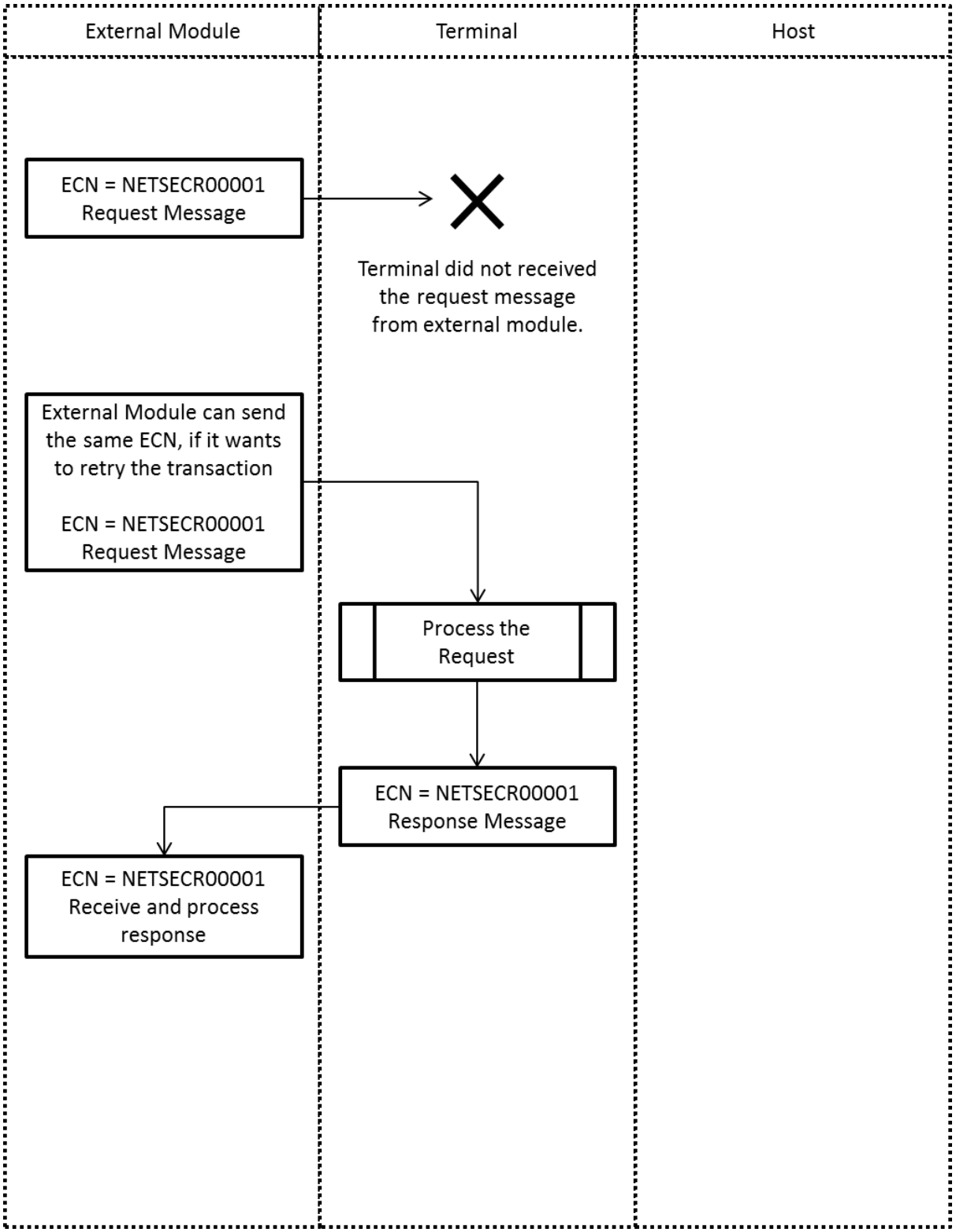
3.1.1 ECR Control Number (ECN) Use Mechanism

ECN shall be incremented to ensure data exchange integrity. This number is used by the terminal and ECR devices to keep track for request and response message pair.



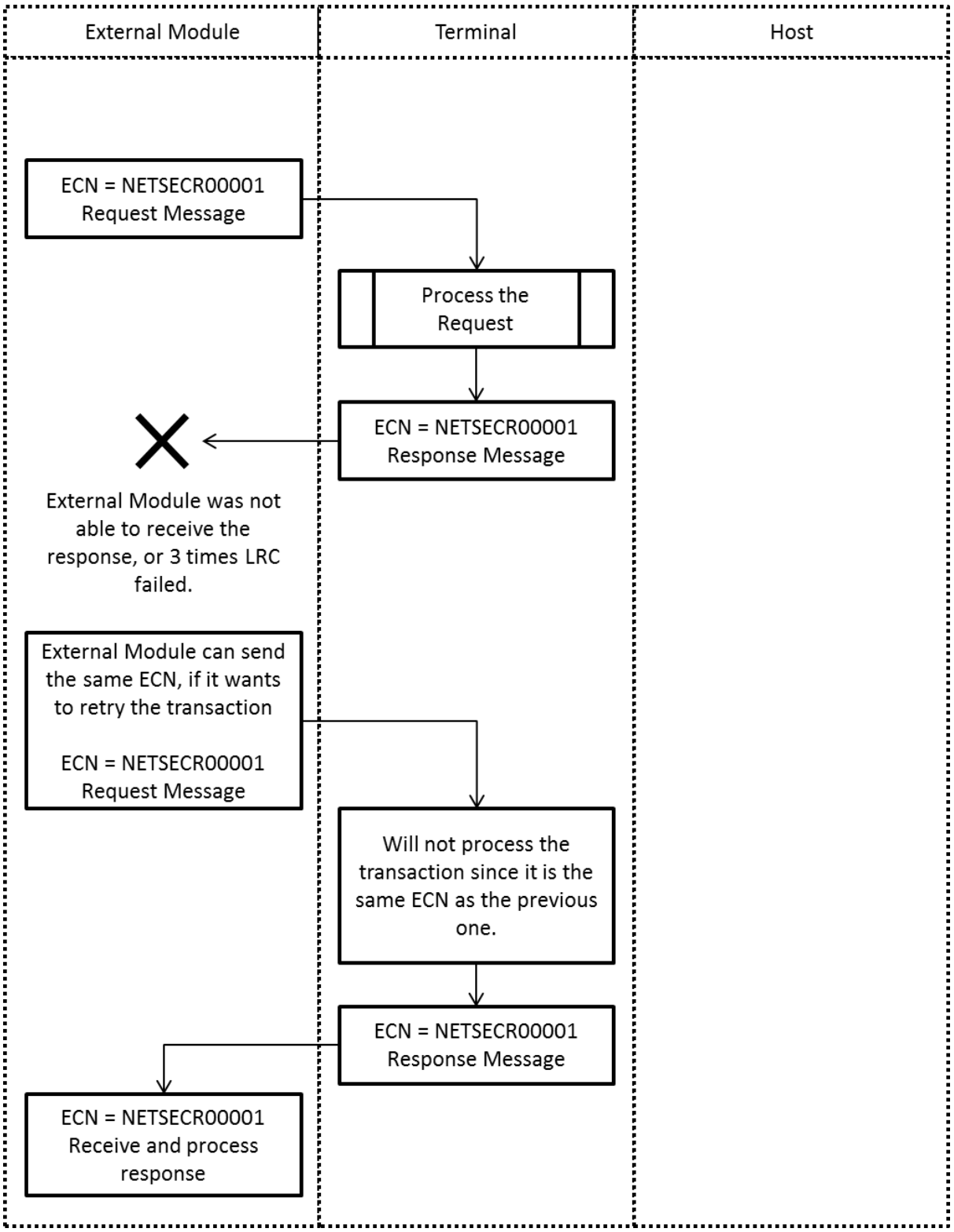
3.1.2 Exception Handling – Transaction Was Not Processed

If external module send a command to terminal but terminal was not able to receive. External module will encounter timeout. The external module may retry the same transaction by sending the same ECN.

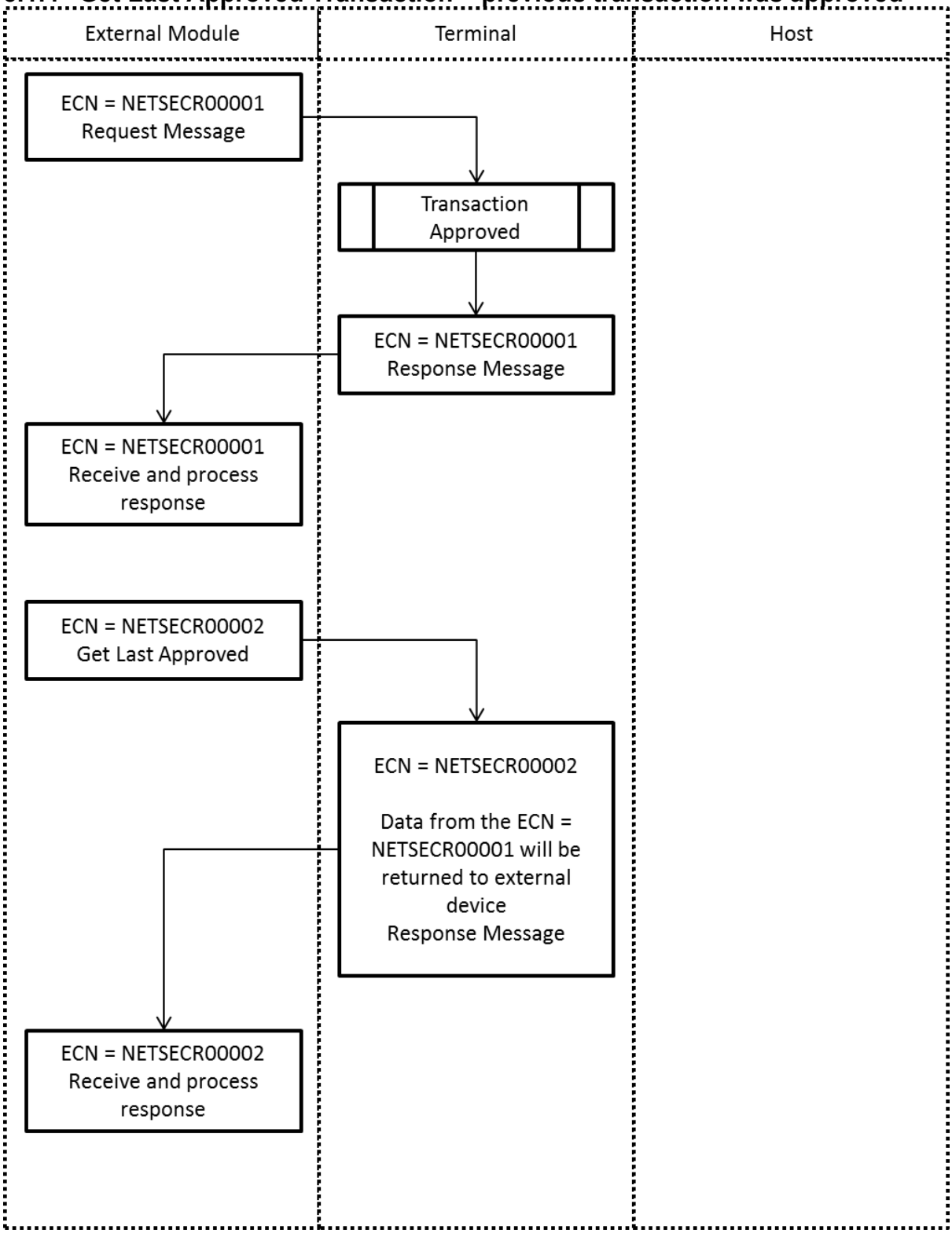


3.1.3 Exception Handling – Transaction Was Already Processed

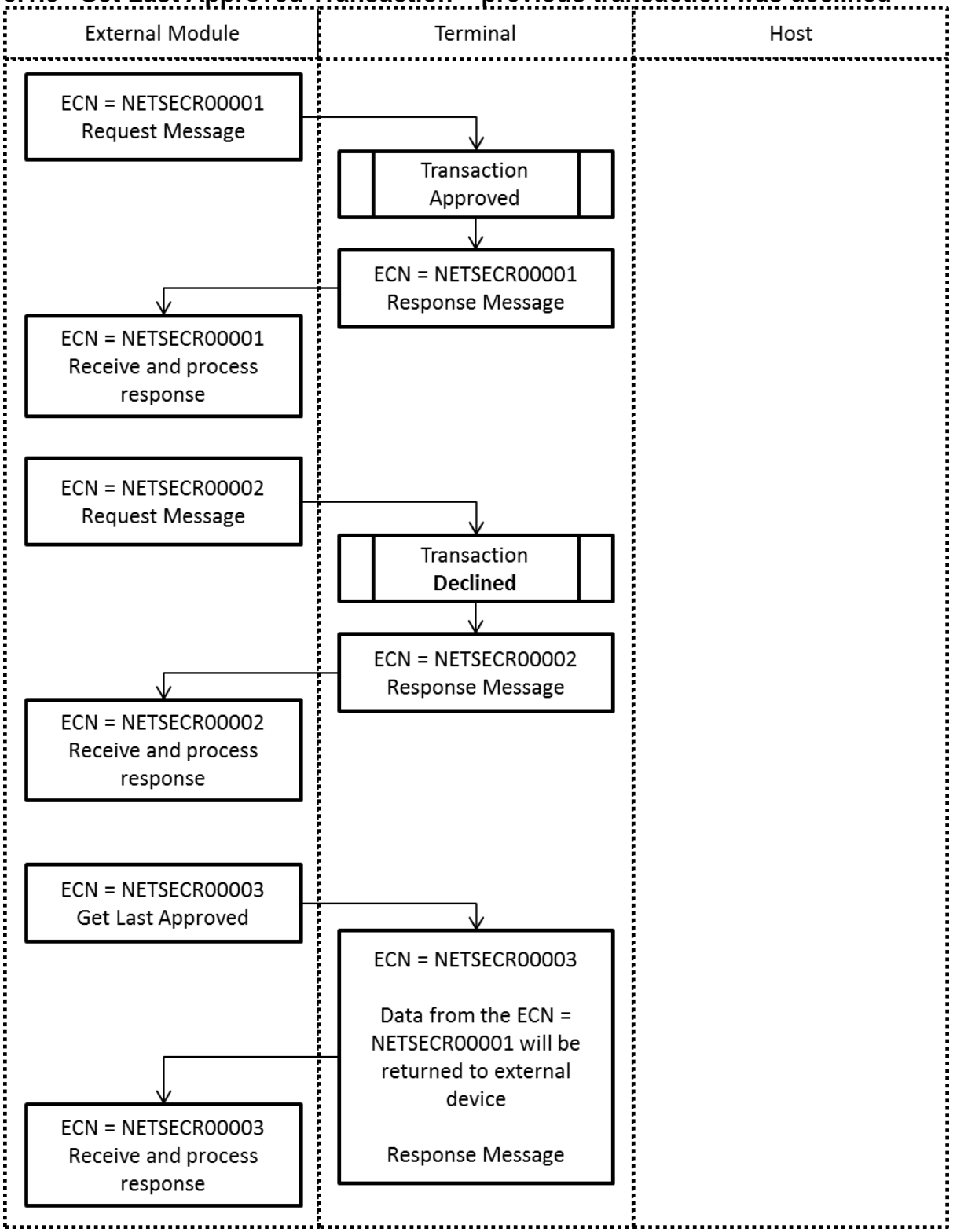
In the event where external module does not receive a response from the terminal, the external module may retry by sending the same ECN. In the event that the transaction was already processed, the terminal would be responding with the transaction details and not perform another transaction. This is to prevent double deduction to customer account.



3.1.4 Get Last Approved Transaction – previous transaction was approved

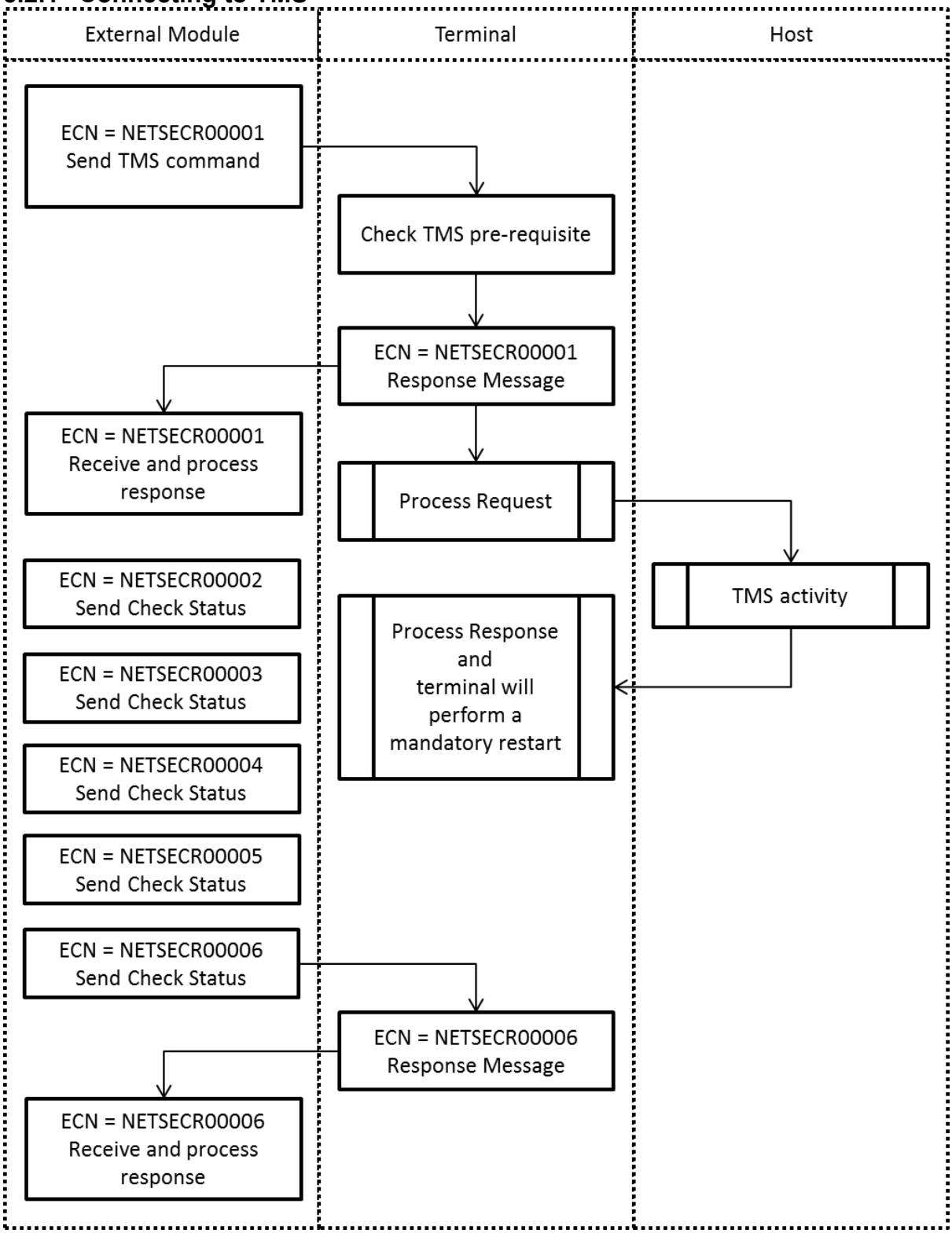


3.1.5 Get Last Approved Transaction – previous transaction was declined



3.2 Administrative Use Cases

3.2.1 Connecting to TMS



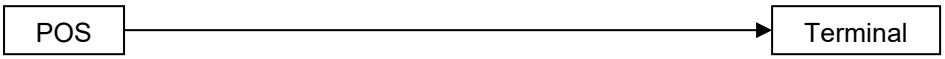
4. Develop Program Specifications

4.1 Message Structure

4.1.1 Request Message Structure

This request message structure is the message that POS have to send to terminal.

Request



Request Transaction Function

STX	LEN	Message Header					Message Data 1				Message Data (..N)				ETX	LRC
		ECN	Function Code	Version Code	RFU	Separator	Field Code	Len (Byte)	Data	Separator	Field Code	Len (Byte)	Data	Separator		

Request's Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request. e.g. 000000000001 (0x30 0x30 0x30 0x30 0x30 0x30 0x30 0x30 0x30 0x30 0x30 0x31)
Function Code	ASC	2	Please refer to individual Function for the value.
Version Code	ASC	2	Please refer to individual Function for the version code.
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Sample Request Message

02001841313233343536373839304243433031301C0334

Request message breakdown:

STX	Length	Message Header	ETX	LRC
02	0018	41313233343536373839304243433031301C	03	34

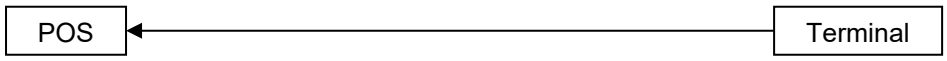
Message header breakdown:

ECN	Function Code	Version Code	RFU	Separator
413132333435363738393042	4343	3031	30	1C

4.1.2 Response Message Structure

This response message structure is the message that POS will be receiving from terminal.

Response



Response Transaction Function

STX	LEN	Message Header					Message Data 1				Message Data (..N)				ETX	LRC
		ECN	Function Code	Response Code	RFU	Separator	Field Code	Len (Byte)	Data	Separator	Field Code	Len (Byte)	Data	Separator		

Response's Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECHO from the Request Message
Function Code	ASC	2	Please refer to individual Function for the value.
Response Code	ASC	2	Please refer to individual Function for the response code available.
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Sample Response Message

0200274131323334353637383930424343303030 1C46430004444154411C0307

Request message breakdown:

STX	Length	Message Header	Message Data	ETX	LRC
02	0027	413132333435363738393042434330303 01C	46430004444154411C	03	07

Message header breakdown:

ECN	Function Code	Response Code	RFU	Separator
413132333435363738393042	4343	3030	30	1C

Message data breakdown:

Field Code	Field Length	Data	Separator
4643	0004	44415441	1C

4.1.3 Message Data

Message Data Description

Name	Attribute	Len (Bytes)	Remark
Field Code	ASC	2	Please refer to the individual function code (transaction) for list of field code for each transaction request and response
Len (Bytes)	BCD	2	The number of bytes of the Data that follows
Data	ASC	2	Variable length and different format. Please refer to the individual function code for data format or structure
Separator	HEX	1	Value is (0x1C)

See sample data from previous section.

Important Notes

A. ***For any Message Request and Message Response, the FIELDs within the Payload can appear in any order dedicated by the Sender. The Receiver shall be programmed to parse the FIELDs correctly and accurately regardless of the order they appear in the Message Payload.***

B. ***For any Message Request and Message Response, additional FIELDs (Field Code) can be added as and when they are required without any changes required at the protocol level. The original application shall be able to work seamlessly without any modification for the current FIELDs that are required to be processed. The application shall be enhanced, if required, to process the new FIELDs available in the Message.***

C. ***For Field Code value, POS should not hardcode the validate to specific values, as the value may updated without prior notice, if any hardcoded validation it could cause POS unable to process as per expected result.***
Ex: "VISA", "Visa", "VISA Debit", "Visa Credit", etc

4.2 Function Codes Supported by Terminal Types

Not all functions listed below will be supported by all terminal models and modes.

Function Code	Function Name	Description
Administrative		
55	Request Terminal Status	Current terminal state, if it is ready to accept payment
56	Get Last Approved Transaction	Retrieve the details of the last approved transaction
80	Logon	Terminal perform NETS logon
81	Settlement	Terminal perform NETS, NFP, NCC, BCA, UPI settlement
85	Get Last Settlement Receipt	Terminal returns the last settlement receipt details
14	Reversal	Manually reverse an online purchase transaction *merchant specific (Transit Link & IUN) T2 – don't have but in ECR1 – FC 89
84	TMS Connect	Used to inform the terminal that it is free to connect to TMS
86	NETS Pre-Settlement	Terminal returns the totals without settling to host
A1	Acquirer Logon	Applicable for NETS and CEPAS acquirers
A2	Acquirer Settlement	Applicable for NETS and CEPAS acquirers
A3	Acquirer Pre-Settlement	Applicable for NETS and CEPAS acquirers
Cash Card		
45	NCC Top Up	Perform NCC topup transaction *Android don't support cash card.
NETS Flash Pay		
46	NFP Top Up	Perform NFP topup transaction
48	NFP Card Enquiry	Perform NFP card enquiry Android doesn't support
4B	NFP ATU Activation	Android doesn't support in the standalone
4C	NFP Refund	*this only in ECR1 FC:84, not in ECR2 Android doesn't support NFP refund standalone.
4D	NFP Card Sale Activation with Mobile Number	*to do in TETRA. Android doesn't support this
CEPAS		
26	CEPAS Debit	Perform CEPAS purchase transaction (Flashpay, EzLink or Concession)
47	CEPAS Top Up	Perform CEPAS top up transaction (Flashpay, EzLink or Concession)
C1	CEPAS VOID	To change in TETRA ECR xml Previously 48
C2	CEPAS Transaction Status	Previously 49
NETS Debit		
30	NETS Purchase	Perform NETS purchase, cashback and NETS loyalty redemption transaction

13	NETS Void	Void a previously approved NETS purchase transaction
UnionPay		
31	UnionPay Purchase	Perform UPI purchase transaction
91	UnionPay Void	Void a previously approved UPI purchase transaction
~~99~~	~~UnionPay Refund~~	~~Perform UPI refund transaction~~
BCA		
65	BCA Purchase	Perform BCA purchase transaction
68	BCA Void	Void a previously approved BCA purchase transaction
Credit Card		
I0	Credit Card Sale	Perform Credit Card sale transaction
I1	Credit Card Void	Void a previously approved credit card purchase transaction
I4	Credit Card Refund	Perform credit card refund transaction
I5	Credit Card Settlement	Perform a credit card settlement
I6	Credit Card Pre-Settlement	Perform a credit card pre-settlement
IA	Signature Image	Retrieve the previous approved transactions signature image
I7	Credit Card IPP Sale	Perform installment multi acquirer transactions
I8	Credit Card IPP Void	To void an installment transaction
IB	Pre-Authorization	To request a pre authorization from host
IC	Pre-Authorization Void	To void pre-auth
ID	Pre-Authorization Completion	To complete the previously approved pre-auth
Specific Functions		
S0	Read Track 2 Data	To retrieve the track2 data from the card
S1	Get Card Info	To retrieve the masked PAN and selected card type for payment
Generic Payment Functions		
00	Single Entry Purchase	A Single API That performs various payment mode purchase command.
NPC Functions		
3A	NPC Balance Inquiry	Perform balance inquiry for NPC
3B	NPC Top Up	To top up amount to NPC
3C	NPC Activation	To activate a NPC
Rupay Functions		
6A	Rupay Purchase	Perform Rupay purchase transaction
MyDebit Functions		
6B	MyDebit Purchase	Perform MyDebit purchase transaction
6C	MyDebit Void	Void a previously approved MyDebit purchase transaction

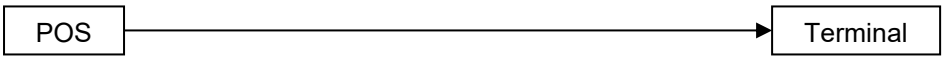
4.3 Administrative Functions

4.3.1 Request Terminal Status (Function code 55)

This function code is for POS to check whether the terminal is connected and ready to accept financial transactions.

Terminal will respond with response code 00 if it is ready to accept financial transactions, otherwise please refer to Annex C for possible actions.

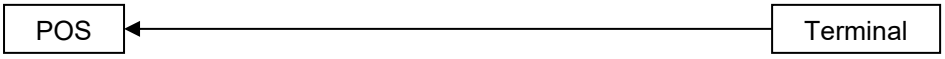
Request



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“55” (0x35 0x35)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response



Message Header Description

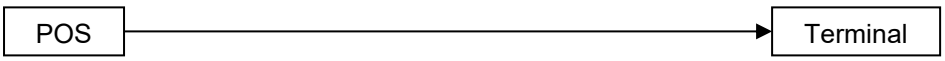
Name	**Attribute**	**Len (Bytes)**	**Remark**
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

4.3.2 Get Last Approved Transaction (Function Code 56)

This function code gets the last approved financial transaction from the terminal.

List of function codes supported by this commands:
1 NETS Purchase (30)
2 Cash Deposit (17)
3 NETS Refund (18)
4 NETS Void (13)
5 NFP Debit (24)
6 NFP Top up (46)
7 NCC Purchase (51)
8 NCC Purchase w/ Prev Balance (52)
9 NCC Top up (45)
10 UPI Purchase (31)
11 UPI Purchase w/ Masked Pan (32)
12 UPI Void (91)
13 UPI Refund (99)
14 BCA Purchase (65)
15 BCA Void (68)
16 Credit Card Sale (I0)
17 Credit Card Void (I1)
18 Credit Card Refund (I4)
19 CEPAS Debit (26)
20 CEPAS Topup (47)
21 NPC Top up (3B)
22 Single Entry API (00)

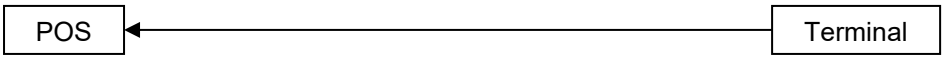
Request



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“56” (0x35 0x36)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

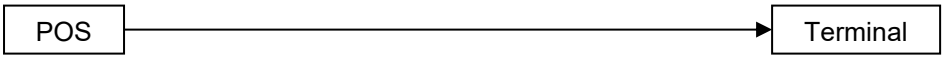
Response Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Function Code	ASC	FC	2	The function code of the last approved transaction.
Transaction Data	VAR	TD	VAR	Contains the data based on the function code response message. See individual response data structure.

4.3.3 Logon (Function Code 80)

This function will initiate the terminal to do a logon to NETS host.

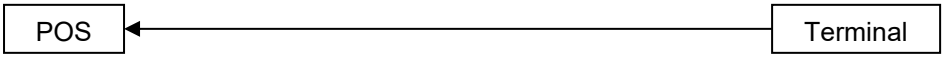
Request



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“80” (0x38 0x30)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

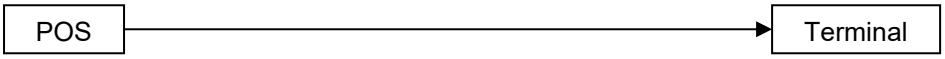
Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
STAN	ASC	65	6	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Host Response Code	ASC	HC	2	Response code received from host. Default to spaces if not relevant

4.3.4 Settlement (Function Code 81)

This function will initiate the terminal to perform settlement to NETS host. When the settlement is approved, the current batch will be cleared.

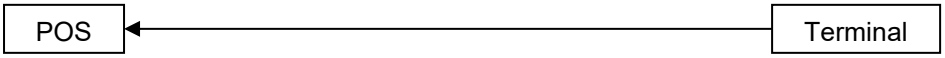
Request



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“81” (0x38 0x31)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
STAN	ASC	H6	6	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Total NETS Purchase Count	BCD	N1	2	
Total NETS Purchase Amount	BCD	G1	6	
Total VOID NETS Purchase Count	BCD	N2	2	

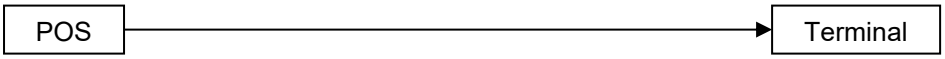
Total VOID NETS Purchase Amount	BCD	G2	6	
Total UnionPay Purchase Count	BCD	N3	2	
Total UnionPay Purchase Amount	BCD	G3	6	
Total VOID UnionPay Purchase Count	BCD	N4	2	
Total VOID UnionPay Purchase Amount	BCD	G4	6	
Total UnionPay Refund Count	BCD	N5	2	
Total UnionPay Refund Amount	BCD	G5	6	
Total UnionPay Pre-Auth Completion Count	BCD	N6	2	
Total UnionPay Pre-Auth Completion Amount	BCD	G6	6	
Total VOID UnionPay Pre-Auth Completion Count	BCD	N7	2	
Total VOID UnionPay Pre-Auth Completion Amount	BCD	G7	6	
Total BCA Purchase Count	BCD	N8	2	
Total BCA Purchase Amount	BCD	G8	6	
Total VOID BCA Purchase Count	BCD	N9	2	
Total VOID BCA Purchase Amount	BCD	G9	6	
Total BCA Pre-Auth Completion Count	BCD	NA	2	
Total BCA Pre-Auth Completion Amount	BCD	GA	6	
Total NFP Count	BCD	NB	2	
Total NFP Amount	BCD	GB	6	
Total NCC Count	BCD	NC	2	
Total NCC Amount	BCD	GC	6	
Total NFP Topup by NETS count	BCD	ND	2	
Total NFP Topup by NETS amount	BCD	GD	6	
Total NCC Topup by NETS count	BCD	NE	2	
Total NCC Topup by NETS amount	BCD	GE	6	
Total Cashback count	BCD	NF	2	For purchase + cashback transactions the purchase will be added to the N1 count and the cashback will be added here.

Total Cashback amount	BCD	GF	6	For purchase + cashback transactions the purchase amount will be added to the G1 and the cashback component will be added here.
Total Cash Deposit count	BCD	NG	2	
Total Cash Deposit amount	BCD	GG	6	
Total NPC Topup by NETS count	BCD	NH	2	
Total NPC Topup by NETS amount	BCD	GH	6	
Total NPC Topup by CASH count	BCD	NI	2	
Total NPC Topup by CASH amount	BCD	GI	6	
Total NPC Activation Count	BCD	NK	2	Number of NPC Activation transaction performed.
Total NFP Refund count	BCD	NL	2	
Total NFP Refund amount	BCD	GL	6	
Total NFP Card Sale Activation count	BCD	NM	2	
Total NFP ATU Activation count	BCD	NO	2	
Total NETS Refund count	BCD	NP	2	
Total NETS Refund amount	BCD	GP	6	
Receipt Text Format	ASC	RP	VAR	

4.3.5 Acquirer Logon (Function Code A1)

This function will initiate the terminal to do a logon.

Request



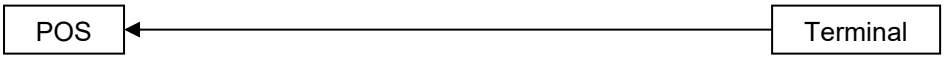
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"A1"
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Acquirer Name	ASC	9G	…20	To specify which acquirer to logon "NETS" "CEPAS" "MPREPAID"

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

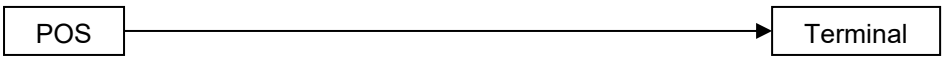
Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
STAN	ASC	65	6	Transaction Sequence Number

Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Host Response Code	ASC	HC	2	Response code returned from acquirer host. Default to spaces if not relevant
Processing Gateway	ASC	D7	20	Terminal returns the processing gateway information. Eg. “NETS” or “CEPAS”
MPP Host Text	ASC	E2	…999	“MPREPAID” host response text
Address 1	ASC	E3	40	Only applicable for “MPREPAID”
Address 2	ASC	E4	40	Only applicable for “MPREPAID”
Address 3	ASC	E5	40	Only applicable for “MPREPAID”
Address 4	ASC	E6	40	Only applicable for “MPREPAID”
Postal Code	ASC	E7	10	Only applicable for “MPREPAID”

4.3.6 Acquirer Settlement (Function Code A2)

This function will initiate the terminal to perform settlement. When the settlement is approved, current batch will be cleared.

Request



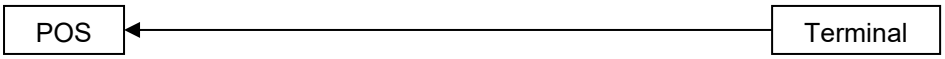
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"A2"
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Acquirer Name	ASC	9G	…20	"NETS" "CEPAS" "MPREPAID"

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

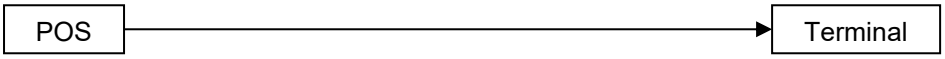
Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
STAN	ASC	H6	6	

Processing Gateway	ASC	D7	20	Terminal returns the processing gateway information
Number of Processing Host	ASC	9J	2	The number of processing host being settled by the terminal
** HOST DATA START ** This entry will be repeated depending on the value at field code 9J (Number of Processing Host)				
Host ID	ASC	9K	2	Host ID is just to indicate the host index based on the number of host reflected in field 9J.
Host Response Code	ASC	HC	2	The response code returned by the host. Default to spaces when not available.
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Type	ASC	9M	..99	Type of Payment Cards
Data Type	ASC	9N	..99	Transaction Type
Sale Count	ASC	N0	4	
Sale Total	ASC	G0	12	
** HOST DATA END **				
MPP Host Text	ASC	E2	…999	“MPREPAID” host response text
Receipt Text Format	ASC	RP	VAR	Optional Printed as is by the POS

4.3.7 Acquirer Pre-Settlement (Function Code A3)

This function will initiate the terminal to perform acquirer pre-settlement.

Request



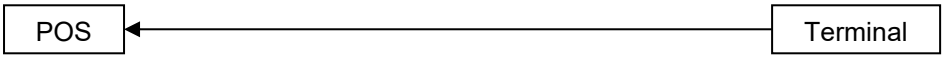
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"A2"
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Acquirer Name	ASC	9G	…20	"NETS" "CEPAS" "MPREPAID"

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
STAN	ASC	H6	6	
Processing Gateway	ASC	D7	20	Terminal returns the processing gateway information

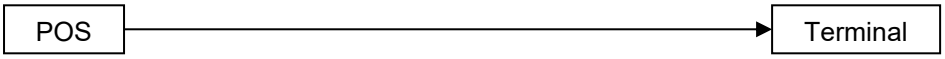
Number of Processing Host	ASC	9J	2	The number of processing host being settled by the terminal
** HOST DATA START ** This entry will be repeated depending on the value at field code 9J (Number of Processing Host)				
Host ID	ASC	9K	2	Host ID is just to indicate the host index based on the number of host reflected in field 9J.
Host Response Code	ASC	HC	2	The response code returned by the host. Default to spaces when not available.
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Type	ASC	9M	..99	Type of Payment Cards
Data Type	ASC	9N	..99	Transaction Type
Sale Count	ASC	N0	4	
Sale Total	ASC	G0	12	
** HOST DATA END **				
MPP Host Text	ASC	E2	…999	“MPREPAID” host response text
Receipt Text Format	ASC	RP	VAR	Optional Printed as is by the POS

4.3.8 Acquirer TMS Update (Function Code A4)

External module will receive a response of acknowledgement first before terminal will perform this function. This function will trigger the terminal to perform TMS update for specific 3rd party application. ECR response code will only tell if the NETS application is able to trigger the 3rd party application TMS update. But the actual response code from their TMS update will not be reflected.

There is a possibility of a reboot when doing the TMS update. External module is advised to send **check status function** before doing a payment transaction to know if the terminal has completed this function and is ready for new command.

Request



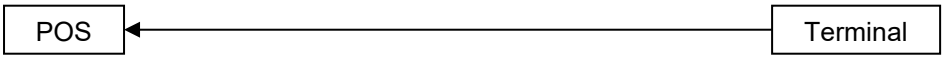
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"A4"
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Acquirer Name	ASC	9G	…20	Ex. "OCBC"

Response



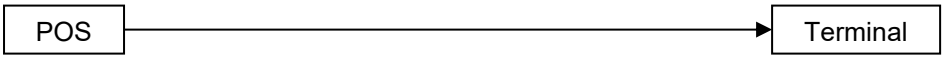
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	"00" – Command received and processed by 3rd party application "05" – 3rd party application not available
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

4.3.9 Pre-Settlement (Function Code 86)

This function will perform NETS pre-settlement for NETS, UPI, BCA, NFP and NCC transactions.

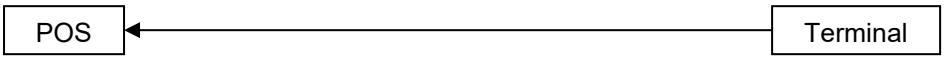
Request



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“86” (0x38 0x36)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
STAN	ASC	H6	6	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Total NETS Purchase Count	BCD	N1	2	
Total NETS Purchase Amount	BCD	G1	6	
Total VOID NETS Purchase Count	BCD	N2	2	

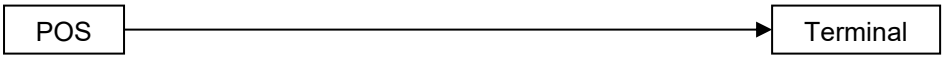
Total VOID NETS Purchase Amount	BCD	G2	6	
Total UnionPay Purchase Count	BCD	N3	2	
Total UnionPay Purchase Amount	BCD	G3	6	
Total VOID UnionPay Purchase Count	BCD	N4	2	
Total VOID UnionPay Purchase Amount	BCD	G4	6	
Total UnionPay Refund Count	BCD	N5	2	
Total UnionPay Refund Amount	BCD	G5	6	
Total UnionPay Pre-Auth Completion Count	BCD	N6	2	
Total UnionPay Pre-Auth Completion Amount	BCD	G6	6	
Total VOID UnionPay Pre-Auth Completion Count	BCD	N7	2	
Total VOID UnionPay Pre-Auth Completion Amount	BCD	G7	6	
Total BCA Purchase Count	BCD	N8	2	
Total BCA Purchase Amount	BCD	G8	6	
Total VOID BCA Purchase Count	BCD	N9	2	
Total VOID BCA Purchase Amount	BCD	G9	6	
Total BCA Pre-Auth Completion Count	BCD	NA	2	
Total BCA Pre-Auth Completion Amount	BCD	GA	6	
Total NFP Count	BCD	NB	2	
Total NFP Amount	BCD	GB	6	
Total NCC Count	BCD	NC	2	
Total NCC Amount	BCD	GC	6	
Total NFP Topup by NETS count	BCD	ND	2	
Total NFP Topup by NETS amount	BCD	GD	6	
Total NCC Topup by NETS count	BCD	NE	2	
Total NCC Topup by NETS amount	BCD	GE	6	
Total Cashback count	BCD	NF	2	
Total Cashback amount	BCD	GF	6	
Total Cash Deposit count	BCD	NG	2	
Total Cash Deposit amount	BCD	GG	6	

Total NPC Topup by NETS count	BCD	NH	2	
Total NPC Topup by NETS amount	BCD	GH	6	
Total NPC Topup by CASH count	BCD	NI	2	
Total NPC Topup by CASH amount	BCD	GI	6	
Total NFP Refund count	BCD	NL	2	
Total NFP Refund amount	BCD	GL	6	
Total NFP Card Sale Activation count	BCD	NM	2	
Total NFP ATU Activation count	BCD	NO	2	
Total NETS Refund count	BCD	NP	2	
Total NETS Refund amount	BCD	GP	6	
Receipt Text Format	ASC	RP	VAR	

4.3.10 Get Last Settlement Receipt (Function Code 85)

This function shall return the last settlement receipt details for printing

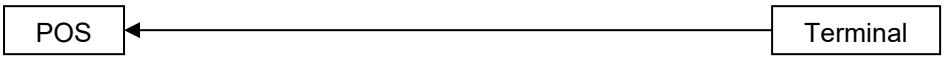
Request



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“85” (0x38 0x35)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Receipt Text Format	ASC	RP	VAR	Settlement details to be printed

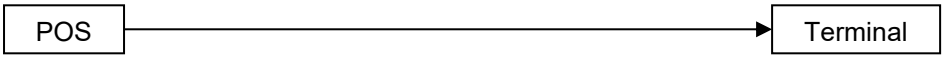
4.3.11 Reversal (Function Code 14)

This function will trigger the terminal to reverse the previous purchase transaction:

1. NETS Purchase
2. UPI Purchase
3. BCA Purchase

Response code 00 means that reversal will be triggered, but terminal may not be sending immediately. Normally reversal is sent to the host on the next online, during idle and before settlement.

Request



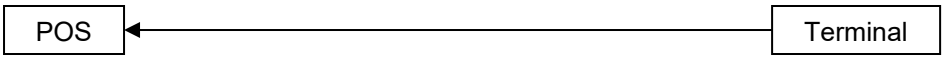
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"14" (0x31 0x34)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	The amount of the purchase to be reversed.
Enhanced ECR Reference Number	ASC	HD	…13	Mandatory field The reference number of the transaction to be reversed.

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available

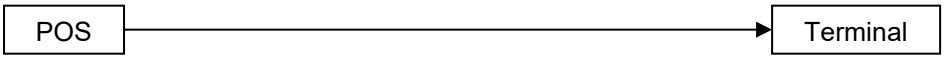
4.3.12 TMS (Function Code 84)

External module will receive a response of acknowledgement first before terminal will perform this function. This function will trigger the terminal to perform TMS logon for parameter, log diagnostic and software update.

This command will perform a mandatory reboot. External module should send **check status function** after receiving the response to this command to know if the terminal has completed this function and is ready for new command.

Note: POS required to have this implementation to allow terminal connects to TMS for necessary update either manual or right after settlement flow.

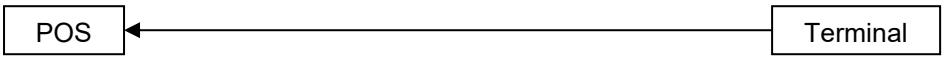
Request



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“84” (0x38 0x34)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response



Message Header Description

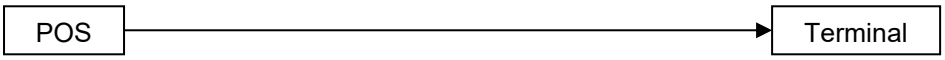
Name	**Attribute**	**Len (Bytes)**	**Remark**
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

4.4 CashCard Functions

4.4.1 NETS CashCard Top Up (Function Code 45)

This function is used to perform top-up / reload a NETS CASHCARD using the NETS Debit ATM card or by cash (Using Merchant Top-Up Card).

Request



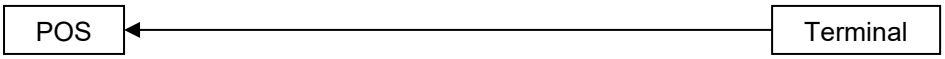
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“45” (0x34 0x35)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	Amount to be top up Mandatory
Top Up Option	HEX	T1	1	1 for Cash Top Up 2 for NETS Top Up Optional Field if this field is not present on the request message, default will be top up via NETS
ECR Reference Number	ASC	HD	..13	Optional Field

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Number	ASC	30	16	CashCard CAN (Card Application Number)
Batch Number	ASC	50	12	In the format of YYMMDDHHMMSS
Terminate Certificate (TTC Cert.)	ASC	H5	6	Default to zeros if not available
STAN	ASC	H6	6	
CashCard transaction Counter (CTC)	ASC	H7	6	Default to zeros if not available
Blacklist Version	ASC	H8	6	
Certificate (Sign Certificate)	ASC	H9	4	Default to zeros if not available
Checksum	ASC	HA	2	
Card Balance	ASC	HE	12	CashCard Balance Default to zeros if not used
Service Fee	ASC	41	12	Total Fee Default to zeros if not in used
Fee due to merchant	ASC	46	12	Default to zeros if not in used
Fee due from merchant	ASC	47	12	Default to zeros if not in used
Host Response Code	ASC	HC	2	Response code received from host.
Card Entry Mode	ASC	CN	2	“10” – chip card transaction “20” – contactless transaction
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

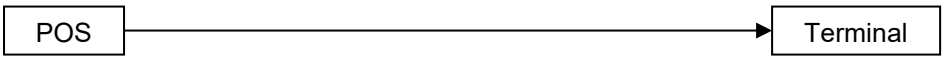
4.5 NETS Functions

4.5.1 All NETS Purchase (Function Code 30)

This function can be used to perform

- NETS purchase
- NFP purchase
- Cashcard purchase
- NETS QR purchase
- Cashback
- NETS with cashback

Request



Message Header Description

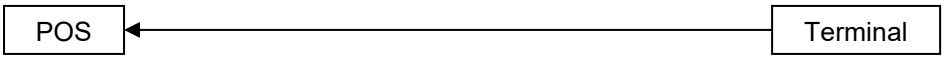
Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“30” (0x33 0x30)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

zz

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Type Indicator	ASC	T2	2	“01” – All NETS Purchase “02” – NETS Purchase with CashBack “03” – CashBack “04” – NETS QR Purchase
Transaction Amount	ASC	40	12	
Cash Back Amount	ASC	42	12	Optional field
Enhanced ECR Reference Number	ASC	HD	…13	Mandatory field

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response Message Data

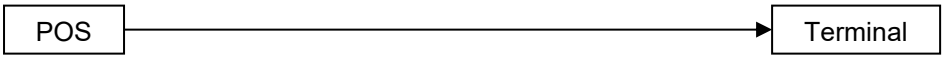
Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available For Flashpay and Cashcard APPROVED\n CCCCCCCCCCCCCCCC\n BAL: NNN.NN where C is the CAN in ASC16 and N is the current balance For NETS Prepaid Card: APPROVED\n xxxxxxxxxxxxCCCC\n BAL: NNNN.NN where C is the masked card number in ASC16 and N is the current balance
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
STAN	ASC	65	6	NFP transactions will return "000000" CC transaction to return "000000"
Approval Code	ASC	01	6	Transaction approval code. NPF transactions to return TRP here. CC transaction to return 00||CC sign cert
Retrieval Reference Number	ASC	D3	12	Only for NETS

Card Name	ASC	L7	20	Default to zeroes if not available. NFP transaction to return "FLASHPAY" CC transaction to return "CASHCARD"
Transaction Amount	ASC	40	12	NET Amount. Set to zeros if not used
Cash Back Amount	ASC	42	12	The cashback amount processed by the terminal. (Optional)
Service Fee	ASC	41	12	Service Fee
POS Messages	ASC	L5	240	Host Data Maximum 240 characters (10 lines message)
Response Message I	ASC	R0	20	Host Data Default to Space
Response Message II	ASC	R1	20	Host Data Default to Space
Loyalty Program Name	ASC	L1	24	Loyalty Data Maximum 24 characters
Loyalty Program Exp Date	ASC	L8	8	Loyalty Data “YYYYMMDD”
Loyalty Type	ASC	L2	1	Loyalty Data “0”(30): No Loyalty Program “1”(31): Points “2”(32): Dollars
Loyalty Marketing Message	ASC	L9	144	Loyalty Data Maximum 144 characters (6 lines message)
Redemption Value	ASC	L3	12	Loyalty Data
Current Loyalty Balance	ASC	L4	12	Loyalty Data
Host Response Code	ASC	HC	2	Response code received from host. NFP & CC transaction returns "00" for successful debit And return "SC" for failed transaction
Card Entry Mode	ASC	CN	2	“10” – chip card transaction “20” – contactless transaction NFP transaction returns "20" for contactless CC transaction returns "10" for chip
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

4.5.2 NETS Void (Function Code 13)

This function is used to perform Void of NETS Purchase before settlement being performed at point of void. This function accepts only NETS EFTPOS card.

Request



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“13” (0x31 0x33)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

There are 2 options to perform NETS void.

Option 1 – provide all the mandatory details to the terminal and the terminal will perform void transaction base on the given information.

Message Data Description

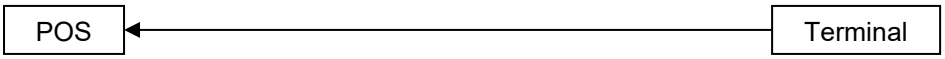
Name	Attribute	Field Code	Len (Bytes)	Remark
Terminal ID	ASC	16	8	
Transaction Date	ASC	03	6	DDMMYY
Transaction Time	ASC	04	6	HHMMSS
Transaction Amount	ASC	40	12	Void Amount Mandatory
Original Transaction STAN	ASC	65	6	Mandatory
ECR Reference Number	ASC	HD	..13	Optional Field

Option 2 – provide the STAN and terminal search in its internal batch for the transaction and void it.

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Original Transaction STAN	ASC	65	6	Mandatory
ECR Reference Number	ASC	HD	..13	Optional field

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

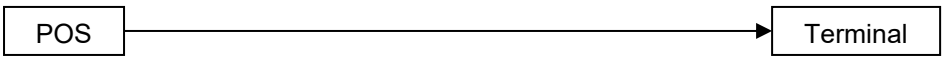
Message Data Descriptions

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Text will be 40 characters long organized as 2 lines of 20 characters. Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Approval Code	ASC	01	6	
STAN	ASC	65	6	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Issuer Name	ASC	D2	10	
Card Number	ASC	30	.. 19	Default to zeros if not available
Retrieval Reference Number	ASC	D3	12	Default to spaces/zeros if not available
Host Response Code	ASC	HC	2	Response code received from host.
Card Entry Mode	ASC	CN	2	“10” – chip card transaction “20” – contactless transaction
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

4.5.3 Cash Deposit (Function Code 17)

This function will perform Cash Deposit authorization request.

Request



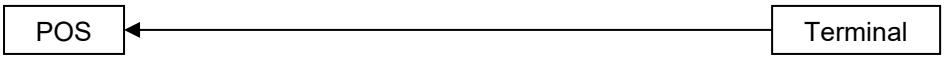
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"17" (0x31 0x37)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	
Enhanced ECR Reference Number	ASC	HD	…13	Mandatory field

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response Message Data

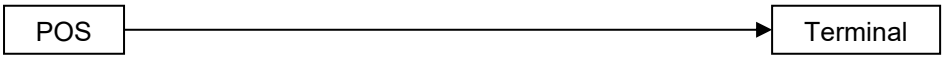
Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Approval Code	ASC	01	6	Taken form NETS HOST
STAN	ASC	65	6	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Issuer Name	ASC	D2	10	Default to space / zeros if not available
Retrieval Reference Number	ASC	D3	12	Default to space / zeros if not available
Cash Deposit Net Fee	ASC	48	12	Amount being deposited into bank account after fee deduction. Default to space / zeros if not available
Cash Deposit Fee	ASC	49	12	Default to space / zeros if not available
Enhanced ECR Reference Number	ASC	HD	…13	Optional field
Host Response Code	ASC	HC	2	Response code received from host.
Card Entry Mode	ASC	CN	2	“10” – chip card transaction “20” – contactless transaction
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

4.5.4 NETS QR Purchase (Merchant's POS Display QR)

This function requires POS to implement 2 messages leg (Q1 and Q2) to complete 1 transaction.

4.5.4.1 Request NETS QR Payload (Function Code Q1)

Request



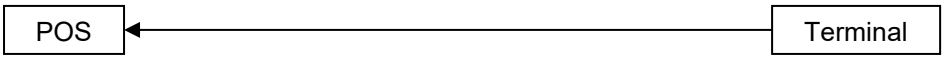
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"Q1" (0x51 0x31)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	Mandatory
Enhanced ECR Reference Number	ASC	HD	…13	Mandatory

Response



Message Header Description

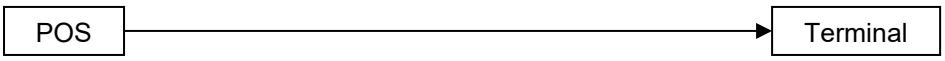
Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Merchant Name And Address	ASC	D0	69	For POS to display on their screen.
QR Payload	ASC	Q0	VAR	Up to 500bytes This QR payload string for POS to convert and display QR code
QR Timer	ASC	Q1	3	Timer in seconds For POS to display QR Code with count down
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available For POS to display together with ECR if available
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Transaction Amount	ASC	40	12	NET Amount. Set to zeros if not used
Enhanced ECR Reference Number	ASC	HD	…13	Optional field

4.5.4.2 NETS QR Payment (Function Code Q2)

Request



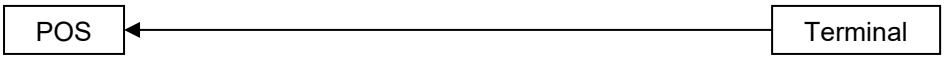
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"Q2" (0x51 0x32)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	
QR Payload	ASC	Q0	VAR	
Enhanced ECR Reference Number	ASC	HD	…13	Mandatory

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

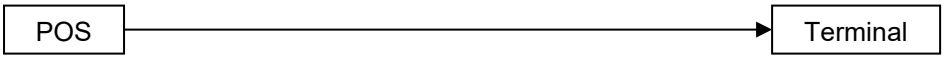
Response Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
STAN	ASC	65	6	
Approval Code	ASC	01	6	Transaction approval code.
Retrieval Reference Number	ASC	D3	12	Only for NETS
Card Name	ASC	L7	20	Default to zeroes if not available.
Transaction Amount	ASC	40	12	NET Amount. Set to zeros if not used
Service Fee	ASC	41	12	Service Fee
POS Messages	ASC	L5	240	Host Data Maximum 240 characters (10 lines message)
Response Message I	ASC	R0	20	Host Data Default to Space
Response Message II	ASC	R1	20	Host Data Default to Space
Host Response Code	ASC	HC	2	Response code received from host.
Card Entry Mode	ASC	CN	2	
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

4.5.5 NETS Refund (Function Code 18)

This function will perform NETS Refund authorization request.

Request



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"18" (0x31 0x33)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

There are 2 options to perform NETS refund.

Option 1 – provide all the mandatory details to the terminal and the terminal will perform refund transaction base on the given information.

Message Data Description

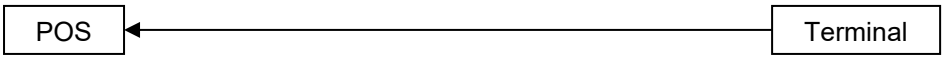
Name	Attribute	Field Code	Len (Bytes)	Remark
Terminal ID	ASC	16	8	
Transaction Date	ASC	03	6	DDMMYY
Transaction Time	ASC	04	6	HHMMSS
Transaction Amount	ASC	40	12	Refund Amount Mandatory
Original Transaction STAN	ASC	65	6	Mandatory
ECR Reference Number	ASC	HD	…13	Optional Field

Option 2 – provide the STAN and terminal search in its internal batch for the transaction and refund it.

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	Refund Amount Mandatory
Original Transaction STAN	ASC	65	6	Mandatory
ECR Reference Number	ASC	HD	..13	Optional field

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Descriptions

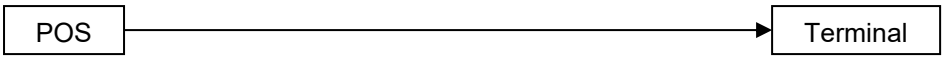
Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Text will be 40 characters long organized as 2 lines of 20 characters. Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Approval Code	ASC	01	6	
STAN	ASC	65	6	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Issuer Name	ASC	D2	10	
Card Number	ASC	30	…19	Default to zeros if not available
Retrieval Reference Number	ASC	D3	12	Default to spaces/zeros if not available
Transaction Amount	ASC	40	12	Refund Amount
POS Messages	ASC	L5	240	Host Data Maximum 240 characters (10 lines message)
Response Message I	ASC	R0	20	Host Data Default to Space
Response Message II	ASC	R1	20	Host Data Default to Space
Host Response Code	ASC	HC	2	Response code received from host.
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

4.6 UPI Functions

4.6.1 UPI Purchase (Function Code 31)

Purpose: This function is to perform a UnionPay Purchase transaction.

Request



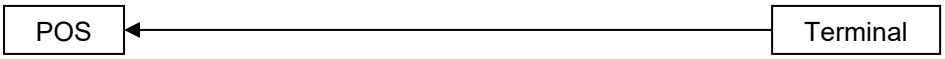
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“31” (0x33 0x31)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	Mandatory
Enhanced ECR Reference Number	ASC	HD	..13	Mandatory

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

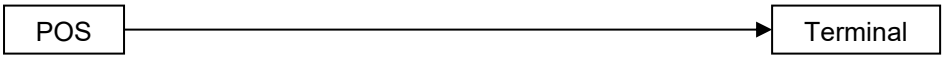
Response Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Approval Code	ASC	01	6	Taken form NETS HOST
Card Issuer Name	ASC	D2	10	Default to space / zeros if not available
Retrieval Reference Number	ASC	D3	12	Default to space / zeros if not available
Transaction Amount	ASC	40	12	Txn Amount. Set to zeroes if not available
Foreign Amount	ASC	FA	12	* Data is unpacked from 6 bytes foreign amount message.
Foreign MID	ASC	F0	15	
AID	ASC	9A	32	Application Identifier (EMV Data)
TC	ASC	9D	16	Transaction Cryptogram (EMV Data)
Response Message I	ASC	R0	20	Host Data Default to Space
Response Message II	ASC	R1	20	Host Data Default to Space
Host Response Code	ASC	HC	2	Response code received from host.
Card Entry Mode	ASC	CN	2	“10” – chip card transaction “20” – contactless transaction “30” – Swipe transaction
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
STAN	ASC	65	6	
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

4.6.2 UPI Void (Function Code 91)

This function is to void a UPI purchase transaction

Request



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“91” (0x39 0x31)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

There are 2 options to perform UPI void.

Option 1 – void a previous UPI Purchase transaction. Information on the UPI Purchase is entered by the user.

Message Data Description

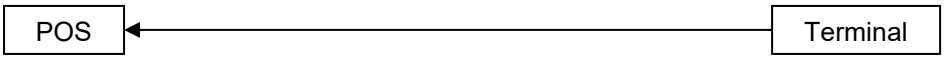
Name	Attribute	Field Code	Len (Bytes)	Remark
Terminal ID	ASC	16	8	
Transaction Date	ASC	03	6	DDMMYY
Transaction Time	ASC	04	6	HHMMSS
Transaction Amount	ASC	40	12	Void Amount Mandatory
STAN	ASC	65	6	Mandatory
Enhanced ECR Reference Number	ASC	HD	13	Mandatory

Option 2 – void a previous UPI Purchase transaction. Information on the UPI Purchase is retrieved from the terminal log.

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
STAN	ASC	65	6	Mandatory
Enhanced ECR Reference Number	ASC	HD	..13	Mandatory

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

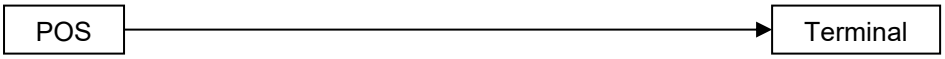
Message Data Descriptions

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	..40	Text will be 40 characters long organized as 2 lines of 20 characters. Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	..69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Approval Code	ASC	01	6	
STAN	ASC	65	6	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Issuer Name	ASC	D2	10	
Card Number	ASC	30	..19	Default to zeros if not available
Retrieval Reference Number	ASC	D3	12	Default to spaces/zeros if not available
Foreign Amount	ASC	FA	12	* Data is unpacked from 6 bytes foreign amount message.
Foreign MID	ASC	F0	15	Default to spaces/zeros if not available
Host Response Code	ASC	HC	2	Response code received from host.
Card Entry Mode	ASC	CN	2	“10” – chip card transaction “20” – contactless transaction
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

4.6.3 UPI Refund (Function Code 99)

This function is to refund a UPI Refund Transaction.

Request



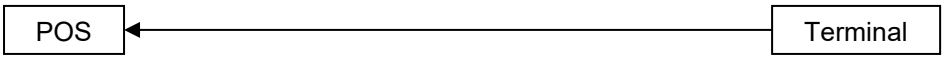
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"99" (0x39 0x39)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	Set to Zero if not used
Original TID	ASC	16	8	
Original Transaction Date	ASC	03	6	DDMMYY
Original Transaction Time	ASC	04	6	HHMMSS
Original Transaction Sequence Number	ASC	65	6	Mandatory
Card Number	ASC	30	..19	Optional field If present, terminal will process as card not present
Enhanced ECR Reference Number	ASC	HD	…13	Mandatory

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Descriptions

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Approval Code	ASC	01	6	Taken form NETS HOST
STAN	ASC	65	6	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Issuer Name	ASC	D2	10	Default to space / zeros if not available
Card Number	ASC	30	..19	Default to zeros if not available
Retrieval Reference Number	ASC	D3	12	Default to space / zeros if not available
Foreign Amount	ASC	FA	12	* Data is unpacked from 6 bytes foreign amount message.
Foreign MID	ASC	F0	15	
Signature Required Indicator	ASC	D5	1	Action Data “0”(30): Not required “1”(31): Required
Response Message I	ASC	R0	20	Host Data Default to Space
Response Message II	ASC	R1	20	Host Data Default to Space
Host Response Code	ASC	HC	2	Response code received from host.
Card Entry Mode	ASC	CN	2	“10” – chip card transaction “20” – contactless transaction
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

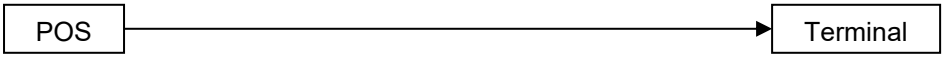
4.7 NFP Functions

4.7.1 NFP Top-up (Function Code 46)

This function is used to perform a top-up or a reload on a NETS Flash Pay Card using the NETS Debit ATM card or by cash (Using Merchant Top-Up Card). It only supports CEPAS 2.0 Cards.

Response Code == “SC” and Transaction Type == 00 indicates that the card exception occurred before the credit command.
Response Code == “SC” and Transaction Type == Ax, indicates that the card exception occurred during/after the credit command and a re-tap scenario has happened.

Request



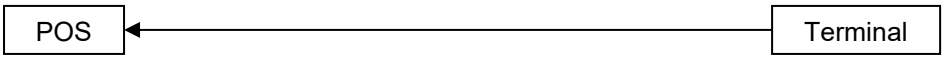
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“46” (0x34 0x36)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	Mandatory
Top-Up Option: Cash or NETS	HEX	T1	1	Set to 1 for Cash Top-Up Set to 2 for NETS Top-Up
Enhanced ECR Reference Number	ASC	HD	13	Mandatory

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Descriptions

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	..40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	..69	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Batch Number	ASC	50	12	In the format of YYMMDDHHMMSS
STAN	ASC	H6	6	
Total Fee (Top Up)	ASC	ZA	6	Default to zeros if not in used
Fee due to merchant (Top Up)	ASC	ZB	6	Default to zeros if not in used
Fee due from merchant (Top Up)	ASC	ZC	6	Default to zeros if not in used
CEPAS Version	ASC	C1	2	CEPAS 2.0 = “20”
Card Number	ASC	30	16	Card Application Number (CAN)
Expiry Date	ASC	C2	8	DDMMYYYY
Host Response Code	ASC	HC	2	Response code received from host.
Card Entry Mode	ASC	CN	2	“10” – Chip transaction “20” – Contactless transaction
*Trans Data	ASC	C0	64	Trans Data is of CEPAS 2.0 format.
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

Trans Data for CEPAS 2.0

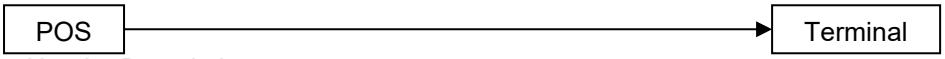
Field Name	Attribute	Len (Bytes)	Remark
Transaction Type	ASC	2	00 == Failed Prior Online 02 == Cash Top Up 03 == NETS Top Up A0 == Not Credited, Post Online A1 == Unconfirmed, Post Online A2 == Host Declined, Post Online
Transaction Amount	ASC	12	
Transaction Date	ASC	6	YYMMDD
Transaction Time	ASC	6	HHMMSS
Prior Card Balance	ASC	12	
Post Card Balance	ASC	12	
Post Auto load Amount	ASC	12	
Post Card Status	ASC	2	“00” – Purse Disabled + ATU Disabled “01” – Purse Enabled + ATU Disabled “02” – Purse Disabled + ATU Enabled “03” – Purse Enabled + ATU Enabled
Total		64	

Note:

1. ECR should identify unconfirmed transactions based on the response code “SC” and Transaction Type “A1”.
2. For unconfirmed top up transactions, no reversal shall be sent from the terminal. If the ECR is printing the receipt, the receipt should be printed with 2 lines of messages to indicate this is an unconfirmed Top Up. The first line states “Transaction Interrupted” and the second line states “Call NETS @ 6274 1212”.
3. To obtain the final card balance for unconfirmed transactions, the ECR should add the Prior Card Balance and the Transaction Amount for the unconfirmed receipt printing.
4. If the ECR receives response code “SC” and the transaction type as 0x00, this would indicate that the card cannot be read before going online.

4.7.2 NFP Card Enquiry (Function Code 48)

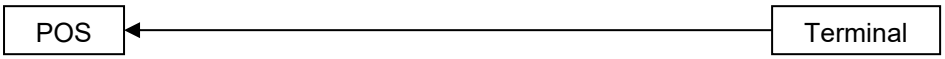
Request



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"48" (0x34 0x38)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

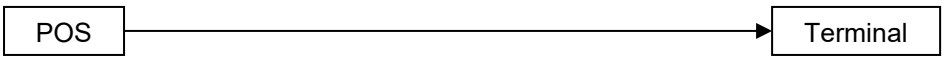
Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
CEPAS Version	ASC	C1	2	CEPAS 2.0 = "20"
Card Number	ASC	30	16	Card Application Number (CAN)
Expiry Date	ASC	C2	8	DDMMYYYY
Card Balance	ASC	HE	12	
Purse Status	ASC	S3	2	00 = disable 01 = enable 02 = not applicable
ATU Status	ASC	S2	2	00 = disable 01 = enable 02 = not applicable
Existing ATU Amount	ASC	C8	12	Current ATU Amount in Card

* This function only supports CEPAS 2 NFP card

4.7.3 NFP Refund (Function Code 4C)

Request



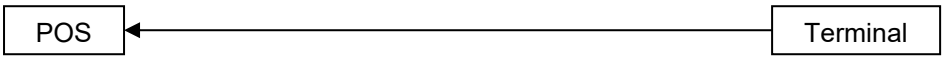
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"4C" (0x34 0x43)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Refund Option: Cash or NETS	HEX	T1	1	Set to 1 for Cash Refund Set to 2 for NETS Refund
Enhanced ECR Reference Number	ASC	HD	13	Mandatory

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Batch Number	ASC	50	12	In the format of YYMMDDHHMMSS

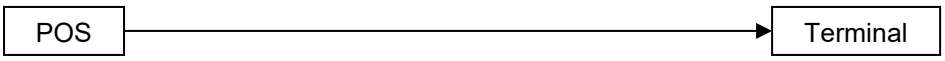
STAN	ASC	H6	6	
CEPAS Version	ASC	C1	2	CEPAS 2.0 = “20”
Card Number	ASC	30	16	Card Application Number (CAN)
Expiry Date	ASC	C2	8	DDMMYYYY
Host Response Code	ASC	HC	2	Response code received from host.
*Trans Data	ASC	C0	64	Trans Data is of CEPAS 2.0 format.
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

Trans Data for CEPAS 2.0

Field Name	Attribute	Len (Bytes)	Remark
Transaction Type	ASC	2	00 == Failed Prior Online 10 == Cash Refunded 20 == NETS Refunded A0 == Not Refunded A1 == Unconfirmed A2 == Host Declined
Transaction Amount	ASC	12	
Transaction Date	ASC	6	YYMMDD
Transaction Time	ASC	6	HHMMSS
Prior Card Balance	ASC	12	
Post Card Balance	ASC	12	
Post Auto load Amount	ASC	12	
Post Card Status	ASC	2	“00” – Purse Disabled + ATU Disabled “01” – Purse Enabled + ATU Disabled “02” – Purse Disabled + ATU Enabled “03” – Purse Enabled + ATU Enabled
Total		64	

4.7.4 NFP Card Sale Activation with Mobile Number (Function Code 4D)

Request



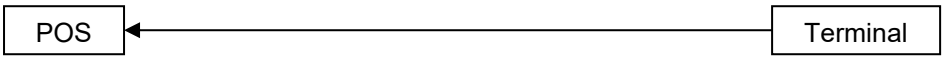
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"4D" (0x34 0x44)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Enhanced ECR Reference Number	ASC	HD	13	Mandatory
Mobile Number	ASC	MN	9	Mandatory field – Right justified with leading spaces, Set to 0x00 if skip

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Batch Number	ASC	50	12	In the format of YYMMDDHHMMSS

STAN	ASC	H6	6	
CEPAS Version	ASC	C1	2	CEPAS 2.0 = “20”
Card Number	ASC	30	16	Card Application Number (CAN)
Expiry Date	ASC	C2	8	DDMMYYYY
Host Response Code	ASC	HC	2	Response code received from host.
*Trans Data	ASC	C0	64	Trans Data is of CEPAS 2.0 format.
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

Trans Data for CEPAS 2.0

Field Name	Attribute	Len (Bytes)	Remark
Transaction Type	ASC	2	00 == Failed Prior Online 51 == Card Activated A0 == Not Credited, Post Online A1 == Unconfirmed, Post Online A2 == Host Declined, Post Online
Transaction Amount	ASC	12	
Transaction Date	ASC	6	YYMMDD
Transaction Time	ASC	6	HHMMSS
Prior Card Balance	ASC	12	
Post Card Balance	ASC	12	
Post Auto load Amount	ASC	12	
Post Card Status	ASC	2	“00” – Purse Disabled + ATU Disabled “01” – Purse Enabled + ATU Disabled “02” – Purse Disabled + ATU Enabled “03” – Purse Enabled + ATU Enabled
Total		64	

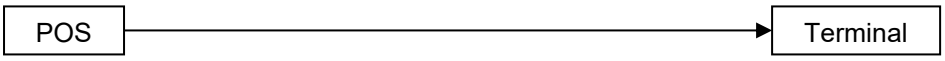
Note:
In the event that the Mobile Number is < 9 numeric asci, it should be NULL Terminated.
E.g. Mobile Number == “12345678”, 0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x00
E.g. Mobile Number == “987654321”, 0x39 0x38 0x37 0x36 0x35 0x34 0x33 0x32 0x31
E.g. Skip Mobile Number, 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00

4.8 Credit Card Functions

This section describes the credit card application.

4.8.1 Credit Card Sale (Function Code I0)

Request



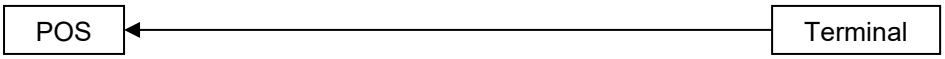
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"I0" (0x49 0x30)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	
Acquirer Name	ASC	9G	..20	Optional field This field will inform the terminal to force all transaction to a specific acquirer. If this field is not present or has an invalid value, the terminal will select the acquirer base on bin processing.
Enhanced ECR Reference Number	ASC	HD	…13	Mandatory
Scheme Category	ASC	9Q	..20	Optional Field Additional info for credit card application to determine specific scheme to process.

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response Message Data

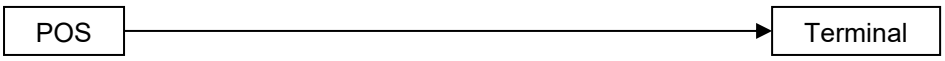
Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	..46	Terminal response message 1 line is consist of 23 characters
Merchant Name And Address	ASC	D0	..92	1 line is consist of 23 characters
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Approval Code	ASC	01	6	
Invoice Number	ASC	9H	6	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Issuer Name	ASC	D2	..20	Card preferred name (EMV tag)
Card Type	ASC	9M	..20	Card type. Eg: “VISA”, “MASTERCARD”, “AMEX”, “UPI”, “JCB”, *etc* **the actual value can differ for sample above.*
Card Number	ASC	30	..19	PAN (masked)
Processing Gateway	ASC	D7	..20	The name of the processing gateway that processed the transaction
Card Description	ASC	D8	..20	The card production name or card description
Expiry Date	ASC	31	4	MMYY Card expiry date
Batch Number	ASC	50	6	Terminal batch number for the transaction
Retrieval Reference Number	ASC	D3	12	
Card Holder Name	ASC	D6	..26	
AID (EMV)	ASC	9A	..16	
Application Profile (EMV)	ASC	9B	..16	

CID (EMV)	ASC	9C	2	
Transaction Certificate (EMV)	ASC	9D	16	Available if transaction is approved
TSI (EMV)	ASC	9F	4	
TVR (EMV)	ASC	9E	10	
Transaction ID	ASC	A1	10	
POS ID	ASC	9I	8	Terminal unique identifier
Host Response Code	ASC	HC	2	The response code returned by the host.
Card Entry Mode	ASC	CN	2	“10” – chip card transaction “20” – contactless transaction “30” – swipe transaction
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Loyalty Type	ASC	93	2	“01” – POSB Everyday “02” – UOB “03” – UOB (SPC) “04” – Diners “05” – DBS Instant Rewards
Loyalty Information	ASC	94	var	Refer to section below for the loyalty data format
Scheme Category	ASC	9Q	..20	Returned if special scheme processing is performed.
Offline Txn Type	HEX	O1	1	Optional Field This field will inform the POS if the transaction was an offline transaction. If field is not present, transaction was done online. 0x00 – Online transaction 0x01 – Offline transaction
Host Card Type	ASC	9R	…20	Card scheme name defined by host
Receipt Text Format	ASC	RP	var	Printed as is by the POS

Please refer to the Annex below for the Loyalty Information.

4.8.2 Credit Card Void (Function Code I1)

Request



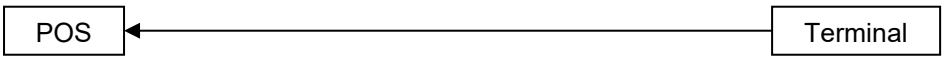
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"I1" (0x49 0x31)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Enhanced ECR Reference Number	ASC	HD	…13	Mandatory
Transaction ID	ASC	A1	…10	Mandatory
Processing Gateway	ASC	D7	..20	The value returned during Credit Card Sale
Scheme Category	ASC	9Q	..20	Optional Field Additional info for credit card application to determine specific scheme to process.

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

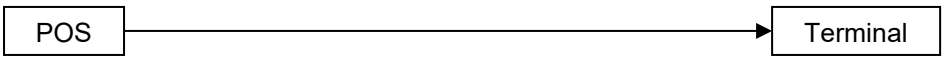
Response Message Data

Name	Attribute	Function Code/ Field Code	Len (Bytes)	Remark

Response Text	ASC	02	40	Terminal response message 1 line is consist of 23 characters
Merchant Name And Address	ASC	D0	69	1 line is consist of 23 characters
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Approval Code	ASC	01	6	
Invoice Number	ASC	9H	6	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Issuer Name	ASC	D2	..20	Card type
Card Number	ASC	30	..19	PAN (masked)
Expiry Date	ASC	31	4	MMYY Card expiry date
Batch Number	ASC	50	6	Terminal batch number for the transaction
Retrieval Reference Number	ASC	D3	12	
Card Holder Name	ASC	D6	..26	
Transaction ID	ASC	A1	10	
POS ID	ASC	9I	8	Terminal unique identifier
Host Response Code	ASC	HC	2	The response code returned by the host.
Card Entry Mode	ASC	CN	2	“10” – chip card transaction “20” – contactless transaction “30” – swipe transaction
Scheme Category	ASC	9Q	..20	Returned if special scheme processing is performed.
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Receipt Text Format	ASC	RP	var	Printed as is by the POS

4.8.3 Credit Card Refund (Function Code I4)

Request



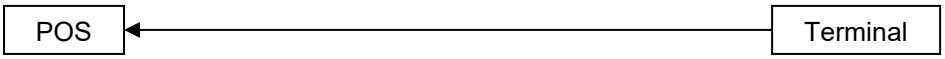
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"I4" (0x49 0x34)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	Set to Zero if not used
ECR Reference Number	ASC	HD	..13	Optional field
Transaction ID	ASC	A1	10	
Processing Gateway	ASC	D7	..20	The value returned during credit card sale
Scheme Category	ASC	9Q	..20	Optional Field. Additional info for credit card application to determine specific scheme to process.

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

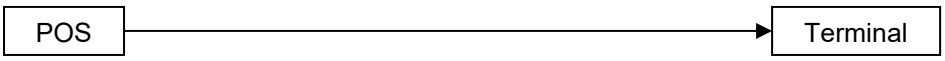
Response Message Data

Name	Attribute	Function Code/ Field Code	Len (Bytes)	Remark
Response Text	ASC	02	..40	Default to Zero
Merchant Name And Address	ASC	D0	..69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Approval Code	ASC	01	6	
Invoice Number	ASC	9H	6	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Issuer Name	ASC	D2	..20	
Card Number	ASC	30	..19	
Expiry Date	ASC	31	4	
Batch Number	ASC	50	6	
Retrieval Reference Number	ASC	D3	12	
Card Holder Name	ASC	D6	26	
AID (EMV)	ASC	9A	..16	
Application Profile (EMV)	ASC	9B	..16	
Cryptogram (EMV)	ASC	9C	2	
Transaction Certificate (EMV)	ASC	9D	16	
TSI (EMV)	ASC	9F	4	
TVR (EMV)	ASC	9E	10	
Transaction ID	ASC	A1	10	
POSID	ASC	9I	8	Terminal unique identifier
Host Response Code	ASC	HC	2	The response code returned by the host.
Card Entry Mode	ASC	CN	2	“10” – chip card transaction “20” – contactless transaction “30” – swipe transaction
Scheme Category	ASC	9Q	..20	Returned if special scheme processing is performed.
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request

Receipt Text Format	ASC	RP	var	Printed as is by the POS

4.8.4 Credit Card Settlement (Function Code I5)

Request



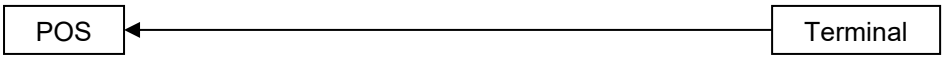
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“I5” (0x49 0x35)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Acquirer Name	ASC	9G	..20	Mandatory field This field will specify which acquirer will perform settlement.

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response Message Data

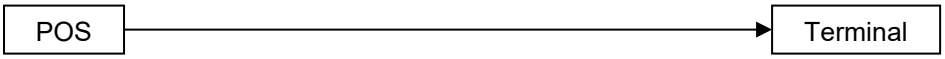
Name	Attribute	Function Code/ Field Code	Len (Bytes)	Remark
Response Text	ASC	02	..40	Terminal response message 1 line is consist of 23 characters
Merchant Name And Address	ASC	D0	..69	1 line is consist of 23 characters
POSID	ASC	9I	8	Terminal unique identifier
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Host Response Code	ASC	HC	2	The response code returned by the host.
Number of Credit Host	ASC	9J	2	The number of Credit Card host being settled by the terminal. If this number is zero the next field code will be Number of IPP Host.
** HOST DATA START ** This entry will be repeated depending on the value at field code 9J Number of Credit Host				
Host ID	ASC	9K	2	Host ID is just to indicate the host data count. The sequence of the host name being returned is based on server response and terminal has no control on it.
Terminal ID	ASC	16	8	Optional, default to zeroes if not available
Merchant ID	ASC	D1	15	Optional, default to zeroes if not available
Card Type	ASC	9M	..99	Type of Credit Card
Data Type	ASC	9N	..99	Transaction Type
Sale Count	ASC	N0	4	
Sale Total	ASC	G0	12	
** HOST DATA END **				

Number of IPP Host	ASC	9P	2	The number of Installment host being settled by the terminal. If this number is zero the next field code will be Number of DCC Host.
** IPP HOST DATA START ** This entry will be repeated depending on the value at field code 9P Number of IPP Host				
Host ID	ASC	9K	2	Host ID is just to indicate the host data count. The sequence of the host name being returned is based on server response and terminal has no control on it.
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Type	ASC	9M	..99	
Data Type	ASC	9N	..99	
Sale Count	ASC	N0	4	
Sale Total	ASC	G0	12	
** IPP HOST DATA END **				
Number of DCC Host	ASC	9L	2	The number of Currency Conversion host being settled by the terminal If this number is zero there will be no more data to follow.
** DCC HOST DATA START ** This entry will be repeated depending on the value at field code 9L Number of DCC Host				
Host ID	ASC	9K	2	The beginning of the host data. Host ID is just to indicate the host data count. The sequence of the host name being returned is based on server response and terminal has no control on it.
Currency Code	ASC	9O	3	
Data Type	ASC	9N	..99	Transaction Type
Sale Count	ASC	N0	4	Total count of the above mentioned transaction
Sale Total	ASC	G0	12	Total amount
** DCC HOST DATA END **				
Receipt Text Format	ASC	RP	var	Printed as is by the POS

4.8.5 Credit Card Pre-Settlement (Function Code I6)

This will perform Credit Card Pre-settlement.

Request



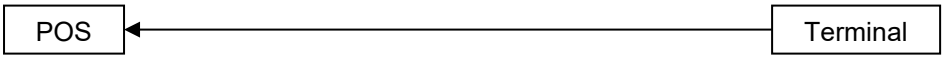
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“I6” (0x49 0x36)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Acquirer Name	ASC	9G	..20	Mandatory field This field will specify which acquirer will perform pre-settlement.

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response Message Data

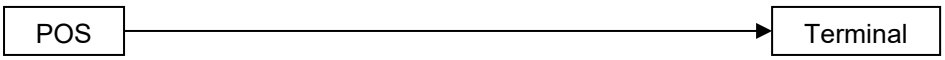
Name	Attribute	Function Code/ Field Code	Len (Bytes)	Remark
Response Text	ASC	02	..40	Terminal response message 1 line is consist of 23 characters
Merchant Name And Address	ASC	D0	..69	1 line is consist of 23 characters
POSID	ASC	9I	8	Terminal unique identifier
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Host Response Code	ASC	HC	2	The response code returned by the host.
Number of Credit Host	ASC	9J	2	The number of Credit Card host being settled by the terminal. If this number is zero the next field code will be Number of IPP Host.
** HOST DATA START ** This entry will be repeated depending on the value at field code 9J Number of Credit Host				
Host ID	ASC	9K	2	Host ID is just to indicate the host data count. The sequence of the host name being returned is based on server response and terminal has no control on it.
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Type	ASC	9M	..99	Type of Credit Card
Data Type	ASC	9N	..99	Transaction Type
Sale Count	ASC	N0	4	
Sale Total	ASC	G0	12	
** HOST DATA END **				
Number of IPP Host	ASC	9P	2	The number of Installment host being settled by the terminal. If this number is zero the next field code will be Number of DCC Host.

** IPP HOST DATA START ** This entry will be repeated depending on the value at field code 9K Number of IPP Host				
Host ID	ASC	9K	2	Host ID is just to indicate the host data count. The sequence of the host name being returned is based on server response and terminal has no control on it.
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Type	ASC	9M	..99	
Data Type	ASC	9N	..99	
Sale Count	ASC	N0	4	
Sale Total	ASC	G0	12	
** IPP HOST DATA END **				
Number of DCC Host	ASC	9L	2	The number of Currency Conversion host being settled by the terminal If this number is zero there will be no more data to follow.
** DCC HOST DATA START ** This entry will be repeated depending on the value at field code 9L Number of DCC Host				
Host ID	ASC	9K	2	The beginning of the host data. Host ID is just to indicate the host data count. The sequence of the host name being returned is based on server response and terminal has no control on it.
Currency Code	ASC	9O	3	
Data Type	ASC	9N	..99	Transaction Type
Sale Count	ASC	N0	4	Total count of the above mentioned transaction
Sale Total	ASC	G0	12	Total amount
** DCC HOST DATA END **				
Receipt Text Format	ASC	RP	var	Printed as is by the POS

4.8.6 Retrieve Signature (Function Code IA)

Function to retrieve signature of the previous approved credit card transaction.
When a new credit card transaction is triggered the signature will be cleared.

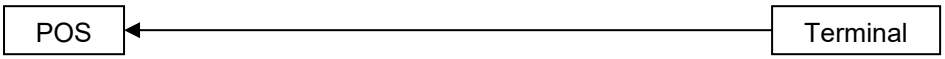
Request



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“IA” (0x49 0x41)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

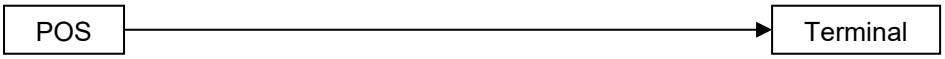
Response Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Signature image	HEX	RQ	..2000	Image format is in bmp

4.8.7 Credit Card IPP SALE (Function Code I7)

This function is used to perform a Credit Card IPP SALE on a transaction using bank cards.

Request



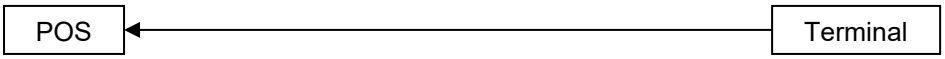
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"I7" (0x49 0x37)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	
Acquirer Name	ASC	9G	..20	Acquirer name Set as space if not used. Or do not send this field code.
ECR Reference Number	ASC	HD	..13	Optional field

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

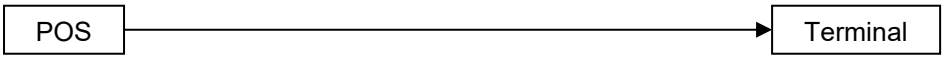
Message Data Descriptions

Name	Attribute	Function Code/ Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to Zero
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Approval Code	ASC	01	6	Taken from HOST
Invoice Number	ASC	9H	6	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Issuer Name	ASC	D2	10	Card Type
Card Number	ASC	30	Up to 19	Masked card number
Expiry Date	ASC	31	4	optional
Batch Number	ASC	50	6	
Retrieval Reference Number	ASC	D3	12	
Card Holder Name	ASC	D6	26	Default to zero if not applicable
AID (EMV)	ASC	9A	16	Default to zero if not applicable
Application Profile (EMV)	ASC	9B	16	Default to zero if not applicable
Cryptogram (EMV)	ASC	9C	2	Default to zero if not applicable
Transaction Certificate (EMV)	ASC	9D	16	Default to zero if not applicable
TSI (EMV)	ASC	9F	4	Default to zero if not applicable
TVR (EMV)	ASC	9E	10	Default to zero if not applicable
Transaction ID	ASC	A1	10	
Plan Code	ASC	F3	Up to 6	optional
Tenure	ASC	F4	Up to 4	In number of months
First Amount	ASC	F5	12	
Monthly Amount	ASC	F6	12	
Last Amount	ASC	F7	12	
Interest Amount	ASC	F8	12	
Receipt Text Format	ASC	RP	var	Printed as is by the POS

4.8.8 Credit Card IPP VOID (Function Code I8)

This function is used to perform a Credit Card IPP VOID on a transaction made using bank cards.

Request



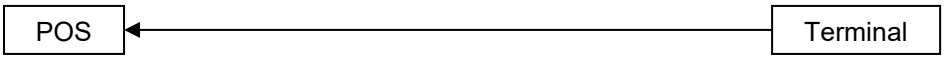
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"I8" (0x49 0x38)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Enhanced ECR Reference Number	ASC	HD	…13	Mandatory
Transaction ID	ASC	A1	…10	Mandatory
Processing Gateway	ASC	D7	..20	The value returned during Credit Card Sale

Response



Message Header Description

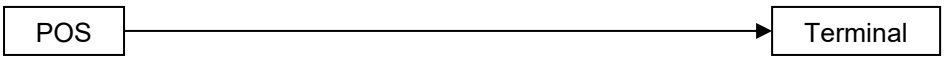
Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Descriptions

Name	Attribute	Function Code/ Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to Zero
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Approval Code	ASC	01	6	Taken from HOST
Invoice Number	ASC	9H	6	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Issuer Name	ASC	D2	10	
Card Number	ASC	30	Up to 19	Masked card number
Expiry Date	ASC	31	4	optional
Batch Number	ASC	50	6	
Retrieval Reference Number	ASC	D3	12	
Card Holder Name	ASC	D6	26	Default to zero if not applicable
AID (EMV)	ASC	9A	16	Default to zero if not applicable
Application Profile (EMV)	ASC	9B	16	Default to zero if not applicable
Cryptogram (EMV)	ASC	9C	2	Default to zero if not applicable
Transaction Certificate (EMV)	ASC	9D	16	Default to zero if not applicable
TSI (EMV)	ASC	9F	4	Default to zero if not applicable
TVR (EMV)	ASC	9E	10	Default to zero if not applicable
Transaction ID	ASC	A1	10	
Plan Code	ASC	F3	Up to 6	optional
Tenure	ASC	F4	Up to 4	In number of months
First Amount	ASC	F5	12	
Monthly Amount	ASC	F6	12	
Last Amount	ASC	F7	12	
Interest Amount	ASC	F8	12	
Receipt Text Format	ASC	RP	var	Printed as is by the POS

4.8.9 Credit Card Offline via UPOS NPX (Function Code IB)

Request



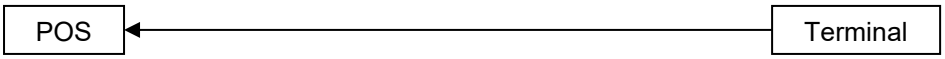
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"IB" (0x49 0x42)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	Mandatory
Approval Code	ASC	01	6	Mandatory
Enhanced ECR Reference Number	ASC	HD	…13	Mandatory

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response Message Data

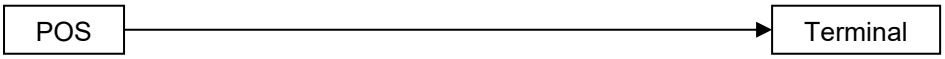
Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	..46	Terminal response message 1 line is consist of 23 characters
Merchant Name And Address	ASC	D0	..92	1 line is consist of 23 characters
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Approval Code	ASC	01	6	
Invoice Number	ASC	9H	6	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Issuer Name	ASC	D2	..20	Card preferred name (EMV tag)
Card Type	ASC	9M	..20	Card type. Eg: “VISA”, “MASTERCARD”, “AMEX”, “UPI”, “JCB”, *etc* **the actual value can differ for sample above.*
Card Number	ASC	30	..19	PAN (masked)
Processing Gateway	ASC	D7	..20	The name of the processing gateway that processed the transaction
Card Description	ASC	D8	..20	The card production name or card description
Expiry Date	ASC	31	4	MMYY Card expiry date
Batch Number	ASC	50	6	Terminal batch number for the transaction
Retrieval Reference Number	ASC	D3	12	
Card Holder Name	ASC	D6	..26	
AID (EMV)	ASC	9A	..16	
Application Profile (EMV)	ASC	9B	..16	

CID (EMV)	ASC	9C	2	
Transaction Certificate (EMV)	ASC	9D	16	Available if transaction is approved
TSI (EMV)	ASC	9F	4	
TVR (EMV)	ASC	9E	10	
Transaction ID	ASC	A1	10	
POS ID	ASC	9I	8	Terminal unique identifier
Host Response Code	ASC	HC	2	The response code returned by the host.
Card Entry Mode	ASC	CN	2	“10” – chip card transaction “20” – contactless transaction “30” – swipe transaction
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Scheme Category	ASC	9Q	..20	Returned if special scheme processing is performed.
Host Card Type	ASC	9R	…20	Card scheme name defined by host
Receipt Text Format	ASC	RP	var	Printed as is by the POS

4.8.10 Credit Card Pre-Authorization (Function Code IE)

This function is used to perform a pre-authorization transaction.

Request



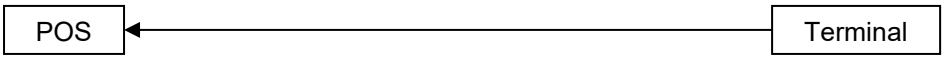
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"IE" (0x49 0x42)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	
Acquirer Name	ASC	9G	..20	Acquirer name Set as space if not used or do not send this field code. This is to be used only if there are multiple applications in the terminal that processes pre-auth.
ECR Reference Number	ASC	HD	..13	Optional field

Response



Message Header Description

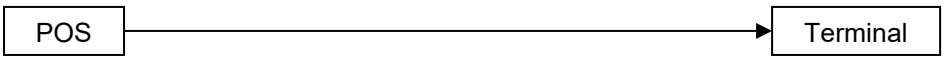
Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Descriptions

Name	Attribute	Function Code/ Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to Zero
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Approval Code	ASC	01	6	Taken from HOST
Expiry Date	ASC	31	4	MMYY Card expiry date
Invoice Number	ASC	9H	6	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Scheme Name	ASC	D2	10	
Card Number	ASC	30	Up to 19	Masked card number
Card Holder Name	ASC	D6	..26	
Batch Number	ASC	50	6	
Retrieval Reference Number	ASC	D3	12	
Transaction ID	ASC	A1	10	
POS ID	ASC	9I	8	Terminal unique identifier
Host Response Code	ASC	HC	2	The response code returned by the host.
Scheme Category	ASC	9Q	..20	Returned if special scheme processing is performed.
Card Entry Mode	ASC	CN	2	“10” – chip card transaction “20” – contactless transaction “30” – swipe transaction
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Receipt Text Format	ASC	RP	var	Printed as is by the POS

4.8.11 Credit Card Pre-Authorization Void (Function Code IF)

Request



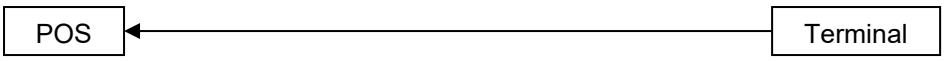
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“IF” (0x49 0x43)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Enhanced ECR Reference Number	ASC	HD	…13	Optional field
Transaction ID	ASC	A1	…10	Mandatory
Scheme Category	ASC	9Q	..20	Optional Field Additional info for credit card application to determine specific scheme to process.

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

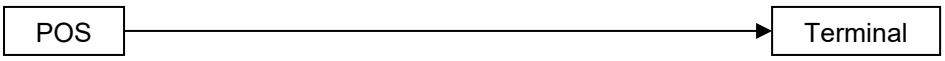
Response Message Data

Name	Attribute	Function Code/ Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Terminal response message 1 line is consist of 23 characters

Merchant Name And Address	ASC	D0	69	1 line is consist of 23 characters
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Approval Code	ASC	01	6	
Invoice Number	ASC	9H	6	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Issuer Name	ASC	D2	..20	Card type
Card Number	ASC	30	..19	PAN (masked)
Expiry Date	ASC	31	4	MMYY Card expiry date
Batch Number	ASC	50	6	Terminal batch number for the transaction
Retrieval Reference Number	ASC	D3	12	
Card Holder Name	ASC	D6	..26	
Transaction ID	ASC	A1	10	
POS ID	ASC	9I	8	Terminal unique identifier
Host Response Code	ASC	HC	2	The response code returned by the host.
Card Entry Mode	ASC	CN	2	“10” – chip card transaction “20” – contactless transaction “30” – swipe transaction
Scheme Category	ASC	9Q	..20	Returned if special scheme processing is performed.
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Receipt Text Format	ASC	RP	var	Printed as is by the POS

4.8.12 Credit Card Pre-Authorization Completion (Function Code IG)

Request



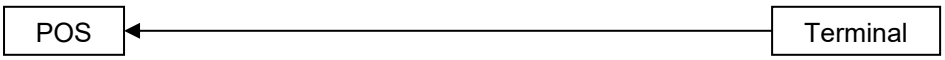
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"IG" (0x49 0x44)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	
Acquirer Name	ASC	9G	..20	Acquirer name Set as space if not used or do not send this field code. This is to be used only if there are multiple applications in the terminal that processes pre-auth.
Enhanced ECR Reference Number	ASC	HD	…13	Optional field
Transaction ID	ASC	A1	…10	Mandatory

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response Message Data

Name	Attribute	Function Code/ Field Code	Len (Bytes)	Remark

Response Text	ASC	02	40	Terminal response message 1 line is consist of 23 characters
Merchant Name And Address	ASC	D0	69	1 line is consist of 23 characters
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Approval Code	ASC	01	6	
Invoice Number	ASC	9H	6	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Issuer Name	ASC	D2	..20	Card type
Card Number	ASC	30	..19	PAN (masked)
Expiry Date	ASC	31	4	MMYY Card expiry date
Batch Number	ASC	50	6	Terminal batch number for the transaction
Retrieval Reference Number	ASC	D3	12	
Card Holder Name	ASC	D6	..26	
Transaction ID	ASC	A1	10	
POS ID	ASC	9I	8	Terminal unique identifier
Host Response Code	ASC	HC	2	The response code returned by the host.
Card Entry Mode	ASC	CN	2	“10” – chip card transaction “20” – contactless transaction “30” – swipe transaction
Scheme Category	ASC	9Q	..20	Returned if special scheme processing is performed.
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Receipt Text Format	ASC	RP	var	Printed as is by the POS

4.9 Callback Functions

This section describes the possible callback to the external module by the terminal. The callback will be returned if terminal is set to **ECR callback flag** enabled. In addition, some callbacks are available only depending on the terminal type and additional flag to be enabled.

4.9.1 Display Info Callback

External module is informed by terminal to display a message. These messages are being returned during a transaction, errors are not included. This callback will only be available if the **ECR Callback Message** flag is enabled.

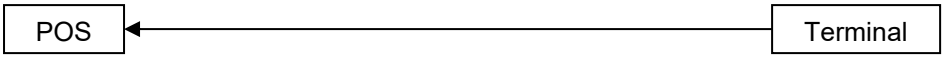
External module can map the index into displaying photo or play a video instead.

Note: In the event that POS does not recognize the index, it shall display the message provided by the terminal instead Field Code M2.

Terminal will still display messages on the screen if this callback is enabled. If terminal display is to be disabled, **ECR Display Callback Message** flag must be disabled.

Terminal is not expecting any response for this callback function.

Request



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the original function code
Function Code	ASC	2	“CB” (0x43 0x42)
Version Code	ASC	2	“01”
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Callback Function Code	ASC	M0	3	Value “DMG”
Message Index	ASC	M1	2	A reference index for the message
Message	ASC	M2	..69	Message to be displayed by the external module. This is the suggested message to be displayed to the external module; the external module however can display message/photo/video with similar meaning.

Message Index	Message	Description
01	Please Insert Card	Prompt the user to insert card on the reader
02	Please Swipe Card	Prompt the user to swipe card on the reader
03	Please Tap Card	Prompt the user to tap the contactless card on the reader

04	Please Insert/ Swipe/Tap Card	Prompt the user to either insert, swipe or tap the contactless card on the reader
05	Please Insert/Tap Card	Prompt the user to either insert or tap the contactless card on the reader
06	Please Swipe/Tap Card	Prompt the user to either swipe or tap the contactless card on the reader
07	Please Insert/Swipe Card	Prompt the user to either insert or swipe the card on the reader
08	Please Remove Card	Prompt the user to remove the card from the reader
09	Please Place NETS Flash Pay Card	Prompt the user to place the NETS Flash Pay on the contactless reader
0A	Please Insert NETS Cash Card	Prompt the user to insert NETS Cash Card on the reader
0B	Please Enter PIN	Prompt the user to enter PIN; Terminal will be requesting for account selection. External module is advised to allow user to select account SAVINGS, CURRENT, DEFAULT before or during this screen. If user does not select any of the account, external module can select as DEFAULT.
0C	Please Enter PIN	Prompt the user to enter PIN; Terminal will not request for account selection.
0D	Please Re-Enter PIN	Prompt the user to re-enter PIN
0E	Processing Please Wait	Prompt the user to please wait while transaction is being processed
0F	Please insert or tap NETS Prepaid Card	Prompt the user to either insert or tap NETS prepaid card.

4.9.2 Info Callback

External module is informed by terminal of a certain action that terminal is performing.

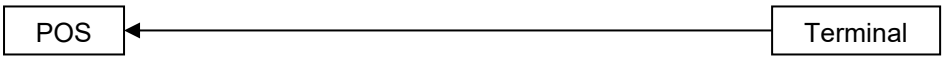
This is typically used to inform external module to display '*' during a pin entry.

This callback will only be available if the **ECR Callback Message** flag is enabled.

Terminal will still display messages on the screen if this callback is enabled. If terminal display is to be disabled, **ECR Display Callback Message** flag must be disabled.

Terminal is not expecting any response for this callback function.

Request



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the original function code
Function Code	ASC	2	"CD" (0x43 0x44)
Version Code	ASC	2	"01"
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

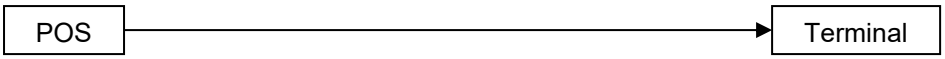
Response Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Callback Function Code	ASC	M0	3	Value "IFO"
Info Index	ASC	M1	2	A reference index for the message 01 – PIN entered indicator (*)
Info	ASC	M2	..69	Eg: "*"

4.10 Specific Functions

4.10.1 Get Card Info (Function Code S1)

Request



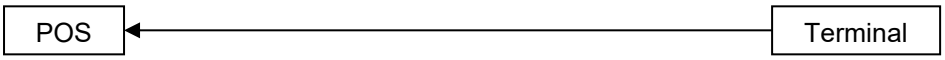
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"S1" (0x53 0x31)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Card Type	ASC	9M	..20	Optional. The POS may request the card number for specific type of card. Options are: • "NETS" • "CREDITCARD" • "NFP" • "CASHCARD" If the field is not available, terminal would get the card number based on card presented and internal selection logic.

Response



Message Header Description

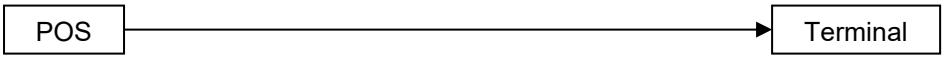
Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Card Type	ASC	9M	..20	The type of card selected for payment. Eg “NETS”, “VISA”, “MASTERCARD”, “NFP”, “CASHCARD”, etc **the actual value can differ for sample above.*
Card Number	ASC	30	..19	PAN (masked)

4.10.2 Terminal NFC and QR Display Function (Function Code S2)

This is non-payment function, selected terminal model – please check with NETS..
Request



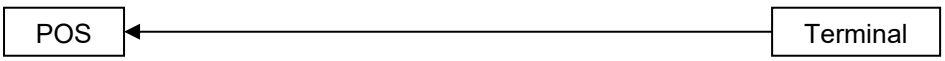
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“S2” (0x53 0x32)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Header Message 1	ASC	T6	38	Mandatory – Alphanumeric
Header Message 2	ASC	T7	38	Mandatory – Alphanumeric
Footer Message 1	ASC	T8	38	Mandatory – Alphanumeric
Display Timer	ASC	T9	2	Mandatory – Numeric If not numeric terminal will reject
NFC Payload	ASC	TA	VAR	Optional, up to 256 bytes This NFC payload strings for Terminal NFC
QR Payload	ASC	TB	VAR	Optional, up to 256 bytes This QR payload strings for Terminal to Display

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark

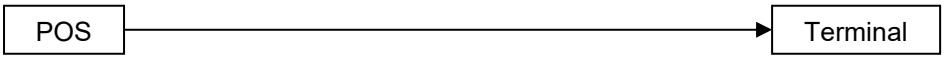
No message data required.

4.11 NETS Prepaid Card (NPC) Functions

4.11.1 NPC Balance Inquiry (Function Code 3A)

This function is used to get the balance of the NPC card.

Request



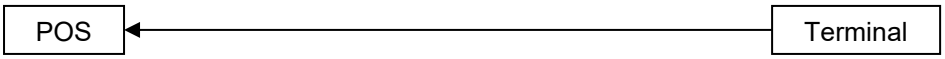
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“3A” (0x33 0x41)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Card Type	ASC	9M	..20	“NPC”

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

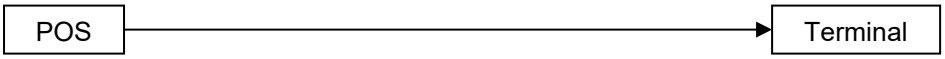
Message Data Descriptions

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
STAN	ASC	H6	6	Default to zeroes if not in used
Host Response Code	ASC	HC	2	Response code received from host.
Card Entry Mode	ASC	CN	2	“10” – chip card transaction “20” – contactless transaction
Current Balance	ASC	EE	12	For NPC Card only, this can contain negative (-) sign depending on Negative Indicator (EF)
Negative Indicator	ASC	EF	1	If value is “1” Previous balance value is negative.
Card Number	ASC	30	..19	PAN (masked)
Expiry Date	ASC	C2	8	DDMMYYYY
Response Message I	ASC	R0	20	Host Data Default to Space
Response Message II	ASC	R1	20	Host Data Default to Space
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

4.11.2 NPC Top-up (Function Code 3B)

This function is to topup NPC card using a NETS EFTPOS card or cash.

Request



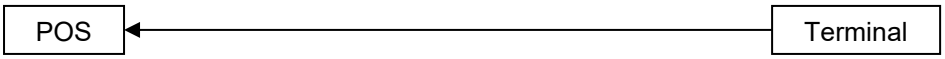
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“3B” (0x33 0x42)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	Mandatory
Top-Up Option: Cash or NETS	HEX	T1	1	Set to 1 for Cash Top-Up Set to 2 for NETS Top-Up
Enhanced ECR Reference Number	ASC	HD	13	Mandatory

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Descriptions

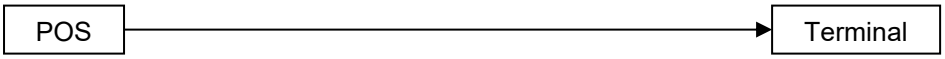
Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	..40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	..69	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
STAN	ASC	H6	6	
Total Fee (Top Up)	ASC	ZA	6	Default to zeros if not in used
Fee due to merchant (Top Up)	ASC	ZB	6	Default to zeros if not in used
Fee due from merchant (Top Up)	ASC	ZC	6	Default to zeros if not in used
Host Response Code	ASC	HC	2	Response code received from host.
Card Number	ASC	30	..19	PAN (masked) of NPC
Expiry Date	ASC	C2	8	DDMMYYYY Expiry date of NPC
Response Message I	ASC	R0	20	Host Data Default to Space
Response Message II	ASC	R1	20	Host Data Default to Space
Card Entry Mode	ASC	CN	2	“10” – Chip transaction “20” – Contactless transaction For the NETS card
Card Entry Mode 2	ASC	CQ	2	“10” – Chip transaction “20” – Contactless transaction For NPC card
Current Balance	ASC	EE	12	For NPC Card only, this can contain negative (-) sign depending on Current Balance Negative Indicator (EG)
Current Balance Negative Indicator	ASC	EG	1	If value is “1” Current balance value is negative.

Previous Balance	ASC	ED	12	This contains the previous balance for NPC Card, this can contain negative (-) sign depending on Previous Balance Negative Indicator (EF)
Previous Balance Negative Indicator	ASC	EF	1	If value is “1” Previous balance value is negative.
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

4.11.3 NPC Activation (Function Code 3C)

This function is to perform NPC Activation.

Request



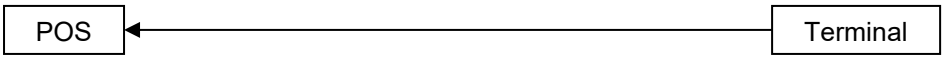
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“3C” (0x33 0x43)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Enhanced ECR Reference Number	ASC	HD	13	Mandatory

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Descriptions

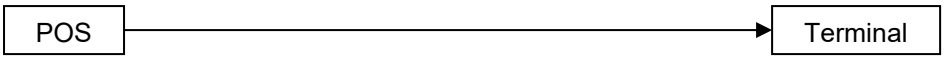
Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	..40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	..69	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
STAN	ASC	H6	6	
Host Response Code	ASC	HC	2	Response code received from host.
Card Number	ASC	30	..19	PAN (masked) of NPC
Expiry Date	ASC	C2	8	DDMMYYYY Expiry date of NPC
Response Message I	ASC	R0	20	Host Data Default to Space
Response Message II	ASC	R1	20	Host Data Default to Space
Current Balance	ASC	EE	12	For NPC Card only
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

4.12 CEPAS Functions

4.12.1 Acquirer Logon (Function Code A1)

This function will initiate specific terminal application to do a logon.

Request



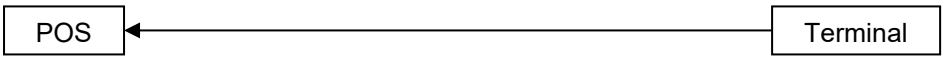
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"A1"
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Acquirer Name	ASC	9G	…20	To specify which acquirer to logon "EZLINK" "CONCESSION"

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

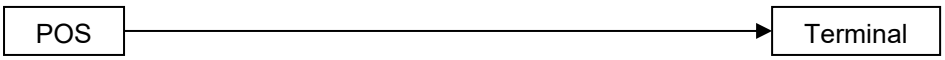
Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS

STAN	ASC	65	6	Transaction Sequence Number
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Host Response Code	ASC	HC	2	Response code returned from acquirer host. Default to spaces if not relevant
Processing Gateway	ASC	D7	20	Terminal returns the processing gateway information. Eg. “NETS” or “CEPAS”

4.12.2 Acquirer Settlement (Function Code A2)

This function will initiate the terminal to perform settlement. When the settlement is approved, current batch will be cleared.

Request



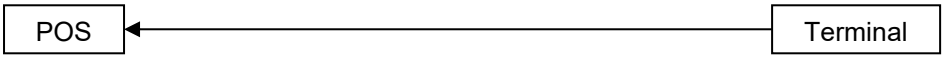
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"A2"
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Acquirer Name	ASC	9G	…20	"EZLINK" "CONCESSION"

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

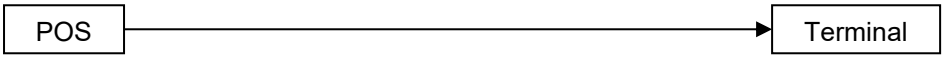
Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
STAN	ASC	H6	6	

Processing Gateway	ASC	D7	20	Terminal returns the processing gateway information “EZLINK“
Number of Processing Host	ASC	9J	2	The number of processing host being settled by the terminal
** HOST DATA START ** This entry will be repeated depending on the value at field code 9J (Number of Processing Host)				
Host ID	ASC	9K	2	Host ID is just to indicate the host index based on the number of host reflected in field 9J.
Host Response Code	ASC	HC	2	The response code returned by the host. Default to spaces when not available.
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Type	ASC	9M	..99	Type of Payment Cards
Data Type	ASC	9N	..99	Transaction Type
Sale Count	ASC	N0	4	
Sale Total	ASC	G0	12	
** HOST DATA END **				

4.12.3 Acquirer Pre-Settlement (Function Code A3)

This function will initiate the terminal to perform acquirer pre-settlement.

Request



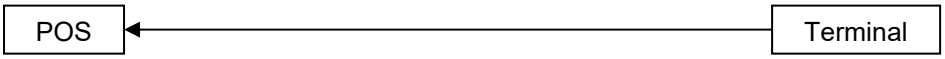
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"A2"
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Acquirer Name	ASC	9G	…20	"EZLINK" "CONCESSION"

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

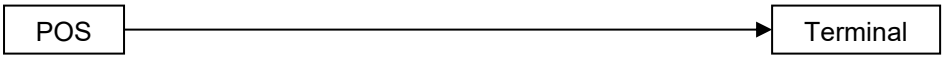
Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
STAN	ASC	H6	6	

Processing Gateway	ASC	D7	20	Terminal returns the processing gateway information “EZLINK“
Number of Processing Host	ASC	9J	2	The number of processing host being settled by the terminal
** HOST DATA START ** This entry will be repeated depending on the value at field code 9J (Number of Processing Host)				
Host ID	ASC	9K	2	Host ID is just to indicate the host index based on the number of host reflected in field 9J.
Host Response Code	ASC	HC	2	The response code returned by the host. Default to spaces when not available.
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Card Type	ASC	9M	..99	Type of Payment Cards
Data Type	ASC	9N	..99	Transaction Type
Sale Count	ASC	N0	4	
Sale Total	ASC	G0	12	
** HOST DATA END **				

4.12.4 CEPAS Debit (Function Code 26)

This function is used to perform a debit transaction for CEPAS cards

Request



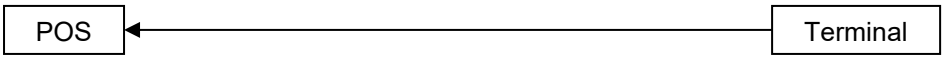
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"26" (0x32 0x36)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	Mandatory
Enhanced ECR Reference Number	ASC	HD	..13	Mandatory
Acquirer Name	ASC	9G	..20	This field will inform the terminal what type of CEPAS card is expected. Supported values: • "EZLINK" • "CONCESSION"

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Descriptions

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Batch Number	ASC	50	12	In the format of YYMMDDHHMMSS
Transaction Amount	ASC	40	12	EFTPOS Transaction Amount without EFTPOS Service Fee
Card Number	ASC	30	16	Card Application Number (CAN)
Expiry Date	ASC	C2	8	DDMMYYYY
Purchase Fee	ASC	ZP	6	Purchase fee where applicable If debit failed, fee will be zeros.
Card Issuer Name	ASC	D2	10	Returns the type of cards: • “EZLINK” • “CONCESSION”
CEPAS Version	ASC	C1	2	CEPAS 2.0 = “20” CEPAS 3.0 = “30”
*Trans Data	ASC	C0	64	
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
STAN	ASC	65	6	Optional
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

Trans Data for CEPAS 2.0

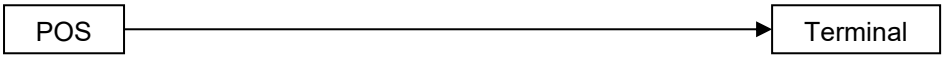
Field Name	Attribute	Len (Bytes)	Remark
Transaction Type	ASC	2	01 == Normal C/L Purchase
Transaction Amount	ASC	12	TT01 == C/L Purchase Amount without Service Fee
Transaction Date	ASC	6	YYMMDD
Transaction Time	ASC	6	HHMMSS
Prior Card Balance	ASC	12	
Post Card Balance	ASC	12	
Post Auto load Amount	ASC	12	

Post Card Status	ASC	2	“00” – Purse Disabled + ATU Disabled “01” – Purse Enabled + ATU Disabled “02” – Purse Disabled + ATU Enabled “03” – Purse Enabled + ATU Enabled
Total		64	

4.12.5 CEPAS Top-up (Function Code 47)

This function is used to perform CEPAS top up by cash (for all CEPAS cards)

Request



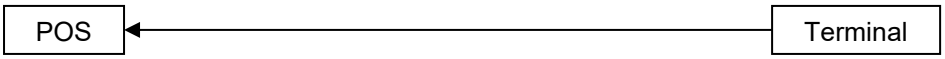
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“47” (0x34 0x37)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	Mandatory
Acquirer Name	ASC	9G	…20	To specify which card to top up “EZLINK“ “CONCESSION“
Top-Up Option	ASC	T1	1	Set to 1 for Cash Top-Up
Enhanced ECR Reference Number	ASC	HD	13	Mandatory

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Descriptions

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	..40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	..69	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Batch Number	ASC	50	12	In the format of YYMMDDHHMMSS
STAN	ASC	H6	6	Trace number
Total Fee (Top Up)	ASC	ZA	6	Default to zeros if not in used
CEPAS Version	ASC	C1	2	CEPAS 2.0 = “20” CEPAS 3.0 = “30”
Card Number	ASC	30	16	Card Application Number (CAN)
Expiry Date	ASC	C2	8	DDMMYYYY
Host Response Code	ASC	HC	2	Response code received from host. “00” – successful
*Trans Data	ASC	C0	64	
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

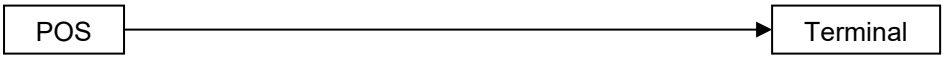
Trans Data for CEPAS 2.0

Field Name	Attribute	Len (Bytes)	Remark
Transaction Type	ASC	2	02 == Cash Top Up
Transaction Amount	ASC	12	
Transaction Date	ASC	6	YYMMDD
Transaction Time	ASC	6	HHMMSS
Prior Card Balance	ASC	12	
Post Card Balance	ASC	12	
Post Auto load Amount	ASC	12	
Post Card Status	ASC	2	“00” – Purse Disabled + ATU Disabled “01” – Purse Enabled + ATU Disabled “02” – Purse Disabled + ATU Enabled “03” – Purse Enabled + ATU Enabled
Total		64	

4.12.6 CEPAS Void (Function Code C1)

This function is used to void a CEPAS debit transaction

Request



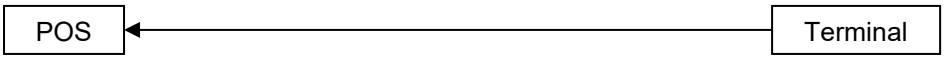
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“C1” (0x43 0x31)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
STAN	ASC	65	6	Mandatory
Enhanced ECR Reference Number	ASC	HD	..13	Mandatory
Acquirer Name	ASC	9G	..20	This field will inform the terminal what type of CEPAS card is expected. Supported values: • “EZLINK” • “CONCESSION”

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer to Annex C
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Descriptions

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
Batch Number	ASC	50	12	In the format of YYMMDDHHMMSS
Transaction Amount	ASC	40	12	EFTPOS Transaction Amount without EFTPOS Service Fee
Card Number	ASC	30	16	Card Application Number (CAN)
Expiry Date	ASC	C2	8	DDMMYYYY
Purchase Fee	ASC	ZP	6	Purchase fee where applicable If debit failed, fee will be zeros.
Card Issuer Name	ASC	D2	10	Returns the type of cards: • “FLASHPAY” • “EZLINK” • “CONCESSION”
CEPAS Version	ASC	C1	2	CEPAS 2.0 = “20” CEPAS 3.0 = “30”
*Trans Data	ASC	C0	64	
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

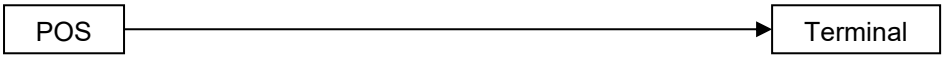
Trans Data for CEPAS

Field Name	Attribute	Len (Bytes)	Remark
Transaction Type	ASC	2	04 == Void Purchase 05 == Void Topup
Transaction Amount	ASC	12	Purchase Amount without Service Fee / Topup amount
Transaction Date	ASC	6	YYMMDD
Transaction Time	ASC	6	HHMMSS
Prior Card Balance	ASC	12	
Post Card Balance	ASC	12	
Post Auto load Amount	ASC	12	

Post Card Status	ASC	2	“00” – Purse Disabled + ATU Disabled “01” – Purse Enabled + ATU Disabled “02” – Purse Disabled + ATU Enabled “03” – Purse Enabled + ATU Enabled
Total		64	

4.12.7 CEPAS Read Balance (Function Code 32)

Request



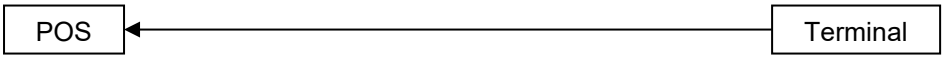
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"32" (0x33 0x32)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Acquirer Name	ASC	9G	..20	This field will inform the terminal what type of CEPAS card is expected. Supported values: • "EZLINK" • "CONCESSION"

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
Header Filler	ASC	12	Defaulted to zero (30 30 …….)
Function Code	ASC	2	Value "32"
Response Code	ASC	2	Refer to Response code list
End of Message	ASC	1	Indicate the end of message Value '0' to indicate no other messages for this Command
Separator	HEX	1	Separator Value is: 1C

Message Data Description

Name	Attribute	Field Code	Len (Bytes)	Remark
Response Text	ASC	02	40	Default to zeroes / spaces if values are not available
Merchant Name And Address	ASC	D0	69	
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	

CEPAS Version	ASC	C1	2	CEPAS 2.0 = “20” CEPAS 3.0 = “30”
Card Issuer Name	ASC	D2	10	Returns the type of cards: • “EZLINK” • “CONCESSION”
Card Number	ASC	30	16	Card Application Number (CAN)
Expiry Date	ASC	C2	8	DDMMYYYY
Card Balance	ASC	HC	12	
Purse Status	ASC	S3	2	00 = disable 01 = enable 02 = not applicable
ATU Status	ASC	S2	2	00 = disable 01 = enable 02 = not applicable
Existing ATU Amount	ASC	C8	12	Current ATU Amount in Card

* This function only supports CEPAS 2 cards

4.13 Consumer Choice Routing (CCR) Functions

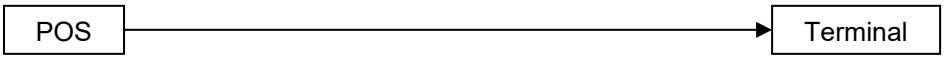
4.13.1 Credit Purchase with CCR (Function Code P1)

This function triggers Credit Purchase with option:

- CCR Selection, if card has both NETS and ICS/CREDIT capability
- ICS/Credit Purchase, if card only has ICS/CREDIT only
- NETS Purchase, if card has NETS only

Please refer to the appendix for implementation flow.

Request



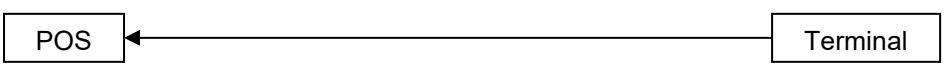
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	“P1” (0x50 0x31)
Version Code	ASC	2	“01” (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	
Enhanced ECR Reference Number	ASC	HD	…13	Mandatory
Acquirer Name	ASC	9G	…20	Optional field This field will inform the terminal to force all transaction to a specific acquirer. If this field is not present or has an invalid value, the terminal will select the acquirer base on bin processing.

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request

Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Consumer Choice Indicator	ASC	MC	2	"00" – No Selection Required. "01" – Routing Selection; Card has NETS and ICS/Credit capability Merchant POS to display routing choice
Consumer Routing Indicator	ASC	MR	2	"00" – Consumer Choice Routing (CCR) Selection POS to continue with P2 for NETS **or** P3 for ICS/Credit. "01" – NETS Purchase "02" – ICS Purchase / CREDIT Purchase. Merchant POS requires saving transaction on respective batch (NETS or ICS / CREDIT) for settlement.
Response Text	ASC	02	46	**For CCR Type "01" NETS Purchase:** Default to zeroes / spaces if values are not available For Flashpay and Cashcard APPROVED\n CCCCCCCCCCCCCCCC\n BAL: NNN.NN where C is the CAN in ASC16 and N is the current balance For NETS Prepaid Card: APPROVED\n xxxxxxxxxxxCCCC\n BAL: NNNN.NN where C is the masked card number in ASC16 and N is the current balance **For CCR Type "02" ICS / CREDIT Purchase:** Terminal response message 1 line is consist of 23 characters
Merchant Name And Address	ASC	D0	92	Up to 4 lines, each line consist of 23 characters
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Terminal ID	ASC	16	8	

Merchant ID	ASC	D1	15	
STAN	ASC	65	6	NFP transactions will return "000000" CashCard transaction to return "000000"
Approval Code	ASC	01	6	Transaction approval code. NPF transactions to return TRP here. CashCard transaction to return 00||CashCard sign cert
Invoice Number	ASC	9H	6	NFP transactions will return "000000" CashCard transaction to return "000000"
Transaction Amount	ASC	40	12	NET Amount. Set to zeros if not used
Service Fee	ASC	41	12	Only applicable for CCR Type "01" NETS Purchase.
Host Response Code	ASC	HC	2	Response code received from host. **For CCR Type "01" NETS Purchase:** NFP / CashCard transaction returns "00" for successful debit And return "SC" for failed transaction
Retrieval Reference Number	ASC	D3	12	**For CCR Type "01" NETS Purchase:** Default to zeroes / spaces if values are not available (NFP Puchase)
Card Entry Mode	ASC	CN	2	"10" – chip card transaction "20" – contactless transaction "30" – swipe transaction NFP transaction returns "20" for contactless CashCard transaction returns "10" for chip
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Field list that applicable for CCR Type "01" NETS Purchase:				
Card Name	ASC	L7	20	Default to zeroes if not available. NFP transaction to return "FLASHPAY" CC transaction to return "CASHCARD"
POS Messages	ASC	L5	240	Host Data, Maximum 240 characters (10 lines message)
Response Message I	ASC	R0	20	Host Data Default to Space
Response Message II	ASC	R1	20	Host Data Default to Space
Loyalty Program Name	ASC	L1	24	Loyalty Data Maximum 24 characters
Loyalty Program Exp Date	ASC	L8	8	Loyalty Data "YYYYMMDD"

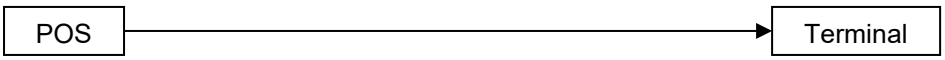
Loyalty Type	ASC	L2	1	Loyalty Data “0”(30): No Loyalty Program “1”(31): Points “2”(32): Dollars
Loyalty Marketing Message	ASC	L9	144	Loyalty Data Maximum 144 characters (6 lines message)
Redemption Value	ASC	L3	12	Loyalty Data
Current Loyalty Balance	ASC	L4	12	Loyalty Data
Field list that applicable for CCR Type "02" ICS / CREDIT Purchase:				
Processing Gateway	ASC	D7	20	The name of the processing gateway that processed the transaction
Card Issuer Name	ASC	D2	20	Card preferred name (EMV tag)
Card Type	ASC	9M	20	Card type. Eg: “VISA”, “MASTERCARD”, “AMEX”, “UPI”, “JCB”, *etc* *the actual value can differ for sample above.
Card Number	ASC	30	19	PAN (masked)
Card Description	ASC	D8	20	The card production name or card description
Expiry Date	ASC	31	4	MMYY - Card expiry date
Batch Number	ASC	50	6	Terminal batch number for the transaction
Card Holder Name	ASC	D6	26	
AID (EMV)	ASC	9A	16	
Application Profile (EMV)	ASC	9B	16	
CID (EMV)	ASC	9C	2	
Transaction Certificate (EMV)	ASC	9D	16	Available if transaction is approved
TSI (EMV)	ASC	9F	4	
TVR (EMV)	ASC	9E	10	
Transaction ID	ASC	A1	10	
POS ID	ASC	9I	8	Terminal unique identifier
Loyalty Type	ASC	93	2	“01” – POSB Everyday “02” – UOB “03” – UOB (SPC) “04” – Diners “05” – DBS Instant Rewards **only applicable for select merchants*
Loyalty Information	ASC	94	var	Refer to section below for the loyalty data format
Scheme Category	ASC	9Q	20	Returned if special scheme processing is performed.
Offline Txn Type	HEX	O1	1	Optional Field This field will inform the POS if the transaction was an offline transaction. If field is not present, transaction was done online. 0x00 – Online transaction 0x01 – Offline transaction

Host Card Type	ASC	9R	…20	Card scheme name defined by host
Applicable for both CCR Type "01" NETS Purchase and "02" ICS / CREDIT Purchase				
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

4.13.2 CCR Selection – Proceed with NETS Purchase (Function Code P2)

This function applicable after P1 triggered with CCR selection. Function code P2 – for Consumer to trigger terminal to proceed with NETS Purchase.
Please refer to the appendix for implementation flow.

Request



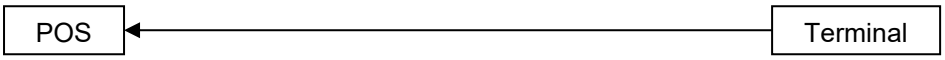
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"P2" (0x50 0x32)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	
Enhanced ECR Reference Number	ASC	HD	…13	Mandatory

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Consumer Routing Indicator	ASC	MR	2	"01" – NETS Purchase Merchant POS requires saving transaction on NETS batch.

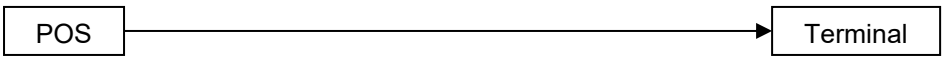
Response Text	ASC	02	46	**For CCR Type "01" NETS Purchase:** Default to zeroes / spaces if values are not available For Flashpay and Cashcard APPROVED\n CCCCCCCCCCCCCCCC\n BAL: NNN.NN where C is the CAN in ASC16 and N is the current balance For NETS Prepaid Card: APPROVED\n xxxxxxxxxxxxCCCC\n BAL: NNNN.NN where C is the masked card number in ASC16 and N is the current balance
Merchant Name And Address	ASC	D0	92	Up to 4 lines, each line consist of 23 characters
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
STAN	ASC	65	6	NFP transactions will return "000000" CashCard transaction to return "000000"
Approval Code	ASC	1	6	Transaction approval code. NPF transactions to return TRP here. CashCard transaction to return 00||CashCard sign cert
Invoice Number	ASC	9H	6	NFP transactions will return "000000" CashCard transaction to return "000000"
Transaction Amount	ASC	40	12	NET Amount. Set to zeros if not used
Service Fee	ASC	41	12	Only applicable for CCR Type "01" NETS Purchase.
Host Response Code	ASC	HC	2	Response code received from host. **For CCR Type "01" NETS Purchase:** NFP / CashCard transaction returns "00" for successful debit And return "SC" for failed transaction
Retrieval Reference Number	ASC	D3	12	**For CCR Type "01" NETS Purchase:** Default to zeroes / spaces if values are not available (NFP Puchase)

Card Entry Mode	ASC	CN	2	“10” – chip card transaction “20” – contactless transaction “30” – swipe transaction NFP transaction returns "20" for contactless CashCard transaction returns "10" for chip
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Field list that applicable for CCR Type "01" NETS Purchase:				
Card Name	ASC	L7	20	Default to zeroes if not available. NFP transaction to return "FLASHPAY" CC transaction to return "CASHCARD"
POS Messages	ASC	L5	240	Host Data, Maximum 240 characters (10 lines message)
Response Message I	ASC	R0	20	Host Data Default to Space
Response Message II	ASC	R1	20	Host Data Default to Space
Loyalty Program Name	ASC	L1	24	Loyalty Data Maximum 24 characters
Loyalty Program Exp Date	ASC	L8	8	Loyalty Data “YYYYMMDD”
Loyalty Type	ASC	L2	1	Loyalty Data “0”(30): No Loyalty Program “1”(31): Points “2”(32): Dollars
Loyalty Marketing Message	ASC	L9	144	Loyalty Data Maximum 144 characters (6 lines message)
Redemption Value	ASC	L3	12	Loyalty Data
Current Loyalty Balance	ASC	L4	12	Loyalty Data
Applicable for both CCR Type "01" NETS Purchase and “02” ICS / CREDIT Purchase				
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

4.13.3 CCR Selection – Proceed with ICS/CREDIT Purchase (Function Code P3)

This function applicable after P1 triggered with CCR selection. Function code P3 – for Consumer to trigger terminal to proceed with ICS/CREDIT Purchase.
Please refer to the appendix for implementation flow.

Request



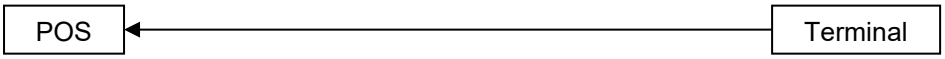
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"P3" (0x50 0x33)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Transaction Amount	ASC	40	12	
Enhanced ECR Reference Number	ASC	HD	…13	Mandatory
Acquirer Name	ASC	9G	…20	Optional field This field will inform the terminal to force all transaction to a specific acquirer. If this field is not present or has an invalid value, the terminal will select the acquirer base on bin processing.

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Consumer Routing Indicator	ASC	MR	2	"02" – ICS Purchase / CREDIT Purchase Merchant POS requires saving transaction on ICS/CREDIT batch.
Response Text	ASC	02	..46	**For CCR Type "02" ICS / CREDIT Purchase:** Terminal response message 1 line is consist of 23 characters
Merchant Name And Address	ASC	D0	..92	Up to 4 lines, each line consist of 23 characters
Transaction Date	ASC	03	6	YYMMDD
Transaction Time	ASC	04	6	HHMMSS
Terminal ID	ASC	16	8	
Merchant ID	ASC	D1	15	
STAN	ASC	65	6	NFP transactions will return "000000" CashCard transaction to return "000000"
Approval Code	ASC	01	6	Transaction approval code. NPF transactions to return TRP here. CashCard transaction to return 00||CashCard sign cert
Invoice Number	ASC	9H	6	NFP transactions will return "000000" CashCard transaction to return "000000"
Transaction Amount	ASC	40	12	NET Amount. Set to zeros if not used
Service Fee	ASC	41	12	Only applicable for CCR Type "01" NETS Purchase.
Host Response Code	ASC	HC	2	Response code received from host. **For CCR Type "01" NETS Purchase:** NFP / CashCard transaction returns "00" for successful debit And return "SC" for failed transaction
Retrieval Reference Number	ASC	D3	12	**For CCR Type "01" NETS Purchase:** Default to zeroes / spaces if values are not available (NFP Puchase)
Card Entry Mode	ASC	CN	2	"10" – chip card transaction "20" – contactless transaction "30" – swipe transaction NFP transaction returns "20" for contactless CashCard transaction returns "10" for chip

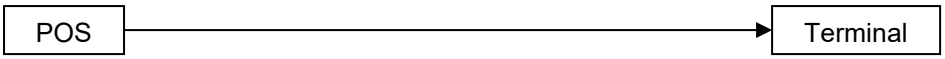
Enhanced ECR Reference Number	ASC	HD	…13	Echo from the request
Field list that applicable for CCR Type "02" ICS / CREDIT Purchase:				
Processing Gateway	ASC	D7	..20	The name of the processing gateway that processed the transaction
Card Issuer Name	ASC	D2	..20	Card preferred name (EMV tag)
Card Type	ASC	9M	..20	Card type. Eg: "VISA", "MASTERCARD", "AMEX", "UPI", "JCB", *etc* *the actual value can differ for sample above.
Card Number	ASC	30	..19	PAN (masked)
Card Description	ASC	D8	..20	The card production name or card description
Expiry Date	ASC	31	4	MMYY - Card expiry date
Batch Number	ASC	50	6	Terminal batch number for the transaction
Card Holder Name	ASC	D6	..26	
AID (EMV)	ASC	9A	..16	
Application Profile (EMV)	ASC	9B	..16	
CID (EMV)	ASC	9C	2	
Transaction Certificate (EMV)	ASC	9D	16	Available if transaction is approved
TSI (EMV)	ASC	9F	4	
TVR (EMV)	ASC	9E	10	
Transaction ID	ASC	A1	10	
POS ID	ASC	9I	8	Terminal unique identifier
Loyalty Type	ASC	93	2	"01" – POSB Everyday "02" – UOB "03" – UOB (SPC) "04" – Diners "05" – DBS Instant Rewards **only applicable for select merchants*
Loyalty Information	ASC	94	var	Refer to section below for the loyalty data format
Scheme Category	ASC	9Q	..20	Returned if special scheme processing is performed.
Offline Txn Type	HEX	O1	1	Optional Field This field will inform the POS if the transaction was an offline transaction. If field is not present, transaction was done online. 0x00 – Online transaction 0x01 – Offline transaction
Host Card Type	ASC	9R	…20	Card scheme name defined by host
Applicable for both CCR Type "01" NETS Purchase and "02" ICS / CREDIT Purchase				
Receipt Text Format	ASC	RP	VAR	Printed as is by the POS

4.14 Non NETS Functions

4.14.1 DFI Read TFM LID (Function Code T0)

This function to read TFM card which applicable to DFI terminal only.

DFI Read TFM - Request



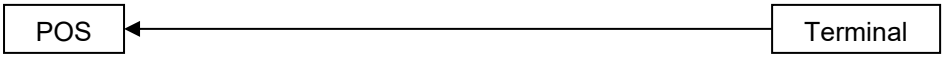
Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	"T0" (0x54 0x30)
Version Code	ASC	2	"01" (0x30 0x31)
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Request Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
Enhanced ECR Reference Number	ASC	HD	…13	Mandatory

Response



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	Please refer below list
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response Message Data

Name	Attribute	Field Code	Len (Bytes)	Remark
TFM Loyalty ID (LID)	ASC	35	..19	TFM Card Loyalty ID Default to zeroes / spaces if values are not available
TFM Card Type	ASC	36	2	01: Ezlink Passion Card 02: POSB MasterCard Debit

Annex A. Host Response Code List

The following table lists the possible response codes from the host that the terminal will send back to the POS in a response message for all online commands via the field code HC (Host Response Code). Each code corresponds to a respective message that can be displayed or printed by the POS.

Only response code **00** is treated as an Approved transaction. The rest is considered as **Not Approved. The host response code is subject to changes without prior notice.**

It's recommended for POS to display a generic error for those transactions that is not approved. E.G. "Transaction Declined"

Response Code	Meaning
00	APPROVED
01	REFER TO NETS
02	REFER TO BANK
03	INVALID TERMINAL
12	INVALID TRANS
13	INVALID AMOUNT
14	INVALID CARD
19	PLS TRY AGAIN
20	INVALID ADDRESS
21	ADDRESS NOT FOUND
22	FOREIGN ADDRESS
23	INVALID SUP CARD
24	INVALID DDA DATA
25	NO RECORD ON FILE
26	AMOUNT ZERO
27	INVALID AMOUNT
28	INVALID ACCOUNT
29	DUPLICATE DDA
30	INVALID TRANSACTION
31	INVALID TRANSACTION
41	LOST/STOLEN CARD
51	INVALID TRANSACTION
54	EXPIRED CARD
55	INCORRECT PIN
58	INVALID TRANSACTION
61	DAILY LIMIT EXCEEDED
62	INVALID TRANSACTION
63	VOID IMPOSSIBLE
64	TXN ALREADY VOID
75	PIN TRIES EXCEEDED
76	INVALID TRANSACTION
78	INVALID CARD

Response Code	Meaning
81	INVALID CARD
85	INVALID CARD
87	DAILY LIMIT EXCEEDED
88	REFUND NOT ALLOWED
91	NO REPLY FROM BANK
V1	REQUEST CRYPTOGRAM DECLINED
V2	ATC DECLINED
V3	CVR DECLINED
V4	TVR DECLINED
V5	FALLBACK DECLINED
V6	PIN REQUIRED
V7	SWITCH TO CONTACT MODE
V8	SUK EXPIRED
C2	CARD LOCKED
C3	EXPIRED CASH CARD
C4	REFUNDED CASH CARD
C5	INVALID TRANSACTION
C6	AMOUNT OVER MAX
C7	INSUFFICIENT FUND
C8	INCORRECT BALANCE
C9	PROBLEM BALANCE
D0	ATU ENABLED
D1	ATU DISABLED
D2	TRANSACTION NOT ALLOWED
D3	INVALID CONTACT NUMBER
D4	INVALID DATE OF BIRTH
D5	AGE LIMIT EXCEEDED
D6	INVALID CARD
D7	REFUND NOT ALLOWED
D8	REFUND AMOUNT GREATER THAN $100
E0	CARD DISABLED
E4	REFER TO NETS
E6	INVALID TRANSACTION
E7	REFER TO NETS
E8	REFER TO NETS
E9	CARD NOT ACTIVATED
F0	REFUNDED CARD
F1	CARD ENABLED
F2	CARD DISABLED
F3	CARD ACTIVATED

Response Code	Meaning
F4	DE-REGISTRATION IN PROGRESS
98	MAC ERROR
IM	UNAUTHORISED RESPONSE
IR	INVALID HOST MESSAGE
IT	INVALID TERMINAL
IA	INVALID HOST AMOUNT
IC	INVALID CARD
IL	INVALID DATA LENGTH
HS	CONNECT PROBLEM TO HOST
TO	TIMEOUT-PLEASE TRY AGAIN
US	CANCELLED BY USER
BF	TRANSACTION BATCH FULL
SC	CASHCARD TRANSACTION UNSUCCESSFUL
N0	DIFFERENT ISSUERS FOR A NETS SALE OR REVALUTION
N1	CASHCARD BLACKLIST
N2	BATCH ALREADY UPLOADED
N3	RESEND BATCH
N4	CASHCARD NOT FOUND
N5	EXPIRED
N6	REFUNDED CASHCARD
N7	CERTIFICATE ERROR
N8	INSUFFICIENT FUNDS/BALANCE
NA	TRANSACTION NOT AVAILABLE
IS	INVALID STAN*
RN	RECORD NOT FOUND
RE	READER NOT READY
T1	INVALID TOP-UP CARD
T2	TERMINAL TOP-UP LIMIT EXCEEDED
T3	RETAILER LIMIT EXCEEDED
LR	MANUAL LOGON REQUIRED**
DK	BLOCKLIST DOWNLOAD REQUIRED
DS	CASHCARD SETTLEMENT REQUIRED
BP	CARD BLOCKED
BA	TRANSACTION BLOCKED, to prevent Autoload
GB	Golden Bullet Card

*Cater for invalid STAN input into POS such as alphanumerical number.

** This response code only catered for ECR command Logon (function code 80) and Settlement (function code 81). It is returned from the terminal if manual logon has not been performed on the terminal and the ECR command 80/ 81 is sent to the terminal.

Annex B. Field Code List

Each field code is identified with a two-character code. ASCII Characters are available for use in the field code.

Field code	Description
01	Approval Code
02	Response Text
03	Transaction Date
04	Transaction Time
16	Terminal ID
17	Authorization ID
21	Host Random Number
22	Sequence Counter
23	SAM Random Number
24	SAM Cryptogram
25	Key Data
26	Host Cryptogram
27	C-MAC
28	Acquirer ID
29	CSN
30	Card Number
31	RFU4
32	LTA SAM ID
33	MID
34	Track 2 Data
35	TFM Loyalty ID (LID)
36	TFM Card Type
40	Transaction Amount
41	Service Fee
42	Cash Back Amount
43	Loyalty Redemption
46	Fee Due to Merchant
47	Fee Due from Merchant
48	Cash Deposit Net Fee
49	Cash Deposit Fee
50	Batch Number
65	STAN
70	GroupRID Listing
71	GroupRID
72	Promotion Data
73	Item Code
74	Item Count
75	Merchant Dollar Denomination
76	Item Count Before
77	Merchant Dollar Denomination Before
78	Item Count Final
79	Merchant Dollar Denomination Final
7A	Card Data
80	Detect Timeout
81	Status Word
82	Card Random Number

Field code	Description
83	Authentication Data
84	Authentication Key ID
85	File ID
86	Data Read / Write
87	Expected Data Size
93	Loyalty Type
94	Loyalty Information
9A	AID (EMV)
9B	Application Profile (EMV)
9C	CID (EMV)
9D	Transaction Certificate (EMV)
9E	TVR (EMV)
9F	TSI (EMV)
9G	Acquirer Name
9H	Invoice Number
9I	POSID
9J	Number of Credit Host
9K	Host ID
9L	Number of DCC Host
9M	Card Type
9N	Data Type
9O	Currency Code
9P	Number of IPP Host
9Q	Scheme Category
9R	Host Card Type
B0	Block Reason Code
B1	Block Message
C0	Trans Data
C1	CEPAS Version
C2	Expiry Date
C3	CAN Status
C4	Date of Birth
C5	CAN Entry Mode
C6	NRIC
C7	Selected ATU Amount
C8	Existing ATU Amount
C9	Card Presence
CA	Contact Number
CB	Allow Unblock
CC	Allow Payment
CD	Notify of Purpose
CE	Notify of Reason
CF	Comment
CG	Activation Code
CH	Customer ID, Country
CI	Customer ID, Type
CJ	Customer ID
CK	Credit Card Number
CL	Insurance Flag
CM	Retailer Reference Number
CN	Card Entry Mode

Field code	Description
CO	Card Option
CP	Transaction Status
CQ	Card Entry Mode 2
D0	Merchant Name And Address
D1	Merchant ID
D2	Card Issuer Name
D3	Retrieval Reference Number
D4	RFU2
D5	Signature Required
D6	Card Holder Name
D7	Processing Gateway
D8	Card Description
E0	Chip Data
E2	MPP Host Text
E3	Address 1
E4	Address 2
E5	Address 3
E6	Address 4
E7	Postal Code
E8	MPP Card Number
E9	MPP Card Program
EA	Transaction Reference Number (Timestamp)
EB	MPP Final Balance
EC	MPP Expiry Date
ED	Previous Balance
EE	Current Balance
EF	Previous Balance Negative Indicator
EG	Current Balance Negative Indicator
F0	Foreign MID
F1	Issuer Category
F2	Currency Name
F3	Plan Code
F4	Tenure
F5	First Amount
F6	Monthly Amount
F7	Last Amount
F8	Interest Amount
FA	Foreign Amount
G0	Sale Total Amount
G1	Total NETS Purchase Amount
G2	Total VOID NETS Purchase Amount
G3	Total UnionPay Purchase Amount
G4	Total VOID UnionPay Purchase Amount
G5	Total UnionPay Refund Amount
G6	Total UnionPay Pre-Auth Completion Amount
G7	Total VOID UnionPay Pre-Auth Completion Amount
G8	Total BCA Purchase Amount
G9	Total VOID BCA Purchase Amount
GA	Total BCA Pre-Auth Completion Amount
GB	Total NFP Purchase Amount
GC	Total NCC Purchase Amount

Field code	Description
GD	Total NFP Topup (NETS) Amount
GE	Total NCC Topup (NETS) Amount
GF	Total Cashback Amount
GG	Total Cash Deposit Amount
GH	Total NPC Topup (NETS) Amount
GI	Total NPC Topup (CASH) Amount
GK	Reserved For Future Use
GL	Total NFP Refund amount
GP	Total NETS Refund amount
H4	ECR Reference Number
H5	Terminate Certificate (TTC Cert.)
H6	STAN
H7	CashCard Transaction Counter (CTC)
H8	BlackList Version
H9	Certificate (Sing Certificate)
HA	Checksum
HB	RFU8
HC	Host Response Code
HD	Enhanced ECR Reference Number
HE	Card Balance
L1	Loyalty Program Name
L2	Loyalty Type
L3	Redemption Value
L4	Current Balance
L5	POS Message
L6	Loyalty Marketing Message
L7	Card Name
L8	Loyalty Program Exp Date
L9	Loyalty Marketing Message
N0	Sale Total Count
N1	Total NETS Purchase Count
N2	Total VOID NETS Purchase Count
N3	Total UnionPay Purchase Count
N4	Total VOID UnionPay Purchase Count
N5	Total UnionPay Refund Count
N6	Total UnionPay Pre-Auth Completion Count
N7	Total VOID UnionPay Pre-Auth Completion Count
N8	Total BCA Purchase Count
N9	Total VOID BCA Purchase Count
NA	Total BCA Pre-Auth Completion Count
NB	Total NFP Purchase Count
NC	Total NCC Purchase Count
ND	Total NFP Topup (NETS) Count
NE	Total NCC Topup (NETS) Count
NF	Total Cashback Count
NG	Total Cash Deposit Count
NH	Total NPC Topup (NETS) Count
NI	Total NPC Topup (CASH) Count
NK	Total NPC Activation Count
NL	Total NFP Refund count
NM	Total NFP Card Sale Activation count

Field code	Description
NO	Total NFP ATU Activation count
NP	Total NETS Refund count
M0	Callback Function Type
M1	Message Index
M2	Message
M3	Number of Selection
M4	Selection Index
M5	Selection Label
M6	Selection Name (Group name)
M7	Message Payload to be send or received from host
M8	
MN	Mobile Number
MR	Consumer Routing Indicator
MC	Consumer Choice Indicator
MS	Consumer Selection Indicator
O1	Offline Txn Type
P0	Parameter List
P1 – PZ	Reserved for parameter items
Q0	QR Payload
Q1	QR Timer
R0	Response Message I
R1	Response Message II
RP	Receipt Text Format
RQ	Signature image in bitmap format
S0	Card Status
S1	Block Status
S2	ATU Status
S3	Purse Status
S4	Record Index
S5	Number of Record
S6	Maximum Transaction Log
S7	Number of Records Read
S8	Transaction Logs
S9	Max Transaction Log Count
SA	NETS Host status
SB	UPI Host status
SC	UPOS Host status
SD	RSVP Host status
SE	Merchant Prepaid Host status
T1	Top-up Option
T2	Transaction Type Indicator
T3	Transaction Specific Data
T4	ATU Debt Amount
T5	Payment Type
T6	Header message 1
T7	Header message 2
T8	Footer message 1
T9	Display Time
TA	NFC Payload
TB	QR Payload
ZA	Total Fee (Top-up)

Field code	Description
ZB	Fee due to Merchant (Top-up)
ZC	Fee due from Merchant (Top-up)
ZD	Card Expiry
ZE	Advertising Message
ZF	Software Version
ZG	Black List Version
ZH	Parameter Version
ZI	Block List Version
ZJ	Response Code (Top Up)
ZK	Response Text (Top Up)
ZP	Purchase Fee
ZT	Additional GB Information
ZY	Transaction Code

Annex C. Response Code List

The following table lists the possible response codes from the terminal for NETS transactions.

Response Code	Meaning	Action
00	APPROVED	POS to include validation of approval code in response message to ensure it is no “000000”
01	Requested function is not available	The feature is not available for POS to use. Key information: 1. Function code 2. Version code
02	Terminal is in pre-activation mode	The terminal is not ready to receive any function request including administrative. Contact NETS.
03	For iUN terminal The iUC (contactless reader) is not connected or faulty.	Check that the contactless reader is connected. Pull out the cable and connect again then restart the terminal. If problem persist replace the reader.
04	For iUN terminal The iUR (contact reader) is not connected or not activated or faulty.	Check that the contact reader is connected. Check that the contact reader has a green light on and not red. If green light is on, pull out the cable and connect again then restart the terminal. If problem persist replace the reader.
05	3rd Party Application not available	To check with terminal provider on application loading.
20	Mandatory field codes are missing	Check the message being sent to the terminal.

Response Code	Meaning	Action
51	Host has declined the authorization request from terminal	POS should refer to field code HC for more information on the decline. The list of possible value for HC is at Annex A. POS can also refer to the field code 02 response texts for more details. Key information: 1. Function code 2. Version code 3. HC value
C0	Unable to detect or read card	Please check if the card is inserted, tapped or swiped correctly
C1	Invalid card	Please make sure that the card used to perform the transaction is valid.
HS	Terminal encountered connection problem	Please check terminal communication connectivity, e.g. SIM card, Ethernet cable, phone line, etc.
LR	Logon is required	POS should perform logon function or if available manually press logon button on the terminal
TO	Timeout occurred during authorization request to host	Terminal is able to connect to host but unable to receive a response. Please contact NETS.
Z0	Callback: Display Message	Terminal is able to connect to host but unable to receive a response. Please contact NETS.
Z1	Callback: Credit Card Application Selection	External module should prompt the user to confirm which credit card scheme application in the card should be used for the transaction.
SF	Terminal requires settlement	Perform Settlement for all acquirers

Annex E. Acquirer Name

Name	Value
UOB	UOB
AMEX	AMEX
OCBC	OCBC

Annex F. Scheme Category

Gateway Name: UOB

Name	Value
VISA 1	V1
VISA 2	V2
MASTERCARD 1	M1
AMEX	A1

Gateway Name: AMEX

Not Applicable

Annex G. Loyalty Information

Loyalty Information – Loyalty Type "01" – POSB Everyday

Field Description	Data Type	Len (Bytes)	Remarks
Loyalty Awarded	ASC	12	
Loyalty Redeemed	ASC	12	
Loyalty Balance	ASC	12	

Loyalty Information – Loyalty Type "02" – UOB

Field Description	Data Type	Len (Bytes)	Remarks
Trans Amount	ASC	12	
Nett Amount	ASC	12	
Redeem Amount	ASC	12	
No of pools	ASC	1	
Club Id	ASC	3	Repeat for the no. of pools
Pool Id	ASC	3	Repeat for the no. of pools
Redeem Amount	ASC	12	Repeat for the no. of pools If Club id isUNI : Redeem UNI$, see IRR Redeem Amt ($) for actual $ value.
Award Amount	ASC	12	Repeat for the no. of pools
Point Balance	ASC	1	Repeat for the no. of pools
Balance Amount	ASC	12	Repeat for the no. of pools
Pool Description	ASC	30	Repeat for the no. of pools
IRR Club ID	ASC	3	To be matched against the Club ID above and printed before the Club Redeem Amount, Award Amount and Bal Amount
IRR Qty	ASC	2	
IRR Description	ASC	40	
IRR Redeem Amount ($)	ASC	12	The actual amount ($) value for the IRR that is redeemed
Loyalty Message 1	ASC	42	
Loyalty Message 2	ASC	42	
Loyalty Message 3	ASC	42	
Loyalty Message 4	ASC	42	
Loyalty Message 5	ASC	42	
Loyalty Message 6	ASC	42	

Loyalty Information – Loyalty Type "03" – UOB (SPC)

Field Description	Data Type	Len (Bytes)	Remarks
Discount Amount	ASC	12	
Base Amount	ASC	12	
Nett Amount	ASC	12	
Redeem Amount	ASC	12	
No of pools	ASC	1	
Club Id	ASC	3	
Pool Id	ASC	3	
Redeem Amount	ASC	12	
Redeem ($ Value)	ASC	12	Example: $ redeem for UNI$
Award Amount	ASC	12	
Point Balance	ASC	1	
Balance Amount	ASC	12	
Pool Description	ASC	30	
IRR Club ID	ASC	3	To be matched against the Club ID above and printed before the Club Redeem Amount, Award Amount and Bal Amount

IRR Qty	ASC	2	
IRR Description	ASC	40	
IRR Redeem Amount ($)	ASC	12	The actual amount ($) value for the IRR that is redeemed

Loyalty Information – Loyalty Type “04” – Diners

Field Description	**Data Type**	**Len (Bytes)**	**Remarks**
Loyalty Awarded	ASC	12	
Loyalty Redeemed	ASC	12	
Loyalty Balance	ASC	12	
Loyalty Card No	ASC	16	
Loyalty Message	ASC	50	

Loyalty Information – Loyalty Type “05” – DBS Instant Rewards

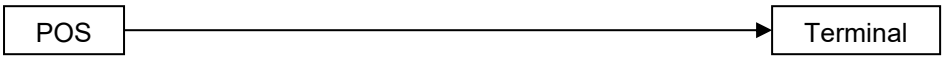
Field Description	**Data Type**	**Len (Bytes)**	**Remarks**
Loyalty Awarded	ASC	12	
Loyalty Redeemed	ASC	12	
Loyalty Balance	ASC	12	

Annex H. 3rd Party Command Pass-Through

In order for the terminal to recognize that the incoming transaction is not for NETS services, please specify the following:

Request

STX	LEN	Message Header					Message Data 1	ETX	LRC
		ECN	Function Code	Version Code	RFU	Separator	3rd Party Command Request		

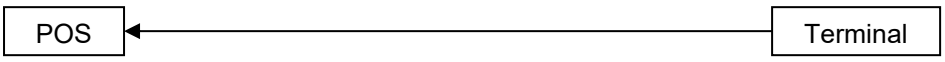


Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	ECR Control Number A numbering sequence that the ECR define. The number must be unique to every Function request.
Function Code	ASC	2	3rd Party Function Code
Version Code	ASC	2	"02" (0x30 0x32) – UOB Commands "03" (0x30 0x33) – Citibank Commands "04" (0x30 0x34) – Auresys Commands "05" (0x30 0x34) – Qwikcilver Commands "06" (0x30 0x36) – OCBC Commands "07" (0x30 0x37) – EZLink Commands "08" (0x30 0x38) – Diners Commands
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Response

STX	LEN	Message Header					Message Data 1	ETX	LRC
		ECN	Function Code	Response Code	RFU	Separator	3rd Party Command Response		



Message Header Description

Name	Attribute	Len (Bytes)	Remark
ECN	ASC	12	Echoes from the request
Function Code	ASC	2	Echoes from the request
Response Code	ASC	2	"00" – Command received and processed by 3rd party application "05" – 3rd party application not available
RFU	ASC	1	Value is 0 (0x30)
Separator	HEX	1	Value is (0x1C)

Annex I. List of Function Code (FC) for Merchant POS Implementation

FC	Function Name	Retails	Remarks
55	Request Terminal Status	Mandatory	
56	Get Last Approved Transaction	Mandatory	
80	NETS Logon	Mandatory	
81	NETS Settlement	Mandatory	
84	TMS / CTS Update	Mandatory	
30	ALL NETS Purchase	Mandatory	
I5	Credit Card Settlement	Conditional	If Credit Card acceptance with NETS, this function is mandatory

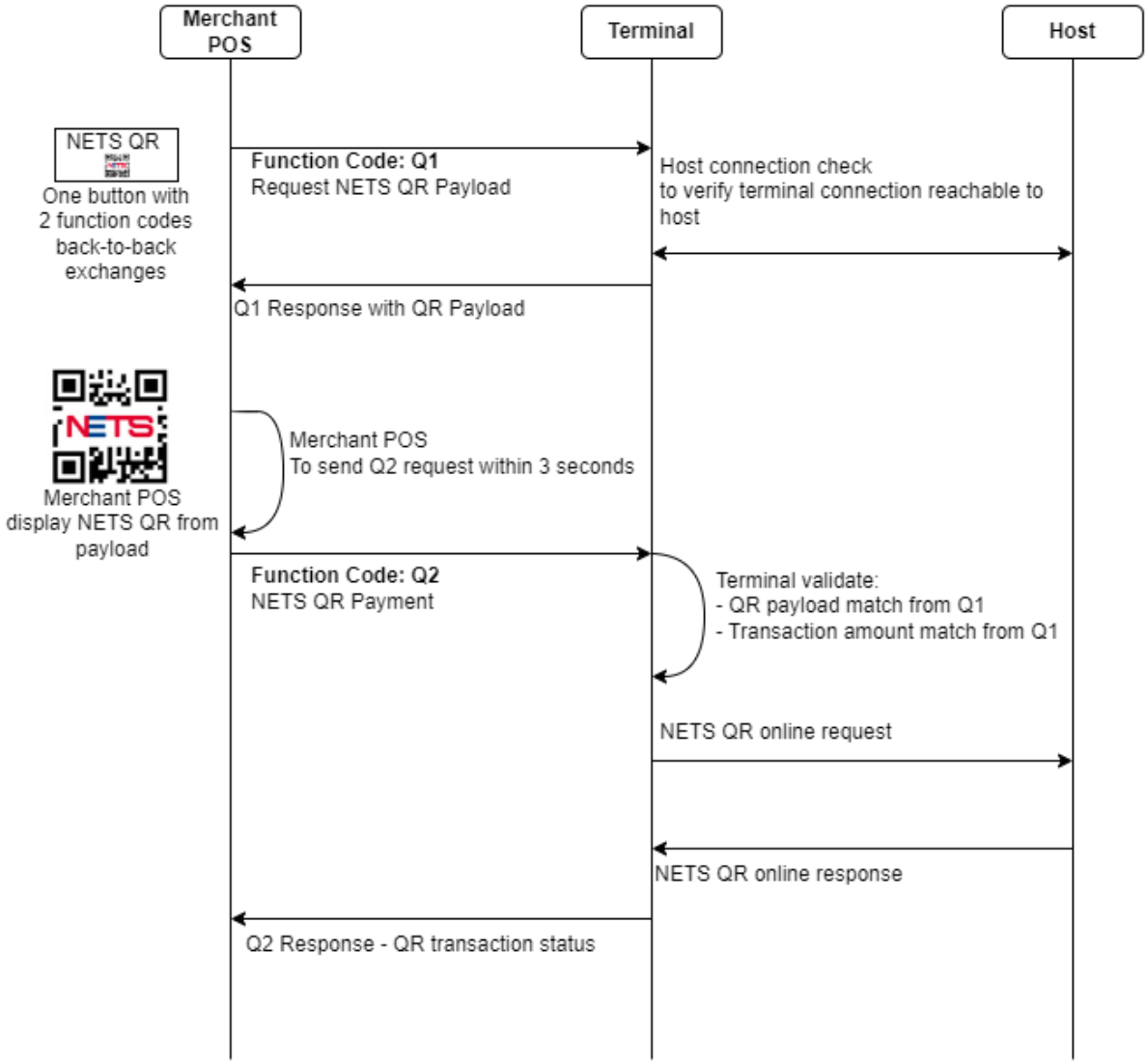
Notes:

- Merchant POS can implement follow logic to trigger automatic command for end of business day in following sequences:
  1. Credit Card Settlement
  2. NETS Settlement
  3. NETS Logon
  4. TMS / CTS Update

Annex J. QR on Merchant's POS – Implementation Flow

Function codes for QR implementation:

1. Request NETS QR Payload (Function Code Q1)
2. NETS QR Payment (Function Code Q2)



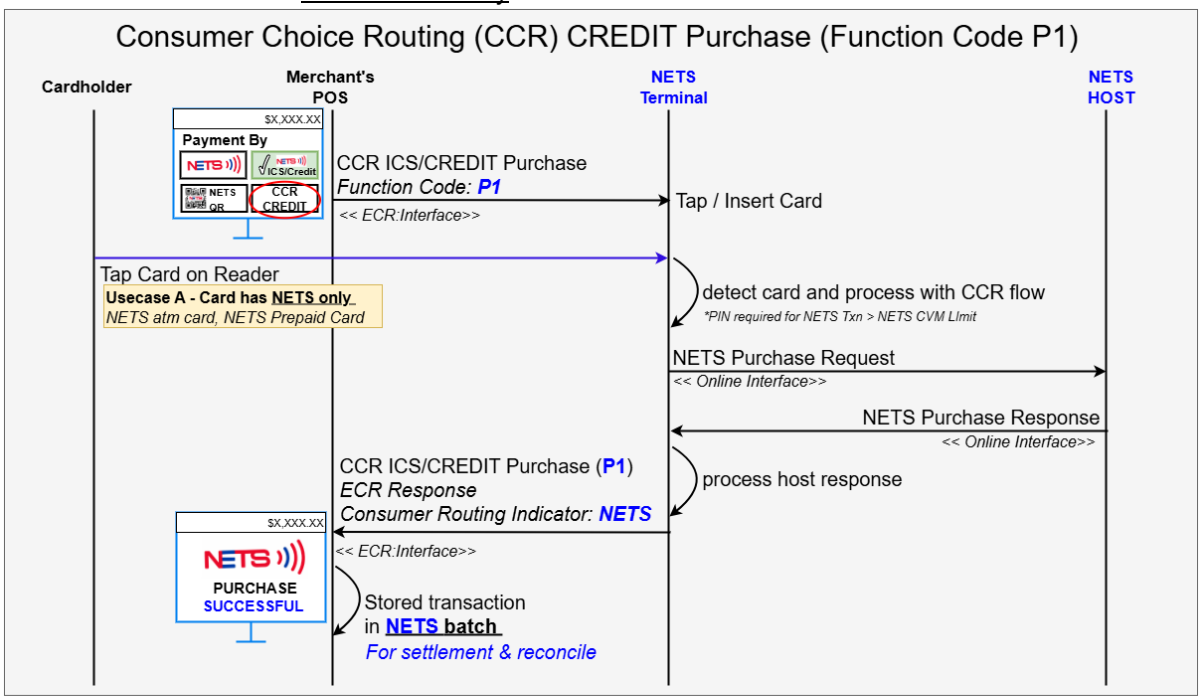
Annex K. Consumer Choice Routing (CCR) – Implementation Flow

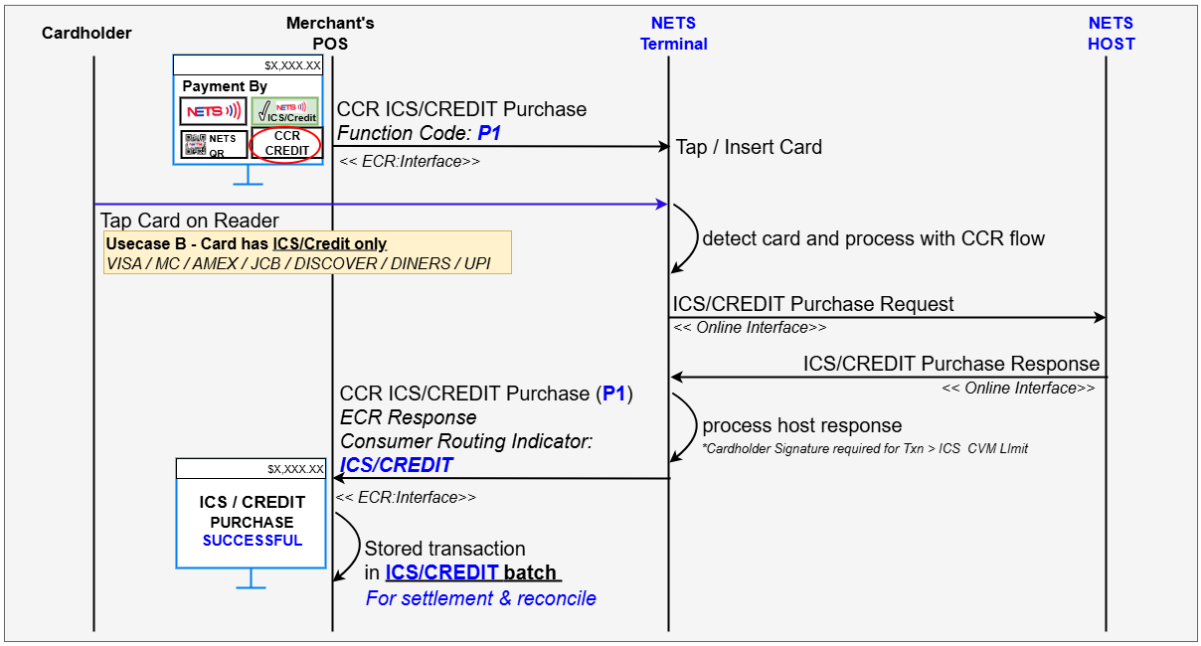
Credit Purchase with CCR (Function Code P1, P2, P3)

This function triggers Credit Purchase with option:

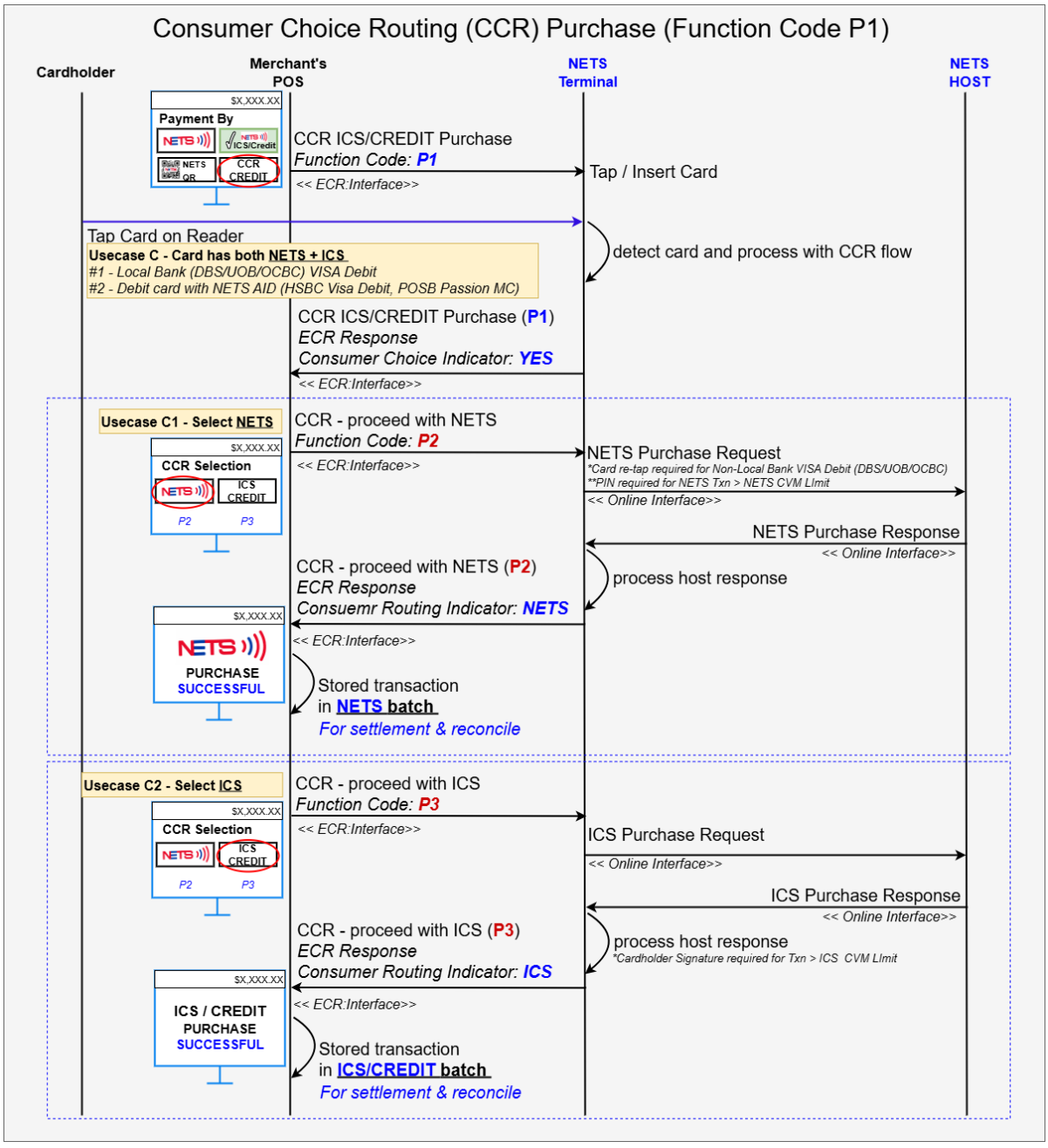
- CCR Selection, if card has both NETS and ICS/CREDIT capability
- ICS/Credit Purchase, if card only has ICS/CREDIT only
- NETS Purchase, if card has NETS only

Use Case #A – Card has NETS only
Use Case #B – Card has ICS/CREDIT only





Use Case #C – Card has both NETS and ICS/CREDIT



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